



COMPLETE STUDY NOTES

Data Communications and Networking

Units 1 & 2 — Exam Preparation Guide



Chapters 1, 2, 3, 6 & 7

Theory, Visual Diagrams, Worked Examples & Practice Questions

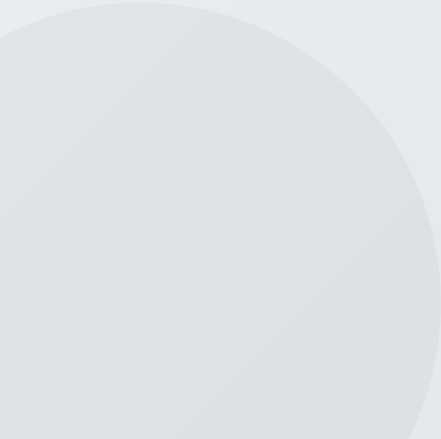


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Chapter 1: Data Communications

Data communications forms the foundation of modern networking. It refers to the exchange of data between two devices via some form of transmission medium, such as a wire cable. For communication to occur, the devices must participate within a communication system made up of hardware (physical equipment) and software (programs). The effectiveness of any data communications system depends on four fundamental characteristics that govern how well information flows from source to destination.

Four Fundamental Characteristics

Delivery. The system must deliver data to the correct destination and recipient only.

Accuracy. Data must be delivered exactly as sent. Corrupted data that cannot be recovered is useless.

Timeliness. Data must arrive on time. Real-time transmission (for video and audio) requires delivery as data is produced, without significant delay.

Jitter. This refers to variation in packet arrival time. If video packets sent every 30ms arrive at varying intervals (some at 30ms, others at 40ms), the resulting video quality becomes uneven and unwatchable.

1.1 Components of Data Communication

A data communications system comprises five distinct components that work together to ensure successful information transfer. Understanding each component and its role is essential for grasping how networks function at a fundamental level.

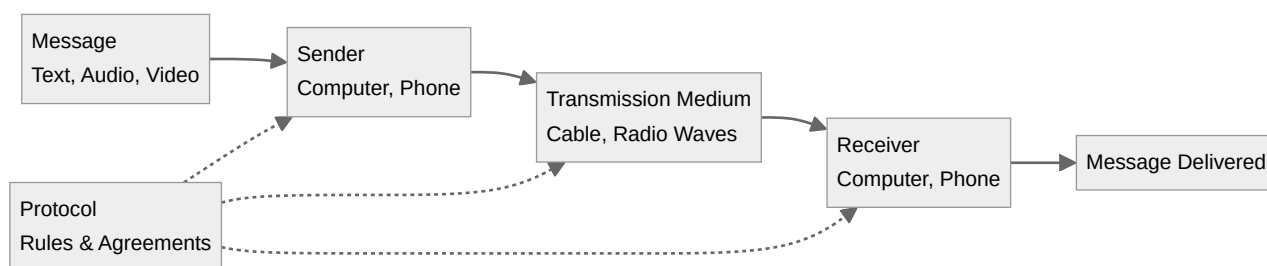


Figure 1.1 The five components of a data communication system

Table 1.1 Five Components of Data Communication

Component	Description	Examples
Message	The information to be communicated	Text, numbers, pictures, audio, video
Sender	The device that sends the data	Computer, phone, camera, workstation
Receiver	The device that receives the message	Computer, phone, printer
Transmission Medium	The physical path the message travels on	Twisted-pair wire, fiber-optic cable, radio waves
Protocol	A set of rules governing communication	TCP/IP, HTTP, FTP

Analogy: A protocol is like a language agreement. If two devices are connected but have no protocol, it is like a French speaker trying to communicate with a Japanese speaker. They can physically hear each other, but they cannot understand one another.

1.1.1 Data Representation

Information exists in various forms, and computers represent each form differently. All data eventually becomes bit patterns (sequences of 0s and 1s) for transmission.

Text. Text is represented as bit patterns. The prevalent coding system today is **Unicode**, which uses 32 bits to represent symbols from virtually any language. **ASCII** is an older system that now forms the first 127 characters of Unicode, providing backward compatibility with legacy systems.

Numbers. Unlike text which requires coding systems, numbers are directly converted to binary. This direct conversion makes mathematical operations simpler and more efficient.

Images. Images consist of a matrix of pixels (picture elements). Higher pixel counts yield better resolution but require more memory. A black-and-white image uses 1 bit per pixel, while grayscale and color images use more bits. Color images typically use the **RGB** method, combining Red, Green, and Blue components to produce the full spectrum of visible colors.

Audio. Audio represents a recording of sound. Unlike text or numbers which are discrete, audio is continuous and analog in nature. It must be sampled and converted to digital form for computer processing.

Video. Video represents recording of pictures or movies. It can be continuous (as from a television camera) or a combination of discrete images arranged sequentially to create the illusion of motion.

1.1.2 Data Flow (Communication Modes)

Communication between devices can occur in three distinct modes, each with specific characteristics and use cases. The choice of mode depends on the application's requirements for directionality and simultaneity.

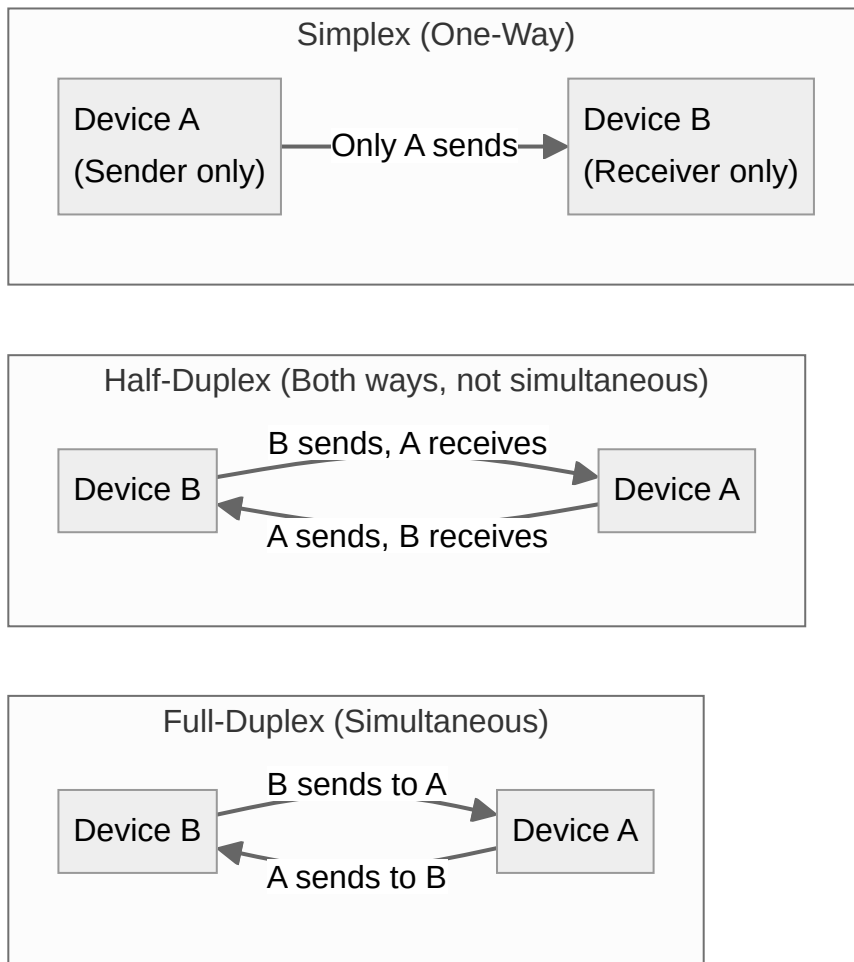


Figure 1.2 The three modes of data flow

Table 1.2 Comparison of Data Flow Modes

Mode	Direction	Analogy	Example
Simplex	Unidirectional: only one device transmits, the other only receives	A one-way street where traffic flows in a single direction only	Keyboard to monitor; radio broadcast
Half-Duplex	Bidirectional, but not simultaneous: devices take turns	A one-lane road allowing traffic in both directions, but only one direction at a time	Walkie-talkies (must say "Over" to switch)
Full-Duplex	Bidirectional and simultaneous: both devices transmit and receive at the same time	A two-way street with traffic flowing in both directions simultaneously	Telephone network

1.2 Networks

A network is the interconnection of a set of devices capable of communication. Two key terms define the participants in a network: a **host** (or end system) refers to devices like computers, desktops, laptops, phones, or security systems; while a **connecting device** refers to hardware that links the network together, such as routers (which connect networks to other networks), switches (which connect devices together), and modems (which change the form of data between analog and digital).

1.2.1 Network Criteria

For a network to be effective, it must satisfy three key criteria. First, **performance** is measured by transit time (the time for data to travel from source to destination) and response time (the time between a request and its response). An important trade-off exists between throughput and delay: sending more data increases throughput but may cause congestion, which in turn increases delay. Second, **reliability** is measured by the frequency of failure, recovery time after failure, and the network's robustness during catastrophic events. Third, **security** involves protecting data from unauthorized access, preventing damage to data, and ensuring against data loss.

1.2.2 Physical Structures

Type of Connection

A link represents the communications pathway that transfers data between devices. Two fundamental connection types exist.

Point-to-Point connection. A dedicated link connects exactly two devices, with the entire capacity of the link reserved for their exclusive use. This is analogous to a television remote: when you change the channel, you establish a direct line between the remote and the TV that no one else shares.

Multipoint (Multidrop) connection. More than two specific devices share a single link. The capacity is shared among multiple devices, either spatially (everyone communicates simultaneously, requiring sufficient capacity) or temporally (devices take turns). This resembles a conference call where multiple participants share the same line.

Physical Topology

Topology refers to the geometric representation of the relationship between devices (called nodes) and their connecting links. Four fundamental topologies form the basis of all network designs.

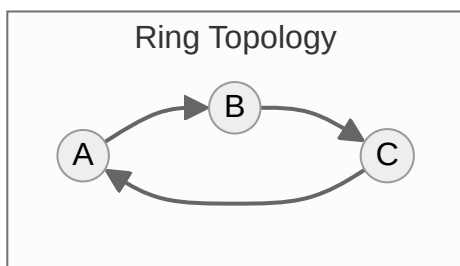
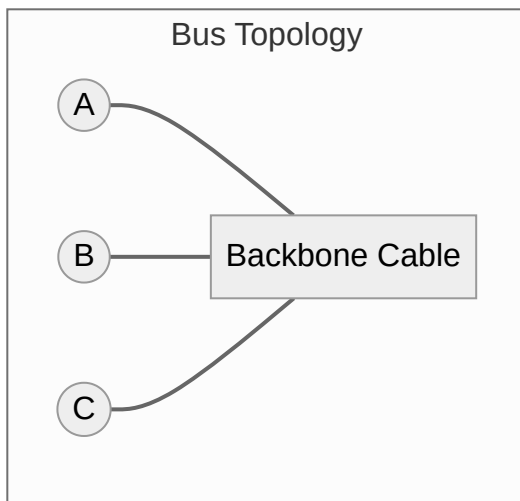
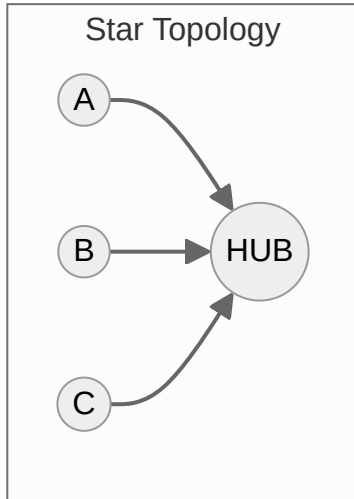
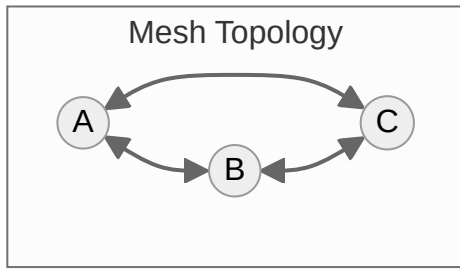


Figure 1.3 *The four fundamental network topologies*

Table 1.3 Comparison of Network Topologies

Topology	Structure	Advantages	Disadvantages	Use Case
Mesh	Every device has a dedicated point-to-point link to every other device. Formula: $n(n - 1)/2$ duplex links for n nodes.	No traffic problems; robust (if one link fails, others work); privacy guaranteed.	Expensive and bulky (massive cabling); difficult installation.	Regional telephone offices
Star	Each device connects only to a central hub. Devices do not link directly.	Cheaper than mesh (less cable); robust (one link failure affects only that device); easy to manage.	Single point of failure: if the hub fails, the entire network fails.	Local-Area Networks (LANs)
Bus	Multipoint configuration with one long backbone cable linking all devices via drop lines and taps.	Easiest to install; uses the least amount of cable.	Signal reflection at taps; fragile (a break stops all transmission).	Legacy LANs
Ring	Each device connects to two immediate neighbors. Signal travels in one direction; each device acts as a repeater.	Easy to install; simple fault isolation (device sounds alarm if signal not received).	Unidirectional; a break can disable the entire network (unless dual ring is used).	IBM Token Ring (legacy)

1.3 Network Types and Switching

1.3.1 Local Area Network (LAN)

A LAN is a privately owned network connecting hosts within a single office, building, or campus. Each host in a LAN has a unique address that identifies it on the network. LAN technology has evolved significantly over time.

In the past, all hosts on a LAN shared a common cable. If Host A sent a message to Host B, every host received the message, but only Host B would keep it. This approach was inefficient because bandwidth was wasted delivering messages to hosts that did not need them.

Modern LANs use a **switch**, which is an intelligent device that recognizes the destination address and guides packets only to the specific host that needs them. This reduces unnecessary traffic and allows multiple simultaneous conversations between different pairs of hosts.

1.3.2 Wide Area Network (WAN)

A WAN spans a wider geographical area, potentially covering a town, state, country, or even the entire world. Three key differences distinguish WANs from LANs: scope (LAN covers a building, WAN covers the world), connectivity (LAN connects hosts directly, WAN connects connecting devices like switches and routers), and ownership (LANs are privately owned while WANs are typically leased from communication companies).

Two types of WANs exist. A **point-to-point WAN** connects exactly two devices through a transmission medium. A **switched WAN** is a network with more than two ends, used in the global communication backbone, and is essentially a combination of several point-to-point WANs connected by switches.

Internetwork

It is rare to see a LAN or WAN in isolation. When two or more networks connect, they form an **internetwork** (or *internet* with a lowercase 'i'). For example, an office on the East Coast (LAN) might connect to an office on the West Coast (LAN) using a leased service (WAN). The router plays a critical role here: if you send a message to a colleague in the same office, the router blocks it and keeps it local; if you send it to the other office, the router passes it to the WAN.

1.3.3 Switching

An internet is a switched network, where a switch's primary job is to forward data from one network to another. Two fundamental switching approaches exist.

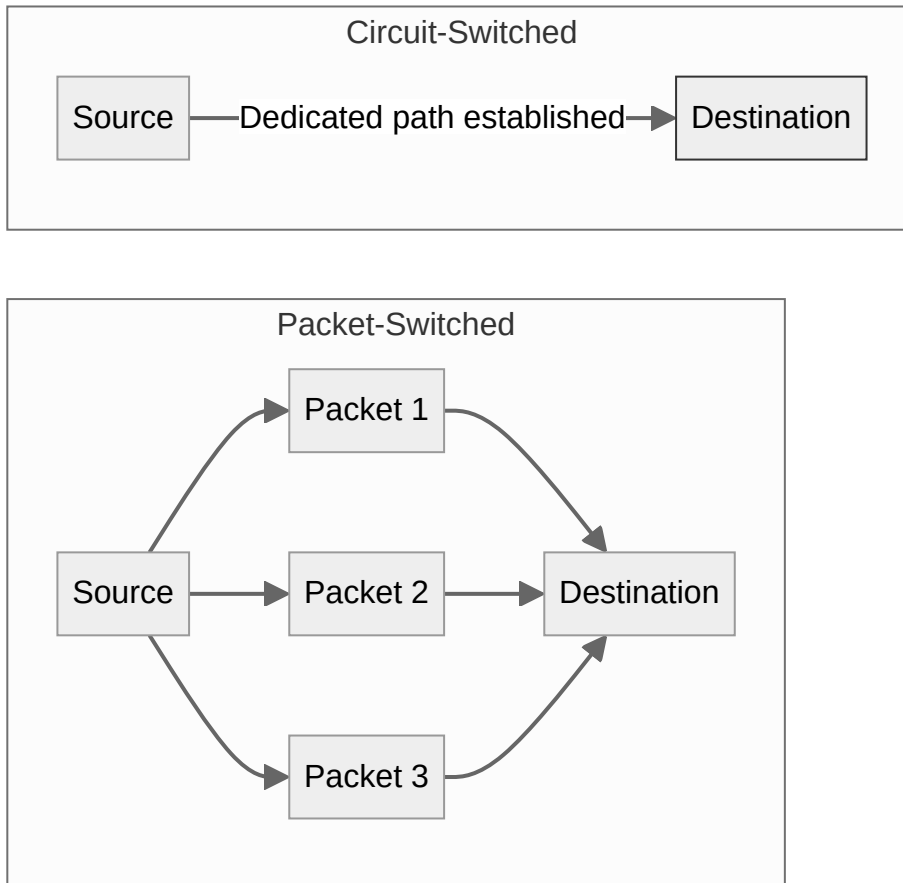


Figure 1.4 *Circuit switching vs. Packet switching*

Circuit-Switched Network. A dedicated connection (circuit) is established between two ends and remains reserved for the duration of the communication. This is analogous to a traditional telephone call: when you call someone, a specific line is reserved for you. Even if you stop talking, that line remains yours and no one else can use it. This approach guarantees a connection but is inefficient because unused capacity goes to waste.

Packet-Switched Network. Communication occurs in independent blocks of data called packets. This resembles postal mail: you do not reserve the entire road to send a letter; instead, you drop a letter (packet) in a box and it travels independently along with everyone else's mail. Routers use a **store-and-forward** mechanism, holding packets in a queue and sending them when the line becomes free. This approach is more efficient because lines are shared, though packets may encounter delays while waiting in queues.

1.3.4 The Internet and Its Hierarchy

The distinction between *internet* (lowercase, meaning two or more networks communicating) and *Internet* (uppercase, referring to the specific global collaboration of thousands of interconnected networks) is important. The Internet operates as a hierarchy with three levels: **backbones** (international ISPs like Sprint and AT&T, connected by peering points), **provider networks** (national or regional ISPs that pay to use

1.3.5 Accessing the Internet

Users must physically connect to an ISP, usually via a point-to-point WAN. Four common access methods exist: telephone networks (dial-up using a modem, or DSL which allows simultaneous voice and data), cable networks (using cable TV infrastructure, though speed varies based on neighbor usage), wireless networks (access via wireless WANs), and direct connections (large organizations leasing high-speed WANs directly from carriers).

Chapter 2: Network Models

Network models provide a structured framework for understanding how data moves through communication systems. While a simple protocol suffices for basic communication, complex systems like the Internet require dividing tasks into different layers, each with specific responsibilities. This layered approach, called protocol layering, is the architectural foundation of all modern networks.

2.1 Protocol Layering

To understand why layering is necessary, consider two scenarios involving friends Maria and Ann communicating with each other.

Scenario 1: Simple Face-to-Face Communication. When Maria and Ann are neighbors talking directly, communication occurs in a single layer. The protocol is straightforward: greet when meeting, use a common vocabulary, take turns speaking, and exchange parting words. No intermediaries are needed.

Scenario 2: Complex Remote Communication with Encryption. When Ann moves to a distant city, they must communicate via mail while keeping their ideas secret. This requires three layers: Layer 3 (High Level) where Maria speaks to a machine that writes plaintext; Layer 2 (Security) where the machine encrypts plaintext into ciphertext; and Layer 1 (Transmission) where the machine places the ciphertext in an envelope and mails it. At the receiving end, the process reverses: receive mail, decrypt, then read.

2.1.1 Why Use Layering?

Modularity through Independent Layers. A layer functions as a "black box": we care about its inputs and outputs, not its internal implementation. If Maria and Ann want to change their encryption method (Layer 2), they simply replace the Layer 2 machine. Layer 1 (mail) and Layer 3 (writing) remain completely untouched. This separation of service from implementation means you can change how a layer works without breaking the system, as long as the service it provides remains the same.

Intermediate Systems. In complex networks like the Internet, intermediate devices such as routers do not need to understand all layers. They only need the lower layers to pass data along, making the system cheaper and more efficient.

EXAM TIP

The main disadvantage of layering is that a single super-machine would theoretically be faster and easier than multiple interacting machines. However, if that super-machine breaks or needs updating, the entire system must be replaced, which is costly. Layering trades some efficiency for maintainability and flexibility.

2.1.2 Principles of Protocol Layering

When designing layered protocols, two fundamental principles must be followed.

First Principle: Bidirectional Capability. If communication is two-way, every layer must be able to perform opposite tasks. Layer 3 must both listen and talk; Layer 2 must both encrypt and decrypt; Layer 1 must both send and receive.

Second Principle: Identical Objects. The object passing between Layer X at the sender and Layer X at the receiver must be identical. Layer 3 at both sites handles plaintext; Layer 2 at both sites handles ciphertext; Layer 1 at both sites handles the physical mail object.

2.1.3 Logical Connections

Because of the principles above, we can imagine a logical connection directly between corresponding layers at sender and receiver. The physical flow of data moves vertically: down from Layer 3 to Layer 1 at the sender, across the physical medium, then up from Layer 1 to Layer 3 at the receiver. However, logically, we pretend there is a direct horizontal line between Layer 2 at the sender and Layer 2 at the receiver. The encryption layer does not care about envelopes or mail carriers; it effectively "talks" directly to the decryption layer at the other end.

2.2 TCP/IP Protocol Suite

The TCP/IP (Transmission Control Protocol/Internet Protocol) suite represents the standard protocol suite used for the Internet today. It is hierarchical, meaning each upper-level protocol relies on services provided by lower-level protocols. While originally defined as a 4-layer model, TCP/IP is today viewed as a 5-layer model for better understanding and teaching.

2.2.1 Layered Architecture: Who Uses Which Layers?

Different network devices operate at different depths within the layer stack. Understanding which devices process which layers is fundamental to network design and troubleshooting.

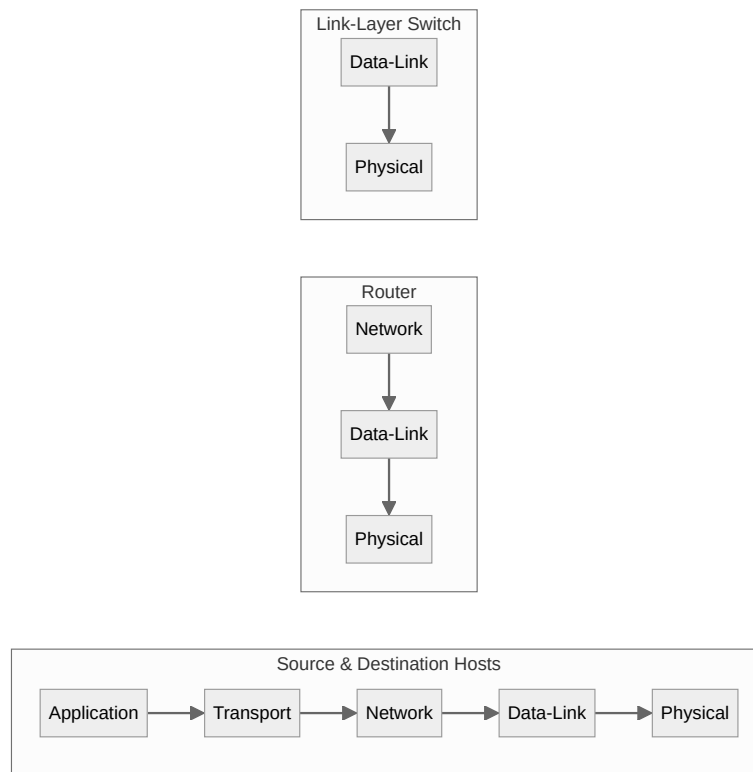


Figure 2.2 *Devices and the layers they process in the TCP/IP model*

2.2.2 The Five Layers of TCP/IP

The TCP/IP protocol suite is discussed from bottom (hardware) to top (user application).

Physical Layer. The Physical layer is responsible for carrying individual bits across the link. It connects devices via a transmission medium (cable or air), converting bits into signals (electrical or optical) that the medium can carry. The logical unit at this layer is a single bit.

Data-Link Layer. The Data-Link layer moves the packet (called a **frame** at this layer) across a specific link. It ensures hop-to-hop delivery from one node to the very next node, such as from your laptop to the Wi-Fi router. TCP/IP does not define a specific Data-Link protocol; it supports all standard protocols including Ethernet and Wi-Fi.

Network Layer. The Network layer is responsible for host-to-host communication and routing. The packet at this layer is called a **datagram**. The key protocol is IP (Internet Protocol), which is connectionless and provides no flow or error control, relying on higher layers for reliability. Auxiliary protocols include ICMP (which reports routing problems), DHCP (which assigns IP addresses), and ARP (which finds the link-layer MAC address when given a network-layer IP address). The Network layer functions like a postal sorting center: it examines the destination address and determines which route the packet should take next.

Transport Layer. The Transport layer handles process-to-process delivery, separating network duties from application duties. It creates an end-to-end logical connection where the Transport layer at the source "talks" directly to the Transport layer at the destination. Two key protocols exist: **TCP** (Transmission Control Protocol) is connection-oriented and reliable, providing flow control and error correction; **UDP** (User Datagram Protocol) is connectionless, unreliable, but fast, making it suitable for small messages where retransmission would be too slow. Think of TCP as certified mail with a receipt and delivery guarantee, and UDP as regular mail that is faster and cheaper but offers no delivery confirmation.

Application Layer. The Application layer enables the user (human or software) to access the network. Communication occurs between two processes (running programs). Common protocols include HTTP for the World Wide Web, SMTP for email, FTP for file transfer, and DNS for translating domain names into IP addresses.

2.2.3 Encapsulation and Decapsulation

Encapsulation and decapsulation describe how data is packaged and unpackaged as it moves vertically through the layers. This process is fundamental to understanding how data travels through a network.



Figure 2.3 *The encapsulation and decapsulation process*

At the **source host**, data moves downward through the layers. Each layer treats data from the layer above as payload and adds its own header. The Application layer creates the message. The Transport layer takes the message as payload and adds a transport header containing source and destination port numbers along with flow control information, producing a segment (TCP) or user datagram (UDP). The Network layer takes the segment as payload and adds a network header with source and destination IP addresses, producing a datagram. The Data-Link layer takes the datagram as payload and adds a link-layer header with source and destination MAC addresses, producing a frame.

At a **router**, the device decapsulates by receiving the frame and removing the header to read the datagram. It checks the IP address to find the best path (routing), generally without changing the datagram content. Then it encapsulates by wrapping the datagram into a new frame suitable for the next link.

At the **destination host**, data moves upward. Each layer removes its specific header, checks for errors, and hands the pure payload to the layer above until the Application layer receives the original message.

Analogy (Russian Nesting Doll): Think of encapsulation like nested containers. The innermost doll is the letter (Message). It goes inside an envelope with apartment numbers (Port Numbers), which goes inside a mailbag with the building address (IP Address), which goes inside a delivery truck container (MAC Address).

2.2.4 Addressing in TCP/IP

Communication requires identifying who is talking to whom. Different layers use different address types, each serving a specific purpose in the delivery process.

Table 2.2 Addressing at Each TCP/IP Layer

Layer	Packet Name	Address Type	Scope	Example
Application	Message	Names (user-specific)	Global	email@domain.com, google.com
Transport	Segment/Datagram	Port Numbers (identifies the program)	Local	Port 80 (Web), Port 25 (Email)
Network	Datagram	Logical Addresses (IP)	Global (identifies the connection)	192.168.1.1
Data-Link	Frame	Physical Addresses (MAC)	Local (identifies the device on the link)	00:1A:2B:3C:4D:5E
Physical	Bits	None	N/A	Bits do not need addresses

2.2.5 Multiplexing and Demultiplexing

Because the TCP/IP suite allows multiple protocols to run simultaneously, mechanisms are needed to mix and un-mix data traffic.

Multiplexing occurs at the source: a protocol at a lower layer accepts packets from several higher-layer protocols. For example, the IP protocol at the Network layer can carry both TCP and UDP packets simultaneously.

Demultiplexing occurs at the destination: a protocol decapsulates a packet and delivers it to the correct higher-layer protocol. When IP receives a packet, it examines a protocol field in the header to determine whether the packet should go to TCP or UDP.

2.3 The OSI Model

While TCP/IP serves as the practical standard for the Internet, the International Organization for Standardization (ISO) established the Open Systems Interconnection (OSI) model in the late 1970s as a comprehensive reference standard. An **open system** is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture. The OSI model is not itself a protocol but rather a blueprint for understanding and designing flexible, interoperable network architectures.

2.3.1 OSI versus TCP/IP

When comparing the two models, a distinct difference emerges in how they organize layers. The TCP/IP suite lacks distinct Session and Presentation layers. Instead, TCP/IP combines the duties of Session, Presentation, and Application layers into a single Application Layer.

TCP/IP combined these layers for good reasons. Some Session layer functions, like tracking connections, are already handled by the Transport layer (TCP). The Application layer in TCP/IP is not monolithic: if a specific application needs session or presentation features like encryption or dialogue control, those features can be built directly into that application's software rather than requiring separate, strict layers. This design offers greater flexibility.

2.3.2 Why OSI Did Not Succeed

Originally, experts believed OSI would replace TCP/IP, but this did not happen. Three main factors contributed to OSI's failure to gain dominance. First, **timing and cost**: TCP/IP was already fully deployed, and changing everything to OSI would have been prohibitively expensive and time-consuming. Second, **undefined layers**: the Session and Presentation layers were never fully defined. Their services were listed conceptually, but the actual protocols and software to implement them were never fully developed. Third, **performance**: initial implementations of OSI did not demonstrate high enough performance to convince Internet authorities to abandon the already-functional TCP/IP suite.

Chapter 3: Data and Signals

While previous chapters discussed "logical" communication between applications, at the Physical Layer communication becomes physical. We do not send files or pictures through wires; we send signals. The transmission medium (cable or air) carries electromagnetic signals, so data must be transformed into these signals for transmission.

3.1 Analog and Digital Signals

Both data (the information being conveyed) and signals (the transmission mechanism) can be either analog or digital. Understanding this distinction is fundamental to all of data communications.

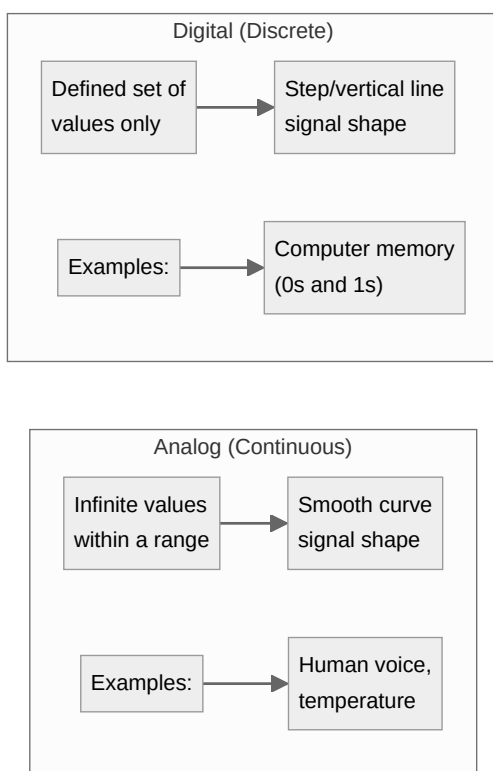


Figure 3.1 *Analog vs. Digital: continuous vs. discrete representation*

Analog signals are continuous, taking on infinite values within a range. An analog clock illustrates this concept perfectly: the hands move continuously, flowing smoothly from 8:05 to 8:06, passing through every microsecond in between. Human voice and temperature readings are naturally analog.

Digital signals are discrete, limited to a defined set of values. A digital clock demonstrates this: the display jumps suddenly from 8:05 to 8:06 with no "in-between" state shown. Computer memory, represented as 0s and 1s, is fundamentally digital.

repeating pattern. In data communications, we typically use periodic analog signals (like sine waves for carrier signals) and nonperiodic digital signals (for actual data transmission).

3.2 Periodic Analog Signals

Periodic analog signals are classified into two types: simple signals (a single sine wave that cannot be broken down further) and composite signals (made of multiple sine waves combined). The sine wave is the most fundamental periodic analog signal, appearing as a smooth, oscillating curve.

3.2.1 Sine Wave Parameters

A sine wave is completely described by three parameters: peak amplitude, frequency, and phase.

Peak Amplitude. The peak amplitude is the absolute value of the signal's highest intensity, proportional to the energy the signal carries. For electric signals, it is measured in volts. The voltage in US homes is stated as 110-120 V, but this is the RMS (Root Mean Square) value; the actual peak amplitude is higher (approximately 155-170 V). The relationship is:

Peak Amplitude from RMS

$$\text{Peak} = \sqrt{2} \times \text{RMS}$$

A battery provides constant voltage (e.g., 1.5 V), which is technically a sine wave with zero frequency and constant amplitude.

Period and Frequency. Period and frequency describe the same characteristic from different perspectives. The **period (T)** is the time in seconds required to complete one cycle. The **frequency (f)** is the number of periods (cycles) in one second. They are inverses of each other.

Fundamental Relationship

$$f = \frac{1}{T} \quad \text{and} \quad T = \frac{1}{f}$$

Period is formally expressed in seconds (s), while frequency is expressed in Hertz (Hz), meaning cycles per second. Frequency also measures the rate of change: high frequency means change happens over a short span of time, while low frequency means change happens over a long span.

Example: House Power Frequency to Period

Scenario: The power used in US homes has a frequency of 60 Hz. What is the period?

Solution:

$$T = \frac{1}{f} = \frac{1}{60} = 0.0166 \text{ s} = 16.6 \text{ ms}$$

The period is 16.6 ms. Our eyes cannot detect a light bulb flickering at this speed, so it appears constant.

Example: Time Conversion

Task: Express a period of 100 ms in microseconds (μs).

Solution: $100 \text{ ms} = 100 \times 1000 \mu\text{s} = 100,000 \mu\text{s}$

Frequency Unit Conversions**Table 3.1 Frequency Unit Prefixes**

Unit	Equivalent	Unit	Equivalent
1 Hz	1 cycle/second	1 kHz (kilohertz)	$10^3 \text{ Hz} = 1,000 \text{ Hz}$
1 MHz (megahertz)	$10^6 \text{ Hz} = 1,000,000 \text{ Hz}$	1 GHz (gigahertz)	10^9 Hz
1 THz (terahertz)	10^{12} Hz		

Two Extremes of Frequency

Zero Frequency (0 Hz): The signal does not change at all, maintaining a constant voltage level forever. It never completes a cycle. Example: a battery providing DC current.

Infinite Frequency (∞ Hz): The signal changes instantaneously, jumping from one level to another in zero time. Since $f = 1/T$, if $T = 0$, then $f = \infty$. This is theoretical, but it helps describe sudden spikes or digital transitions.

3.2.2 Phase

Phase (or phase shift) describes the position of the waveform relative to time zero ($t = 0$). It indicates the status of the first cycle, essentially showing where in the cycle the wave starts when the clock begins.

Table 3.2 Phase Shift Descriptions

Phase	Starting Position	Visual Description
0°	Starts at 0 amplitude, increases	Normal, unshifted wave
90°	Starts at peak amplitude, decreases	Shifted left by 1/4 cycle
180°	Starts at 0 amplitude, decreases (goes negative)	Shifted left by 1/2 cycle

Phase Calculation

$$\text{Phase} = \frac{\text{Offset in cycles}}{1 \text{ cycle}} \times 360^\circ$$

Example: Phase Calculation

Problem: A sine wave is offset 1/6 cycle with respect to time 0. What is the phase?

Solution:

$$\text{Phase} = \frac{1/6}{1} \times 360^\circ = 60^\circ$$

3.2.3 Wavelength

Wavelength (λ) characterizes a signal traveling through a transmission medium. It binds the period (T) or frequency (f) to the propagation speed of the medium. Wavelength is defined as the distance a simple signal can travel in one period.

Wavelength Formula

$$\lambda = c \times T = \frac{c}{f}$$

where c is the propagation speed of the medium

Frequency is independent of the medium (a 60 Hz signal is 60 Hz whether traveling through copper or air), but wavelength depends on both frequency and medium because propagation speed changes across different media. In a vacuum, light travels at 3×10^8 m/s, while in a cable the speed is slower, producing a shorter wavelength.

3.2.4 Time and Frequency Domains

To fully understand a signal, we can view it from two different perspectives. The **time-domain plot** shows amplitude versus time, illustrating how the signal changes moment by moment. Its limitation is that phase is not explicitly shown, and reading frequency from a wavy line is difficult. The **frequency-domain plot** shows peak amplitude versus frequency, revealing the "ingredients" of the signal. A complete sine wave appears as a single vertical spike whose position indicates frequency and whose height indicates peak amplitude. The frequency domain is more compact and allows immediate reading of values without tracing curves.

When multiple sine waves are mixed together, the time domain shows a messy, complex waveform, while the frequency domain displays clean, distinct spikes that make analysis much easier.

3.2.5 Composite Signals and Fourier Analysis

While simple sine waves are useful for power delivery or alarm signals, they cannot carry data. If you sent a single sine wave over a phone line, the receiver would hear only a continuous buzz. To convey meaningful information (like a conversation), we need variety, which is achieved through **composite signals** made of many simple sine waves combined.

In the 1900s, Jean-Baptiste Fourier proved a critical concept: *any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases*. The decomposition depends on the signal type:

Table 3.3 Fourier Decomposition by Signal Type

Signal Type	Decomposition Result	Frequency Domain Appearance
Periodic composite	A series of discrete frequencies (integer multiples like $f, 2f, 3f$)	Distinct spikes (harmonics)
Nonperiodic composite	A combination of continuous frequencies (all real values)	A continuous curve

For example, a periodic square wave decomposes into a fundamental frequency (f) plus odd harmonics ($3f, 5f, 7f, \dots$) with progressively decreasing amplitudes. A human voice is nonperiodic and contains an infinite number of frequencies between 0 and 4 kHz, appearing as a continuous curve in the frequency domain.

3.2.6 Bandwidth

The **bandwidth** of a composite signal is the range of frequencies it contains, calculated as the difference between the highest and lowest frequencies.

Bandwidth Formula

$$B = f_h - f_l$$

where f_h is the highest frequency and f_l is the lowest frequency

For a periodic signal, the bandwidth includes all discrete frequencies (spikes) between the minimum and maximum. For a nonperiodic signal, the bandwidth encompasses the entire continuous range between minimum and maximum.

Example: Discrete Spikes Bandwidth

Scenario: A periodic signal has sine waves at frequencies 100, 300, 500, 700, and 900 Hz.

Calculation: $B = 900 - 100 = 800$ Hz

The spectrum shows exactly 5 spikes at these frequencies.

Example: Finding the Lowest Frequency

Scenario: Bandwidth is 20 Hz. Highest frequency is 60 Hz. What is the lowest?

Solution: $f_l = f_h - B = 60 - 20 = 40$ Hz

Example: Nonperiodic Signal Bandwidth

Scenario: Bandwidth = 200 kHz. Middle frequency = 140 kHz.

Logic: The bandwidth extends 100 kHz above and 100 kHz below the middle.

$$f_l = 140 - 100 = 40 \text{ kHz}$$

$$f_h = 140 + 100 = 240 \text{ kHz}$$

Table 3.4 Real-World Bandwidth Examples

Application	Bandwidth	Total Frequency Range
AM Radio (per station)	10 kHz	530 to 1700 kHz
FM Radio (per station)	200 kHz (higher quality than AM)	88 to 108 MHz
Analog TV (B&W video)	4 MHz	Varies by channel (6 MHz per channel)

3.3 Digital Signals

Information can be represented by a digital signal with discrete values instead of an analog signal. The basic form uses two levels: a positive voltage represents 1 and zero voltage represents 0. However, digital signals can also have more than two levels, allowing us to send more than 1 bit per level.

Bits per Level

$$\text{Bits per level} = \log_2(L)$$

where L is the number of signal levels

Table 3.5 Bits per Level Examples

Levels (L)	Bits per Level	Notes
2	$\log_2(2) = 1$ bit	Basic binary (0 or 1)
4	$\log_2(4) = 2$ bits	Can represent 00, 01, 10, 11
8	$\log_2(8) = 3$ bits	Can represent 000 through 111
9	$\log_2(9) \approx 3.17$	Must round up to 4 bits (which can support up to 16 levels)

3.3.1 Bit Rate

Since digital signals are rarely periodic, we do not use frequency to describe them. Instead, we use **bit rate**, defined as the number of bits sent in one second, measured in bits per second (bps).

Example: Text Download Bit Rate

Scenario: Downloading 100 pages per second. Each page = 24 lines x 80 characters. Each character = 8 bits.

Calculation:

$$\text{Bit Rate} = 100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bps} = 1.536 \text{ Mbps}$$

Example: Digitized Voice

Scenario: 4 kHz analog voice, sampled at twice the frequency (2×4000 Hz), using 8 bits per sample.

Calculation:

$$\text{Bit Rate} = 2 \times 4000 \times 8 = 64,000 \text{ bps} = 64 \text{ kbps}$$

Example: HDTV Bit Rate

Scenario: 1920 x 1080 pixels, refreshed 30 times per second, 24 bits per color pixel.

Calculation:

$$\text{Bit Rate} = 1920 \times 1080 \times 30 \times 24 \approx 1.5 \text{ Gbps}$$

This is enormous. TV stations use compression to reduce this to 20-40 Mbps for transmission.

3.3.2 Bit Length

Just as analog signals have wavelength (the distance of one cycle), digital signals have bit length, defined as the distance one bit occupies on the transmission medium.

Bit Length Formula

$$\text{Bit Length} = \text{Propagation Speed} \times \text{Bit Duration}$$

3.3.3 Digital Signal as a Composite Analog Signal

According to Fourier analysis, a digital signal is actually a composite analog signal with infinite bandwidth. This may seem surprising, but the reasoning is clear: a digital signal has vertical lines representing instant jumps from one level to another. A vertical line implies a frequency of infinity (instantaneous change), while horizontal segments imply zero frequency (no change). To create this shape theoretically requires all frequencies between 0 and infinity.

3.3.4 Transmission of Digital Signals

Since a digital signal theoretically requires infinite bandwidth, we need practical approaches to transmit it. Two fundamental approaches exist: baseband and broadband transmission.

Baseband Transmission. Baseband sends a digital signal over a channel without converting it to an analog signal (no modulation). It requires a **low-pass channel** (a channel where bandwidth starts from 0 Hz). When the low-pass channel has wide bandwidth (dedicated medium like coaxial cable or fiber in a LAN), the signal shape is preserved accurately. However, when bandwidth is limited, we must approximate: the signal will look curved rather than perfectly square. The harmonics rule applies: a rough approximation uses bandwidth $B = N/2$ (first harmonic only), producing a sine-like wave; better approximation uses $B = 3N/2$ or $5N/2$ (including third or fifth harmonics), producing a squarer waveform. The key takeaway: required bandwidth is proportional to bit rate.

Baseband Bandwidth Requirements (for bit rate N)

$$\text{Rough (minimum): } B = \frac{N}{2} \quad \text{Better: } B = \frac{3N}{2} \quad \text{Best: } B = \frac{5N}{2}$$

Example: Required Bandwidth for 1 Mbps

Goal: Send 1 Mbps (1,000,000 bps).

Solutions:

Minimum (rough): $B = 1 \text{ Mbps} / 2 = 500 \text{ kHz}$

Better (1st + 3rd harmonics): $B = 3 \times 500 = 1.5 \text{ MHz}$

Best (1st + 3rd + 5th harmonics): $B = 5 \times 500 = 2.5 \text{ MHz}$

Example: Maximum Bit Rate

Scenario: Low-pass channel with bandwidth of 100 kHz. What is the maximum speed?

Solution: Using rough approximation ($B = N/2$):

$$100 \text{ kHz} = \frac{N}{2} \quad \Rightarrow \quad N = 200 \text{ kbps}$$

Broadband Transmission (Modulation). When a low-pass channel is unavailable (bandwidth does not start at 0 Hz), we cannot use baseband. Instead, we use broadband transmission: converting the digital signal to an analog signal through **modulation**. The channel used is a **bandpass channel** (bandwidth starts at a non-zero frequency). The process involves taking a single-frequency analog signal called a **carrier** and modifying one of its properties (amplitude, frequency, or phase) to encode the digital data. The receiver then converts the analog signal back to digital through **demodulation**.

Real-World Example: Telephone Modem

Telephone lines were built for voice (analog) with a narrow bandwidth of 0 to 4 kHz. If treated as a low-pass channel for digital data, the maximum speed would be only 8 kbps. Instead, modems treat the line as a bandpass channel, converting the computer's digital signal into an analog signal that fits within the phone line's frequency range, achieving much higher speeds.

3.4 Transmission Impairment

Transmission media are not perfect. As signals travel, they suffer from impairment, meaning the received signal is rarely identical to the transmitted signal. Three primary causes of impairment exist: attenuation, distortion, and noise.

3.4.1 Attenuation

Attenuation refers to a loss of energy. When a signal travels through a medium, it must overcome resistance, and some electrical energy converts into heat (just as a wire warms when current passes through). The signal becomes weaker as distance increases. To compensate, we use **amplifiers** to boost the signal strength back to usable levels.

The Decibel (dB). Engineers use decibels to measure relative signal strength between two points.

Decibel Formulas

$$dB = 10 \log_{10} \left(\frac{P_2}{P_1} \right) \quad (\text{for power})$$

$$dB = 20 \log_{10} \left(\frac{V_2}{V_1} \right) \quad (\text{for voltage})$$

Decibel Key Rules

Negative dB: The signal is attenuated (weaker at the destination).

Positive dB: The signal is amplified (stronger at the destination).

Example: A signal travels through three segments with dB values of -3 dB, $+7$ dB, and -3 dB. Total: $dB = -3 + 7 - 3 = +1$ dB, meaning the signal has gained power overall.

3.4.2 Distortion

Distortion means the signal changes its form or shape. This typically occurs in composite signals made of different frequencies. Each frequency component travels at its own propagation speed through the medium, arriving at the destination with different delays. These different delays cause phase shifts, and when the components are recombined, the signal shape is deformed compared to what was sent.

Consider a sender transmitting a clean composite wave made of frequencies A, B, and C. In the medium, frequency B travels slightly slower than A. At the receiver, when the components recombine, the wave appears lumpy or flattened because component B arrived later than the others.

3.4.3 Noise

Noise is external energy that corrupts the signal by adding unwanted information. Four types of noise affect communication systems.

Table 3.6 Types of Noise

Type	Cause	Characteristics
Thermal Noise	Random motion of electrons in the wire	Creates background hum or static; always present
Induced Noise	External sources like motors or appliances	The motor acts as a transmitting antenna, the wire as receiving antenna
Crosstalk	Effect of one wire on another	Like hearing a faint conversation on an old phone line because wires are too close
Impulse Noise	High-energy spikes in very short time	Caused by lightning strikes, power surges; sudden and severe

3.4.4 Signal-to-Noise Ratio (SNR)

To understand the theoretical limit of data transmission speed, we measure channel quality using SNR, the ratio of average signal power to average noise power. It represents the ratio of what is wanted (signal) to what is not wanted (noise).

SNR Formulas

$$\text{SNR} = \frac{\text{Average Signal Power}}{\text{Average Noise Power}} \quad (\text{ratio})$$

$$\text{SNR}_{dB} = 10 \log_{10}(\text{SNR}) \quad (\text{in decibels})$$

Analogy: *High SNR* is like talking to a friend in a quiet library: your voice (signal) is clear and the background (noise) is near zero. *Low SNR* is like talking at a loud rock concert: the music (noise) drowns out your voice (signal), making communication difficult.

EXAM TIP

Do not confuse SNR (the raw ratio) with SNR_{dB} (the decibel version). If $\text{SNR} = 10,000$, then $\text{SNR}_{dB} = 40$ dB. Calculations for data limits typically require converting dB back to the standard ratio first.

3.5 Data Rate Limits

A critical question in data communications asks: "How fast can we send data?" The data rate (in bps) depends on three factors: the bandwidth available, the number of signal levels used, and the quality of the channel (noise level). Two theoretical formulas provide answers: Nyquist for ideal noiseless channels and Shannon for real noisy channels.

3.5.1 Nyquist Bit Rate (Noiseless Channel)

Assuming a perfectly noiseless channel, the maximum bit rate depends on bandwidth and the number of signal levels.

Nyquist Bit Rate Formula

$$\text{BitRate} = 2 \times \text{Bandwidth} \times \log_2(L)$$

where L = number of signal levels

Theoretically, any bit rate is achievable by increasing L . However, increasing levels imposes a burden on the receiver: distinguishing between 2 levels is easy (voltage is either 0 or 5 V), but distinguishing between 64 levels requires incredibly sophisticated receivers to tell the difference between adjacent levels like 32 and 33. Increasing levels reduces system reliability.

3.5.2 Shannon Capacity (Noisy Channel)

Noiseless channels do not exist in reality. In 1944, Claude Shannon introduced a formula for the absolute maximum data rate of a noisy channel.

Shannon Capacity Formula

$$C = \text{Bandwidth} \times \log_2(1 + \text{SNR})$$

where C = capacity in bps, and SNR is the raw ratio (not in dB)

Notice that signal levels (L) do not appear in this formula. No matter how many levels are used or how clever the encoding, the data rate cannot exceed the Shannon Capacity. This formula defines the characteristic of the channel itself, not the transmission method.

3.5.3 Using Both Limits in Practice

In practice, engineers use both methods together. First, Shannon's formula determines the upper limit (the "speed limit of the road"). Then, Nyquist's formula determines how many signal levels are needed to achieve that speed. If Shannon says the maximum is 10 Mbps and Nyquist says we need 8 levels to achieve it, we know both the theoretical limit and the practical implementation requirement.

Example: Applying Both Limits

Scenario: A channel has bandwidth 1 MHz and SNR = 20 dB.

Step 1: Convert SNR from dB to ratio:

$$20 = 10 \log_{10}(\text{SNR}) \Rightarrow \text{SNR} = 10^2 = 100$$

Step 2: Apply Shannon's formula:

$$C = 10^6 \times \log_2(1 + 100) = 10^6 \times \log_2(101) \approx 10^6 \times 6.66 = 6.66 \text{ Mbps}$$

Step 3: Use Nyquist to find levels needed:

$$6.66 \times 10^6 = 2 \times 10^6 \times \log_2(L) \Rightarrow \log_2(L) = 3.33 \Rightarrow L = 2^{3.33} \approx 10 \text{ levels}$$

Chapter 6: Bandwidth Utilization

Bandwidth represents one of the most precious resources in data communications. Rather than installing new physical cables every time additional capacity is needed, we employ multiplexing techniques that allow the simultaneous transmission of multiple signals across a single data link. Multiplexing is practical whenever the bandwidth of the medium (the pipe) exceeds the bandwidth needs of the individual devices connected to it. The goal is efficiency: maximizing the use of high-bandwidth media like fiber optics or microwaves by sharing them among many lower-speed users.

6.1 Multiplexing Techniques

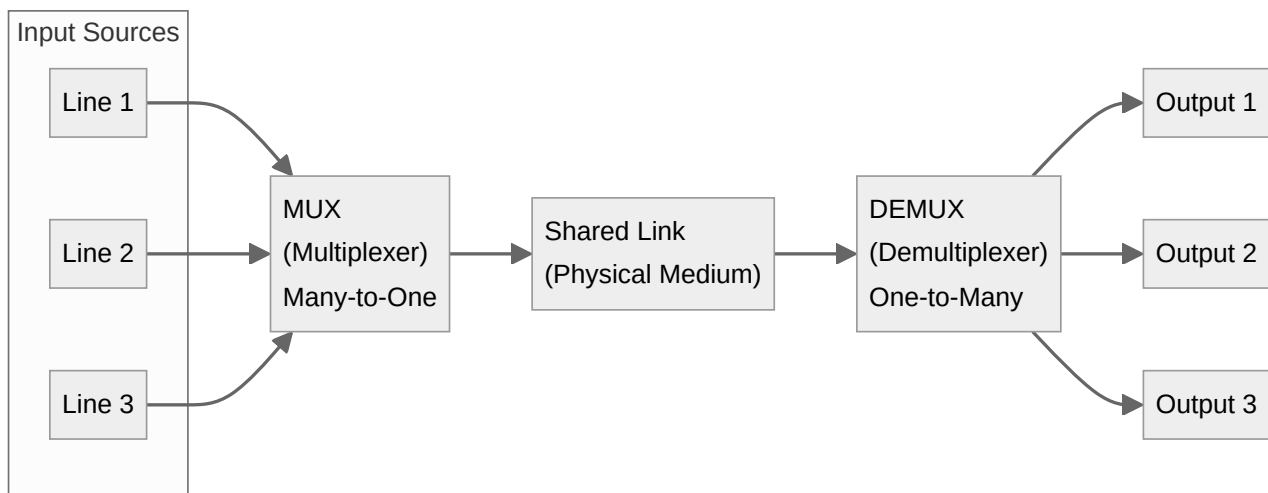


Figure 6.1 The multiplexing system: MUX combines inputs, DEMUX separates outputs

The multiplexing system consists of n input lines on the left, a **MUX (Multiplexer)** that combines the inputs into a single stream (many-to-one), a shared physical link (the wire), and a **DEMUX (Demultiplexer)** that separates the combined stream back into individual transmissions (one-to-many). It is important to distinguish between a **link** (the physical path, the cable itself) and a **channel** (the portion of a link carrying a specific transmission). One link can carry n channels simultaneously.

Three basic multiplexing techniques exist: **FDM (Frequency-Division Multiplexing)** for analog signals, **WDM (Wavelength-Division Multiplexing)** for analog light signals, and **TDM (Time-Division Multiplexing)** for digital signals.

6.1.1 Frequency-Division Multiplexing (FDM)

FDM is an analog technique used when the bandwidth of the link (in Hz) is wider than the combined bandwidths of the signals to be transmitted. Each sender's signal modulates a different carrier frequency.

Guard bands, which are strips of unused bandwidth, separate channels to prevent signals from overlapping or interfering with each other.

Analogy: FDM works like FM Radio stations. All stations transmit through the air simultaneously without interfering because Station A broadcasts at 98.5 MHz while Station B broadcasts at 101.5 MHz. Each occupies a different "channel" within the same medium.

The FDM Process. At the sender side (multiplexing), each source generates a signal, each signal modulates a different carrier frequency (f_1 , f_2 , f_3), the modulated signals are combined into one composite signal, and that composite signal is sent across the link. At the receiver side (demultiplexing), the demultiplexer receives the composite signal, uses filters to decompose it into its component parts, a demodulator separates each signal from its carrier frequency, and the original signals are delivered to their output lines.

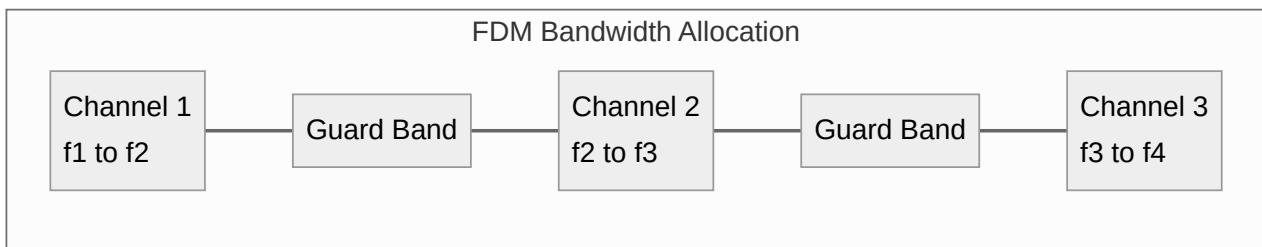


Figure 6.2 FDM divides bandwidth into channels separated by guard bands

Example: Voice Channels (No Guard Bands)

Scenario: Three voice channels, each occupying 4 kHz, combined onto one link with bandwidth 12 kHz (20 to 32 kHz). No guard bands.

Solution (Frequency Shift):

Channel 1: Shifted to 20-24 kHz

Channel 2: Shifted to 24-28 kHz

Channel 3: Shifted to 28-32 kHz

At Receiver: Filter 1 listens to 20-24 kHz, Filter 2 listens to 24-28 kHz, Filter 3 listens to 28-32 kHz.

The Analog Carrier System Hierarchy

Telephone companies use a hierarchical system to multiplex many low-bandwidth voice lines onto fewer high-bandwidth channels. FDM builds up from individual voice channels into massive "Jumbo Groups."

Table 6.1 Analog Carrier System Hierarchy

Level	Composition	Bandwidth
Voice Channel	1 voice line	4 kHz
Group	12 voice channels	48 kHz
Supergroup	5 groups (60 voice channels)	240 kHz
Master Group	10 supergroups (600 voice channels)	2.52 MHz
Jumbo Group	6 master groups (3600 voice channels)	16.984 MHz

Other FDM Applications

FDM serves as the backbone of broadcasting. AM radio stations each receive 10 kHz within the total band of 530 to 1700 kHz. FM radio offers higher quality with 200 kHz per station across 88 to 108 MHz. Television broadcasting assigns 6 MHz per channel. First-generation cellular phones (1G) used FDM, giving each user two 30-kHz channels (one for sending voice, one for receiving), with voice signals modulated using FM to occupy the full channel width.

Example: Cellular Capacity (AMPS)

Scenario: AMPS uses two 25 MHz bands: send (824-849 MHz) and receive (869-894 MHz). Each user requires 30 kHz.

Solution:

$$\text{Total channels} = \frac{25 \text{ MHz}}{30 \text{ kHz}} = \frac{25,000}{30} \approx 833 \text{ channels}$$

(In practice, the band is divided into 832 channels.)

42 channels are reserved for control signals. Available for users: $832 - 42 = 790$ channels.

6.1.2 Wavelength-Division Multiplexing (WDM)

WDM is designed specifically to utilize the immense data rate capability of fiber-optic cables. Using a single fiber for one signal wastes capacity; WDM combines several optical lines into one. Conceptually, WDM is identical to FDM: it combines signals of different frequencies. The difference is that WDM deals with extremely high frequencies (light waves).

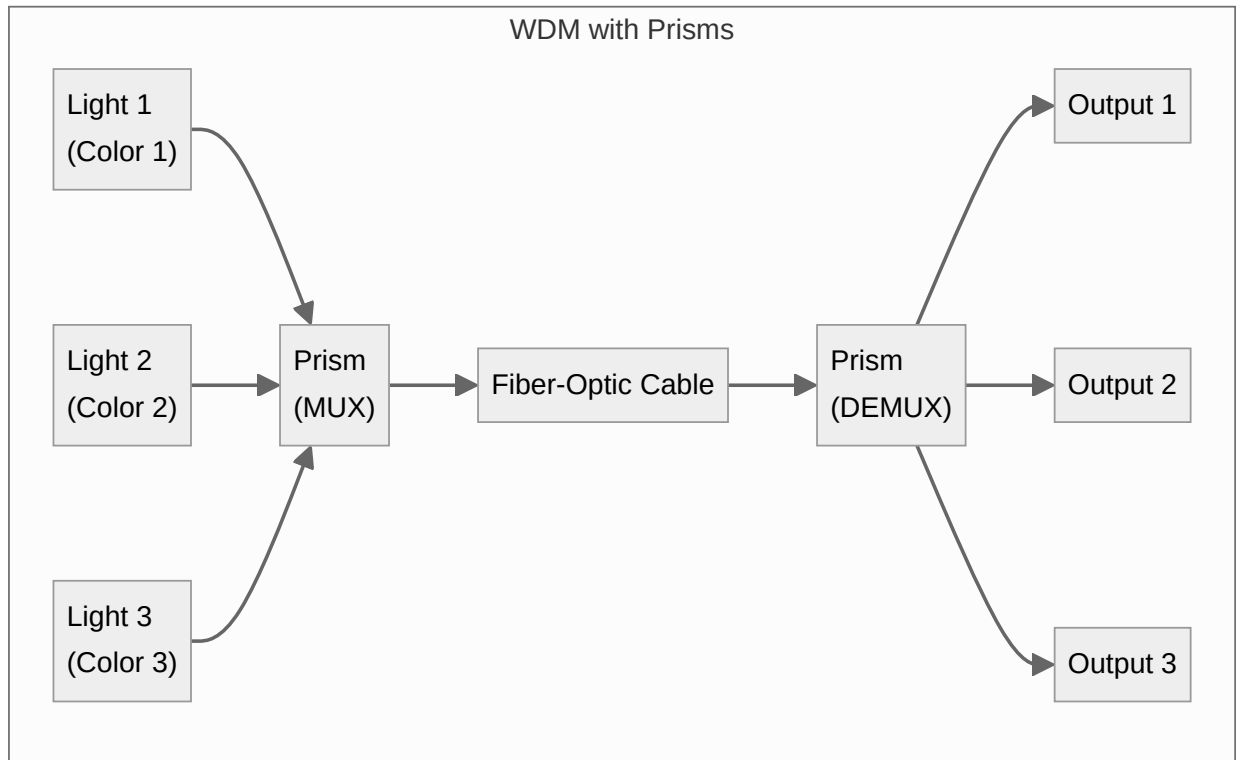


Figure 6.3 WDM uses prisms to combine and separate light signals

Although the technology is complex, the underlying physics is simple: a prism combines multiple input light beams (each a narrow band or color) into one wider beam at the multiplexer, and another prism separates the combined beam back into individual narrow beams at the demultiplexer. **Dense WDM (DWDM)** is a newer method that packs channels very close together for even greater efficiency.

6.1.3 Time-Division Multiplexing (TDM)

TDM is a digital process. Instead of sharing portions of bandwidth (like FDM/WDM), users share portions of time. The high bandwidth of the link functions as a single fast lane where users take turns occupying the link sequentially.

Synchronous TDM

In synchronous TDM, the multiplexer allocates the same slot to the same device at all times. Each input connection has an allotment in the output even if it is not sending data. This means if a device is silent, its time slot is sent empty. The bandwidth is reserved strictly for that device whether used or not.

Data flow is divided into units (1 bit, 1 character, or 1 block). The multiplexer takes one unit from each line and packs them into a **frame**, which consists of one complete cycle of slots (one slot for every input line). Since we must fit n items into the time normally used to send 1, the output time slot duration becomes shorter and the unit travels faster. The frame duration is usually the same as the duration of one input unit (T).

Data Rate Relationship

$$\text{Output Data Rate} = n \times \text{Input Data Rate}$$

where n = number of input lines

For smooth flow, the speed of the main link must equal the combined speed of all inputs.

Chapter 7: Transmission Media

Transmission media are the physical pathways that carry information from source to destination. While the Physical Layer (Layer 1) controls transmission, the media itself conceptually lies below it at "Layer 0." In the broadest sense, anything that carries information qualifies as transmission media, including air for conversation or trucks for mail. In data communications, the term specifically refers to free space, metallic cable, or fiber-optic cable.

7.1 Introduction and Taxonomy

Computers transmit signals as electromagnetic energy, a combination of electric and magnetic fields vibrating in relation to one another. The electromagnetic spectrum includes power, radio waves, infrared, visible light, ultraviolet, X-rays, gamma rays, and cosmic rays. Not all portions are usable for telecommunications; we are currently limited primarily to radio frequencies and light.

Transmission media divide into two broad categories based on how the signal is constrained. **Guided media (wired)** contain the signal within a physical boundary, including twisted-pair cable, coaxial cable, and fiber-optic cable. **Unguided media (wireless)** allow the signal to travel through free space without a physical conductor, including air, vacuum, and water.

7.2 Guided Media

Guided media provide a physical conduit from one device to another, directing and containing the signal within the physical limits of the medium. Three main types exist: twisted-pair and coaxial cables (both metallic copper conductors transporting electric current) and fiber-optic cable (glass or plastic transporting light).

7.2.1 Twisted-Pair Cable

A twisted pair consists of two copper conductors, each with its own plastic insulation, twisted together. One wire carries the signal to the receiver; the second wire serves as a ground reference. The receiver uses the **difference** between the two wires to interpret data.

Why twist the wires? Signals are affected by external interference (noise) and crosstalk (signals leaking from nearby wires). If wires ran parallel, the wire closer to a noise source would pick up more interference than the farther one, creating a disparity the receiver cannot filter out. Because the wires are twisted, they swap positions regularly. Over the cable's length, both wires are equally affected by noise. Since the receiver

examines the difference between the wires and the noise is now equal on both, the noise effectively cancels itself out. The rule is simple: more twists per inch means better quality and better noise cancellation.

Unshielded Twisted-Pair (UTP) is the most common type, consisting of just twisted wires in a plastic casing. It is cheap, flexible, and easy to install, though more susceptible to noise than shielded varieties.

Shielded Twisted-Pair (STP), originally produced by IBM, encases each pair in metal foil or braided mesh that prevents noise and crosstalk penetration, but it is bulkier, heavier, and more expensive.

The EIA classifies UTP into seven categories (1 is lowest quality, 7 is highest), with each category suitable for specific uses. The most common connector is the RJ45, which features a keyed design allowing insertion in only one orientation. Performance is measured by comparing attenuation against frequency and distance. Twisted-pair cable can pass a wide range of frequencies, but attenuation increases sharply as frequency increases, especially above 100 kHz. Thicker wire gauge generally offers less resistance and better performance.

7.2.2 Coaxial Cable

Coaxial cable carries signals of higher frequency ranges than twisted-pair due to its unique layered construction. It consists of a central core (solid or stranded copper wire carrying the signal), an insulating sheath enclosing the core, an outer conductor of metal foil or braid (serving dual roles as shield against noise and second conductor completing the circuit), and a plastic outer cover.

Coaxial cables are categorized by Radio Government (RG) ratings, with each number denoting specific physical specifications. The BNC (Bayonet Neill-Concelman) connector family connects coax to devices, including the standard BNC connector for device connection, BNC T-connector for Ethernet network branching, and BNC terminator for preventing signal reflection at cable ends.

While coaxial cable offers much higher bandwidth than twisted-pair, attenuation is surprisingly higher, meaning signals weaken rapidly and require frequent repeaters. Applications include telephone networks (historically carrying up to 10,000 voice signals, now mostly replaced by fiber), cable TV (traditionally using coax throughout, now typically hybrid networks with fiber backbone and coax at the boundaries), and Ethernet LANs.

7.2.3 Fiber-Optic Cable

Fiber-optic cable, made of glass or plastic, transmits signals as light. Its operation depends on two key optical properties: **refraction** (light bending when moving from dense to less dense substances) and **reflection** (light turning back into the denser substance when hitting the boundary at a steep enough angle, called the critical angle). The fiber's core (high-density glass or plastic) is surrounded by cladding (lower density). Light traveling through the core hits the boundary and reflects back through **total internal reflection**, guiding light down the fiber like water in a pipe.

Propagation Modes

Table 7.1 Fiber-Optic Propagation Modes

Mode	Core	Light Path	Distortion	Use Case
Multimode Step-Index	Large diameter, constant density	Straight lines bouncing at edges	High distortion	Short distances
Multimode Graded-Index	Large diameter, varying density	Smooth curved paths	Less distortion	Medium distances
Single-Mode	Extremely small diameter	Nearly horizontal (one path)	Negligible	Long distances

Single-mode fiber uses a highly focused laser source. The critical angle approaches 90 degrees, so the beam travels almost horizontally with all beams arriving together, making it ideal for long-distance communication.

Fiber-optic cable composition includes the outer jacket (PVC or Teflon), Kevlar strands (the same strong material used in bulletproof vests, strengthening the cable during installation), plastic coating (cushioning the fiber), and the cladding and core (the actual optical transmission vehicle). Connectors include SC (push/pull locking, used for cable TV), ST (bayonet twist-and-lock, used for networking), and MT-RJ (same size as RJ45).

Table 7.2 Advantages and Disadvantages of Fiber-Optic

Advantages	Disadvantages
Higher bandwidth than copper	Requires specialized expertise and equipment for installation
Less signal attenuation (50 km vs 5 km without repeaters)	Unidirectional (two fibers needed for bidirectional)
Immunity to electromagnetic interference	More expensive interfaces and cable
Lightweight compared to copper	Glass splicing is harder than twisting wire
Very difficult to tap (high security)	

7.3 Unguided Media (Wireless)

Unguided media transport electromagnetic waves without a physical conductor. Signals broadcast through free space and are available to anyone with a receiving device. Signals travel in three ways depending on frequency.

Table 7.3 Signal Propagation Methods

Method	Frequency	Behavior	Range Factors
Ground Propagation	Low frequency	Waves follow earth's curvature, hugging the ground	Depends on transmission power
Sky Propagation	Higher frequency	Waves radiate to ionosphere, which reflects them back	Enables long-distance with lower power
Line-of-Sight	Very high frequency	Waves travel straight from antenna to antenna	Requires tall, facing antennas

7.3.1 Radio Waves

Radio waves operate in the 3 kHz to 1 GHz range. Their key characteristic is being **omnidirectional**: when an antenna transmits, waves travel in all directions, so sending and receiving antennas do not need to be aligned. This omnidirectional nature makes radio susceptible to interference from other antennas. Radio waves can penetrate walls (you can receive AM radio inside your house), but this penetration also means you cannot easily isolate communication to just inside a building. The available bands are relatively narrow, leading to lower digital data rates. Radio is primarily used for **multicasting** (one sender, many receivers), including AM/FM radio, television, and paging systems.

7.3.2 Microwaves

Microwaves operate in the 1 GHz to 300 GHz range. They are **unidirectional**: waves can be narrowly focused, and sending and receiving antennas must be aligned (line-of-sight). Microwaves cannot pass through walls, which makes them secure for in-building communications. The band is very wide (299 GHz), enabling high-speed transmission. However, because they travel in straight lines, the earth's curvature blocks them over long distances, requiring very tall towers and repeaters.

Microwave antennas include the **parabolic dish** (based on parabola geometry, working like a funnel to catch waves and direct them to a focus point) and the **horn antenna** (shaped like a scoop, deflecting outgoing transmissions and collecting incoming ones). Microwaves are used for **unicast** (one-to-one communication) in cellular phones, satellite networks, and wireless LANs (Wi-Fi).

7.3.3 Infrared

Infrared operates in the 300 GHz to 400 THz range and is used for short-range communication. Infrared cannot pass through walls, which prevents interference between rooms (your TV remote will not change your neighbor's channel). However, infrared cannot be used outdoors because the sun emits infrared waves that interfere with signals. The infrared band is massive (400 THz), offering potential for very high data rates.

Applications include remote controls and IrDA ports for wireless keyboards, mice, and printers (requiring line-of-sight pointing).

Table 7.4 Wireless Media Comparison Summary

Type	Frequency	Direction	Wall Penetration	Primary Usage
Radio Waves	3 kHz - 1 GHz	Omnidirectional	Yes	Multicast (Radio, TV)
Microwaves	1 GHz - 300 GHz	Unidirectional	No	Unicast (Cell phones, Wi-Fi)
Infrared	300 GHz - 400 THz	Line-of-sight	No	Short-range (Remotes)

Practice Questions & Answers

This section contains a comprehensive set of practice questions designed to test your understanding of all chapters covered in these notes. The questions range from theoretical concepts that test your understanding to numerical problems that test your ability to apply formulas. Attempt each question before checking the answer.

Theoretical Questions

Q1. What are the four fundamental characteristics of a data communication system? Explain each with an example.

Answer: The four characteristics are: (1) **Delivery** - data must reach the correct destination (e.g., an email must reach the intended recipient, not someone else); (2) **Accuracy** - data must be delivered exactly as sent (e.g., a downloaded file must match the original, bit for bit); (3) **Timeliness** - data must arrive on time (e.g., video streaming requires frames to arrive in sequence without excessive delay); (4) **Jitter** - variation in packet arrival time must be minimal (e.g., if video packets arrive at irregular intervals, playback becomes choppy and unwatchable).

Q2. List and explain the five components of a data communication system.

Answer: (1) **Message** - the information to be communicated (text, audio, video, images); (2) **Sender** - the device that sends the data (computer, phone, camera); (3) **Receiver** - the device that receives the message (computer, phone, printer); (4) **Transmission Medium** - the physical path (twisted-pair, fiber-optic, radio waves); (5) **Protocol** - the set of rules governing communication (TCP/IP, HTTP). Without a protocol, two connected devices are like two people speaking different languages - they can hear each other but cannot understand.

Q3. Compare and contrast simplex, half-duplex, and full-duplex communication modes with real-world examples.

Answer: **Simplex** is unidirectional - only one device transmits, the other only receives (like a keyboard sending to a monitor, or a radio broadcast). **Half-duplex** allows both directions but not simultaneously - devices take turns (like walkie-talkies where you say "Over" to switch). **Full-duplex** allows simultaneous two-way communication (like a telephone conversation where both parties can talk and listen at the same time).

Q4. What is a protocol? Why is it essential for communication even when two devices are physically connected?

Answer: A protocol is a set of rules that governs data communication. It is essential because physical connection alone is insufficient. A protocol defines how data is formatted, transmitted, received, and acknowledged. Without protocols, connected devices are analogous to a French speaker and a Japanese speaker in the same room - they can physically hear each other but cannot exchange meaningful information because they lack agreed-upon rules for communication.

Q5. Compare Mesh and Star topologies in terms of cost, reliability, and use cases.

Answer: **Mesh** provides a dedicated point-to-point link between every pair of devices. It is highly robust (if one link fails, alternatives exist) and private, but extremely expensive requiring $n(n-1)/2$ links for n nodes. Used in regional telephone offices. **Star** connects each device to a central hub only. It is cheaper than mesh (less cable), and a single link failure affects only that device. However, the hub represents a single point of failure. Star is commonly used in LANs.

Q6. What is the difference between a LAN and a WAN? Explain with respect to scope, connectivity, and ownership.

Answer: **Scope:** A LAN covers a single building or campus; a WAN spans towns, states, or countries. **Connectivity:** A LAN connects hosts (computers, phones) directly; a WAN connects connecting devices (routers, switches) across networks. **Ownership:** LANs are privately owned by the organization using them; WANs are typically leased from communication service providers.

Q7. Explain the difference between circuit switching and packet switching. What are the advantages and disadvantages of each?

Answer: **Circuit switching** establishes a dedicated connection reserved for the entire communication duration, like a traditional phone call. It guarantees connection quality but wastes bandwidth during silent periods. **Packet switching** breaks data into independent packets that travel separately, like postal mail. It is more efficient because lines are shared, but packets may encounter delays in queues. Circuit switching is better for real-time, constant-rate communication; packet switching is better for bursty data traffic.

Q8. What is protocol layering and why is it important? Explain with the "black box" concept.

Answer: Protocol layering divides complex communication tasks into independent layers, each with specific responsibilities. The "black box" concept means each layer is treated as a self-contained module where only inputs and outputs matter, not internal implementation. This modularity is important because changes to one layer (e.g., changing encryption methods) do not affect other layers, making the system maintainable and flexible. Without layering, a single change would require rebuilding the entire system.

Q9. State and explain the two principles of protocol layering.

Answer:
Principle 1 (Bidirectional Capability): If communication is two-way, every layer must perform both operations. If Layer 3 writes, it must also read; if Layer 2 encrypts, it must also decrypt.
Principle 2 (Identical Objects): The object exchanged between corresponding layers at sender and receiver must be identical. If Layer 2 at the sender handles ciphertext, Layer 2 at the receiver must also handle ciphertext (not plaintext or any other form).

Q10. Describe the five layers of the TCP/IP model and the function of each layer.

Answer: (1) **Physical:** Transmits individual bits across the medium, converting bits to electrical/optical signals. (2) **Data-Link:** Handles hop-to-hop delivery using frames, moving packets between adjacent nodes. (3) **Network:** Manages host-to-host communication and routing using datagrams and IP addresses. (4) **Transport:** Provides process-to-process delivery using ports; TCP offers reliability while UDP offers speed. (5) **Application:** Enables user access to network services through protocols like HTTP, SMTP, FTP, and DNS.

Q11. What is encapsulation? Describe the process as data moves from the Application layer to the Physical layer.

Answer: Encapsulation is the process of wrapping data with protocol information at each layer. At the **Application** layer, the message is created. At the **Transport** layer, a header with port numbers is added, creating a segment. At the **Network** layer, a header with IP addresses is added, creating a datagram. At the **Data-Link** layer, a header with MAC addresses and a trailer are added, creating a frame. At the **Physical** layer, the frame becomes a stream of bits. Each layer treats the data from above as payload and adds its own control information.

Q12. What is the difference between TCP and UDP? When would you use each?

Answer: **TCP** is connection-oriented, reliable, and provides flow control and error correction. It guarantees delivery but has higher overhead. Use TCP for file transfers, web browsing, and email where data integrity matters. **UDP** is connectionless, unreliable, but faster with lower overhead. Use UDP for streaming video, online gaming, and DNS queries where speed matters more than guaranteed delivery.

Q13. Compare the OSI and TCP/IP models. Why did TCP/IP succeed while OSI did not become dominant?

Answer: OSI has 7 layers (Application, Presentation, Session, Transport, Network, Data-Link, Physical) while TCP/IP has 5, combining OSI's Application, Presentation, and Session into one Application layer. TCP/IP succeeded for three reasons: (1) **Timing:** TCP/IP was already deployed when OSI arrived, making change prohibitively expensive; (2) **Incomplete specification:** OSI's Session and Presentation layers were never fully defined or implemented; (3) **Performance:** Early OSI implementations were slower than the already-working TCP/IP suite.

Q14. What is the difference between analog and digital signals? Give two examples of each.

Answer: **Analog signals** are continuous, taking infinite values within a range. Examples: human voice and temperature readings. **Digital signals** are discrete, limited to defined values (typically 0 and 1). Examples: computer memory and data stored on hard drives. An analog clock with continuously moving hands represents analog, while a digital clock with jumping numbers represents digital.

Q15. Define period, frequency, and phase. How are period and frequency related?

Answer: **Period (T)** is the time in seconds to complete one cycle. **Frequency (f)** is the number of cycles per second (Hz). **Phase** describes the waveform's position relative to time zero. Period and frequency are inverses: $f = 1/T$ and $T = 1/f$. A signal with period 0.01 s has frequency 100 Hz.

Q16. What is bandwidth? How is it calculated for both periodic and nonperiodic signals?

Answer: Bandwidth is the range of frequencies contained in a composite signal, calculated as $B = f_h - f_l$ where f_h is the highest and f_l is the lowest frequency. For periodic signals, bandwidth includes discrete frequency spikes (harmonics). For nonperiodic signals, bandwidth encompasses a continuous range of frequencies.

Q17. Explain the difference between time domain and frequency domain representations of a signal.

Answer: The **time domain** plots amplitude versus time, showing how a signal changes moment by moment. It is intuitive but makes it hard to read frequency. The **frequency domain** plots peak amplitude versus frequency, showing the signal's component frequencies as spikes. It is compact and makes analyzing complex signals easier. A sine wave in time domain is a curve; in frequency domain it is a single spike.

Q18. What is a composite signal? Explain Fourier analysis and its significance in data communications.

Answer: A composite signal is made of multiple simple sine waves combined. Jean-Baptiste Fourier proved that *any* composite signal can be decomposed into simple sine waves with different frequencies, amplitudes, and phases. This is significant because it means any complex signal (including digital signals) can be analyzed as a sum of simple sine waves, which is fundamental to understanding bandwidth requirements and signal processing.

Q19. What is the difference between baseband and broadband transmission? When is each used?

Answer: **Baseband** sends digital signals without converting to analog, requiring a low-pass channel (bandwidth starts at 0). Used in LANs with dedicated cables. **Broadband** converts digital to analog through modulation, allowing use of bandpass channels (bandwidth does not start at 0). Used when low-pass channels are unavailable, such as over telephone lines (modems) or cellular networks.

Q20. Define attenuation, distortion, and noise. List the four types of noise.

Answer: **Attenuation** is loss of signal energy due to medium resistance (fixed with amplifiers). **Distortion** is change in signal shape due to different frequency components arriving at different times. **Noise** is unwanted external energy corrupting the signal. The four noise types are: (1) Thermal noise (electron motion in wires), (2) Induced noise (from motors/appliances), (3) Crosstalk (interference between adjacent wires), (4) Impulse noise (sudden spikes from lightning/power surges).

Q21. What is SNR? Explain the difference between SNR and SNR in decibels.

Answer: SNR (Signal-to-Noise Ratio) is the ratio of average signal power to average noise power ($\text{SNR} = P_{\text{signal}} / P_{\text{noise}}$). SNR in decibels is $\text{SNR}_{dB} = 10 \log_{10}(\text{SNR})$. If $\text{SNR} = 10,000$, then $\text{SNR}_{dB} = 40 \text{ dB}$. The decibel version compresses large ratios into manageable numbers. For Shannon capacity calculations, you must convert dB back to the raw ratio first.

Q22. State the Nyquist Bit Rate formula and the Shannon Capacity formula. What is the key difference between them?

Answer: **Nyquist:** $\text{BitRate} = 2 \times B \times \log_2(L)$ - determines maximum bit rate for a noiseless channel based on bandwidth and signal levels. **Shannon:** $C = B \times \log_2(1 + \text{SNR})$ - determines absolute maximum capacity for a noisy channel. The key difference: Nyquist includes signal levels (L) and assumes no noise; Shannon accounts for noise (via SNR) but does not include signal levels. Nyquist tells us what encoding to use; Shannon tells us the channel's upper limit.

Q23. What is multiplexing? Why is it necessary? Name the three basic techniques.

Answer: Multiplexing allows simultaneous transmission of multiple signals across a single data link. It is necessary because bandwidth is limited and expensive; adding physical cables for every new channel is impractical. The three techniques are: **FDM (Frequency-Division Multiplexing)** for analog signals (shares frequency), **WDM (Wavelength-Division Multiplexing)** for optical signals (shares light wavelengths), and **TDM (Time-Division Multiplexing)** for digital signals (shares time slots).

Q24. Explain FDM and TDM with analogies. How do they differ in their approach to sharing a link?

Answer: **FDM** is like FM radio stations: all stations broadcast simultaneously through the air on different frequencies without interfering. It divides bandwidth horizontally into frequency bands. **TDM** is like a single-lane road where cars take turns: each user gets the entire bandwidth for a short time slot. It divides the link vertically into time slots. FDM is used for analog signals; TDM for digital signals.

Q25. What is WDM and how does it differ from FDM?

Answer: WDM (Wavelength-Division Multiplexing) is conceptually identical to FDM but operates at much higher frequencies (light waves in fiber-optic cables). While FDM combines radio frequencies, WDM combines different wavelengths (colors) of light using prisms. WDM is specifically designed for fiber-optic cables and can carry much more data than FDM due to the enormous bandwidth of light frequencies.

Q26. Compare twisted-pair, coaxial, and fiber-optic cables in terms of bandwidth, attenuation, cost, and immunity to noise.

Answer: **Twisted-pair:** Lowest bandwidth, moderate attenuation, cheapest, susceptible to noise (improved by twisting and shielding). **Coaxial:** Higher bandwidth than twisted-pair, surprisingly high attenuation, moderate cost, better noise immunity due to shielding. **Fiber-optic:** Highest bandwidth, very low attenuation (can go 50 km without repeaters), most expensive, complete immunity to electromagnetic interference.

Q27. Why are the wires in a twisted-pair cable twisted? Explain the noise cancellation mechanism.

Answer: Twisting causes both wires to be equally exposed to external noise over the cable length. In one twist, wire A is closer to the noise source; in the next twist, wire B is closer. Since the receiver interprets the *difference* between the two wires, and noise affects both equally, the noise cancels out. More twists per inch provides better noise cancellation. Without twisting, parallel wires would pick up unequal noise, creating a differential that corrupts the signal.

Q28. Explain how fiber-optic cable uses total internal reflection to guide light signals.

Answer: Fiber-optic cable has a high-density core surrounded by lower-density cladding. When light traveling through the core hits the core-cladding boundary at an angle greater than the critical angle, it reflects back into the core rather than passing through. This **total internal reflection** traps light inside the core, guiding it down the fiber like water in a pipe. In single-mode fiber, the small core diameter ensures the light travels almost straight with minimal bouncing, reducing distortion.

Q29. Compare radio waves, microwaves, and infrared in terms of frequency range, directionality, and applications.

Answer: **Radio waves** (3 kHz - 1 GHz) are omnidirectional, penetrate walls, used for multicasting (AM/FM radio, TV). **Microwaves** (1 GHz - 300 GHz) are unidirectional, require line-of-sight, cannot penetrate walls, used for unicasting (cell phones, Wi-Fi, satellite). **Infrared** (300 GHz - 400 THz) is short-range line-of-sight, cannot penetrate walls, used for remotes and IrDA ports.

Q30. What is the difference between a link and a channel in the context of multiplexing?

Answer: A **link** is the physical pathway (the actual cable or medium). A **channel** is the portion of a link that carries a specific transmission. One link can carry multiple channels simultaneously through multiplexing. For example, a single fiber-optic link might carry 100 channels using WDM, where each channel is a different wavelength of light.

Numerical Problems

N1. A signal has a period of 20 ms. What is its frequency in Hz and kHz?

Solution:

$$T = 20 \text{ ms} = 20 \times 10^{-3} \text{ s} = 0.02 \text{ s}$$

$$f = 1/T = 1/0.02 = 50 \text{ Hz} = 50 \times 10^{-3} \text{ kHz} = 0.05 \text{ kHz}$$

Answer: 50 Hz or 0.05 kHz

N2. A signal has a frequency of 500 kHz. What is its period in microseconds?

Solution:

$$f = 500 \text{ kHz} = 500 \times 10^3 \text{ Hz} = 5 \times 10^5 \text{ Hz}$$

$$T = 1/f = 1/(5 \times 10^5) = 2 \times 10^{-6} \text{ s} = 2 \text{ us}$$

Answer: 2 microseconds

N3. Express a period of 50 ms in microseconds and in seconds.

Solution:

$$50 \text{ ms} = 50 \times 1000 \text{ us} = 50,000 \text{ us}$$

$$50 \text{ ms} = 50 \times 10^{-3} \text{ s} = 0.05 \text{ s}$$

Answer: 50,000 microseconds or 0.05 seconds

N4. If the RMS voltage in a US home is 120 V, what is the peak amplitude?

Solution:

$$\text{Peak} = \sqrt{2} \times \text{RMS}$$

$$\text{Peak} = \sqrt{2} \times 120 = 1.414 \times 120 = 169.7 \text{ V}$$

Answer: Approximately 169.7 V (or about 170 V)

N5. A sine wave is offset by 1/4 cycle with respect to time zero. What is its phase shift?

Solution:

$$\text{Phase} = (1/4) \times 360 \text{ degrees} = 90 \text{ degrees}$$

Answer: 90 degrees

N6. A periodic signal has sine waves at frequencies 200 Hz, 600 Hz, and 1000 Hz. What is its bandwidth?

Solution:

$$f_l = 200 \text{ Hz}, f_h = 1000 \text{ Hz}$$

$$B = f_h - f_l = 1000 - 200 = 800 \text{ Hz}$$

Answer: 800 Hz

N7. A signal has a bandwidth of 50 Hz. The highest frequency is 180 Hz. What is the lowest frequency?

Solution:

$$\begin{aligned} B &= f_h - f_l \\ 50 &= 180 - f_l \\ f_l &= 180 - 50 = 130 \text{ Hz} \end{aligned}$$

Answer: 130 Hz

N8. A periodic composite signal has a bandwidth of 1500 Hz, centered at 2500 Hz. What are the lowest and highest frequencies?

Solution:

$$\begin{aligned} B &= 1500 \text{ Hz, middle frequency} = 2500 \text{ Hz} \\ B \text{ extends } B/2 &= 750 \text{ Hz above and below} \\ f_l &= 2500 - 750 = 1750 \text{ Hz} \\ f_h &= 2500 + 750 = 3250 \text{ Hz} \end{aligned}$$

Answer: Lowest = 1750 Hz, Highest = 3250 Hz

N9. If a signal has 8 levels, how many bits can be sent per level?

Solution:

$$\text{Bits per level} = \log_2(L) = \log_2(8) = 3 \text{ bits}$$

Answer: 3 bits per level

N10. How many signal levels are needed to send 4 bits per level?

Solution:

$$\begin{aligned} L &= 2^n \text{ where } n = \text{bits per level} \\ L &= 2^4 = 16 \text{ levels} \end{aligned}$$

Answer: 16 levels

N11. A digital signal has a bit rate of 2000 bps. If the propagation speed is 2×10^8 m/s, what is the bit length?

Solution:

```
Bit duration = 1 / bit rate = 1 / 2000 = 0.0005 s =  $5 \times 10^{-4}$  s
Bit length = Propagation speed x Bit duration
Bit length =  $2 \times 10^8 \times 5 \times 10^{-4} = 10^5$  m = 100 km
```

Answer: 100 km (or 100,000 meters)

N12. We need to send 200 kbps over a low-pass channel using baseband transmission. What is the minimum bandwidth required (rough approximation)? What bandwidth is needed for better accuracy (using 1st and 3rd harmonics)?

Solution:

```
Rough (minimum):  $B = N/2 = 200/2 = 100$  kHz
Better (1st + 3rd harmonics):  $B = 3N/2 = 3 \times 100 = 300$  kHz
```

Answer: Minimum = 100 kHz; Better = 300 kHz

N13. A signal loses 3 dB of power after passing through a medium. If the input power is 10 mW, what is the output power?

Solution:

```
dB =  $10 \log_{10}(P2/P1)$ 
-3 =  $10 \log_{10}(P2/10)$ 
-0.3 =  $\log_{10}(P2/10)$ 
 $10^{-0.3} = P2/10$ 
0.501 =  $P2/10$ 
 $P2 = 5.01$  mW
```

Answer: Approximately 5 mW (the signal lost about half its power)

N14. A signal passes through three amplifiers with gains of +5 dB, -2 dB, and +8 dB. What is the total gain in dB? Has the signal been amplified or attenuated overall?

Solution:

$$\text{Total dB} = 5 + (-2) + 8 = 11 \text{ dB}$$

Answer: +11 dB total. Since the result is positive, the signal has been **amplified** overall.

N15. A channel has a bandwidth of 2 kHz and an SNR of 30 dB. What is the Shannon capacity?

Solution:

Step 1: Convert SNR from dB to ratio:

$$30 = 10 \log_{10}(\text{SNR})$$

$$\text{SNR} = 10^3 = 1000$$

Step 2: Apply Shannon's formula:

$$C = B \times \log_2(1 + \text{SNR})$$

$$C = 2000 \times \log_2(1 + 1000)$$

$$C = 2000 \times \log_2(1001)$$

$$C = 2000 \times 9.967$$

$$C = 19,934 \text{ bps} = 19.93 \text{ kbps}$$

Answer: Approximately 19.93 kbps (approximately 20 kbps)

N16. A noiseless channel has a bandwidth of 5 kHz. If we use 8 signal levels, what is the maximum bit rate according to Nyquist?

Solution:

$$\text{BitRate} = 2 \times B \times \log_2(L)$$

$$\text{BitRate} = 2 \times 5000 \times \log_2(8)$$

$$\text{BitRate} = 2 \times 5000 \times 3$$

$$\text{BitRate} = 30,000 \text{ bps} = 30 \text{ kbps}$$

Answer: 30 kbps

N17. A channel has a bandwidth of 1 MHz and SNR = 40 dB. (a) Find the Shannon capacity. (b) If we want to achieve this capacity, how many signal levels are needed according to Nyquist?

Solution:

(a) Convert SNR: $40 = 10 \log_{10}(\text{SNR}) \Rightarrow \text{SNR} = 10^4 = 10,000$
 $C = 10^6 \times \log_2(1 + 10000)$
 $C = 10^6 \times \log_2(10001)$
 $C = 10^6 \times 13.29 = 13.29 \text{ Mbps}$

(b) Using Nyquist: $13.29 \times 10^6 = 2 \times 10^6 \times \log_2(L)$
 $\log_2(L) = 6.64$
 $L = 2^{6.64} = 99.4 \approx 100 \text{ levels}$

Answer: (a) 13.29 Mbps; (b) Approximately 100 signal levels

N18. We need to upgrade the bit rate of a channel from 1 Mbps to 4 Mbps. If the original signal used 4 levels, how many levels are needed for the upgraded rate (assuming same bandwidth and noiseless channel)?

Solution:

Original: $1 \times 10^6 = 2 \times B \times \log_2(4) = 2 \times B \times 2$
 So $B = 250 \text{ kHz}$

 New: $4 \times 10^6 = 2 \times 250,000 \times \log_2(L)$
 $4 \times 10^6 = 500,000 \times \log_2(L)$
 $\log_2(L) = 8$
 $L = 2^8 = 256 \text{ levels}$

Answer: 256 levels

N19. Five channels, each with a bandwidth of 200 kHz, are to be multiplexed using FDM. If we need 20 kHz guard bands between channels, what is the minimum bandwidth required for the multiplexed link?

Solution:

```
Total signal bandwidth = 5 x 200 kHz = 1000 kHz
Number of guard bands needed = 5 - 1 = 4
Total guard band bandwidth = 4 x 20 kHz = 80 kHz
Minimum link bandwidth = 1000 + 80 = 1080 kHz = 1.08 MHz
```

Answer: 1.08 MHz

N20. In a synchronous TDM system, 4 input lines each send data at 9600 bps. What is the minimum data rate of the multiplexed output line?

Solution:

```
Output Data Rate = n x Input Data Rate
Output Data Rate = 4 x 9600 = 38,400 bps = 38.4 kbps
```

Answer: 38.4 kbps

N21. A mesh network has 8 devices. How many duplex links are needed?

Solution:

```
Number of links = n(n-1)/2
Links = 8 x 7 / 2 = 56 / 2 = 28 links
```

Answer: 28 duplex links

N22. A fiber-optic cable needs to send data at 1 Gbps. If light travels at 2×10^8 m/s in the fiber, what is the bit length?

Solution:

```
Bit duration = 1 / (1 x 10^9) = 10^-9 s = 1 ns
Bit length = 2 x 10^8 x 10^-9 = 0.2 m = 20 cm
```

Answer: 20 cm per bit

N23. An AM radio station operates at 1000 kHz. If each station is allocated 10 kHz, what is the frequency range of this station's broadcast?

Solution:

Center frequency = 1000 kHz
Bandwidth per station = 10 kHz
Range: $1000 - 5 = 995$ kHz to $1000 + 5 = 1005$ kHz

Answer: 995 kHz to 1005 kHz

N24. A signal travels through a medium and its power drops to one-tenth of its original value. Express this attenuation in decibels.

Solution:

$P_2/P_1 = 1/10 = 0.1$
 $\text{dB} = 10 \log_{10}(0.1) = 10 \times (-1) = -10$ dB

Answer: -10 dB (10 dB of attenuation/loss)

N25. A periodic signal completes 500 cycles in 2 milliseconds. What is its frequency?

Solution:

Frequency = cycles / time = $500 / (2 \times 10^{-3})$
 $f = 500 / 0.002 = 250,000$ Hz = 250 kHz

Answer: 250 kHz



Computer Networks

Complete Exam Notes – Unit 3

Switching • Data Link Layer • Error Control

MAC Protocols • HDLC • PPP

With Diagrams, Explanations & Practice Q&A



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1. Switching Techniques

From the perspective of telephone engineering, the phone system is divided into two principal parts: the **outside plant** (local loops and trunks, physically outside switching offices) and the **inside plant** (switches, inside switching offices). Two switching techniques are used in modern networks: **circuit switching** (traditional telephone system) and **packet switching** (used in computer networks and increasingly in voice-over-IP).

1.1 Circuit Switching

Core Concept

When a telephone call is placed, switching equipment seeks out a **physical path** from the caller's telephone to the receiver's telephone. This technique is called **circuit switching**.

Mechanism (see Fig. 1-1): Each rectangle represents a carrier switching office (end office, toll office, etc.). Each office has incoming and outgoing lines. When a call passes through a switching office, a physical connection is established between the incoming line and an outgoing line. The path is dedicated end-to-end for the duration of the call.

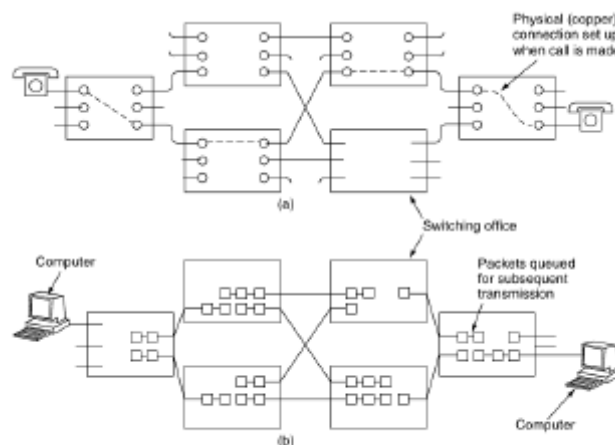


Figure 2-42. (a) Circuit switching. (b) Packet switching.

Figure 1-1: (a) Circuit switching with physical connections through switching offices. (b) Packet switching with packets queued at routers.

Historical Note: Strowger Gear

- Early connections were made manually by operators plugging jumper cables.
- **Almon B. Strowger**, a 19th-century Missouri undertaker, invented automatic circuit switching.
- **Motivation:** The other undertaker in town had his wife as the town telephone operator, who directed all funeral calls to her husband. Strowger chose to invent automatic switching rather than go out of business.
- For nearly 100 years, circuit-switching equipment worldwide was known as **Strowger gear**.

Important Properties of Circuit Switching

Property 1: End-to-end path establishment required

No data can be sent until a complete path is set up. Elapsed time between end of dialing and start of ringing: easily **10 seconds** or more for long-distance/international calls. During setup, the system hunts for a path. Call request signal must propagate to destination and be acknowledged before data transmission begins. Long setup times are undesirable for many computer applications (e.g., point-of-sale credit verification).

Property 2: Reserved path characteristics

Once setup is complete, the only delay is **propagation time** for the electromagnetic signal: approximately **5 msec per 1000 km**. No danger of congestion after call is established — no busy signals once connected. Busy signals may occur during setup due to lack of switching or trunk capacity.

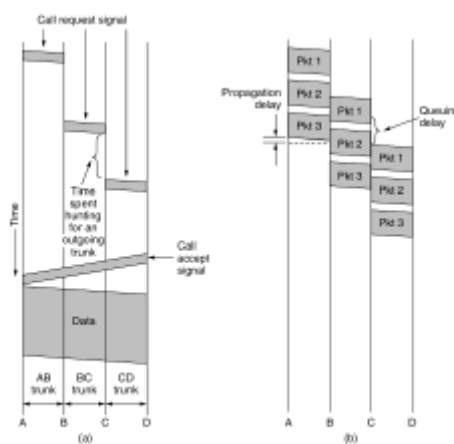


Figure 1-2: Timing of events in (a) circuit switching, (b) packet switching.

Figure 1-2: (a) Circuit switching timing: call request, trunk hunting, call accept, data transmission. (b) Packet switching timing: propagation delay and queuing delay across nodes.

1.2 Packet Switching

Core Concept

Packets are sent as soon as they are available. There is **no need to set up a dedicated path** in advance. Routers use **store-and-forward transmission**. Each packet is sent independently toward the destination. Unlike circuit switching, bandwidth is **not reserved** end-to-end.

Key Properties

1. **No fixed path** — Different packets can follow different paths depending on network conditions. Packets may arrive **out of order**.
2. **Packet size limits** — Networks place a tight upper limit on packet size. This ensures no user monopolizes a transmission line for very long, enables handling of interactive traffic, and reduces delay (first packet of a long message can be forwarded before the second one has fully arrived).
3. **Store-and-forward delay** — Accumulating a packet in router memory before forwarding exceeds the delay of circuit switching. In circuit switching, bits flow continuously through the wire.
4. **Queuing and congestion** — Packets may have to wait to be forwarded (queuing delay). Congestion occurs when many packets are sent simultaneously. No busy signal — network is always accessible, but packets may experience delay. **Congestion timing differs**: circuit switching at setup time; packet switching when packets are sent.
5. **Bandwidth efficiency** — Circuit switching: reserved but unused bandwidth is **wasted** (cannot be used by others). Packet switching: bandwidth is **not wasted** — more efficient from a system perspective. **Fundamental trade-off**: guaranteed service + wasted resources (circuit) vs. no guarantee + no waste (packet).
6. **Fault tolerance** — Packet switching is more fault tolerant — this is why it was invented. If a switch fails in circuit switching, all circuits using it are terminated. In packet switching, packets can be **routed around dead switches**.
7. **Charging algorithms** — **Circuit switching**: historically based on distance and time. **Packet switching**: connect time is irrelevant; **volume of traffic** matters. Home users: typically flat monthly rate. Backbone carriers: charge regional networks based on traffic volume.

1.3 Summary of Differences

Table 1-1: Circuit Switching vs Packet Switching

Aspect	Circuit Switching	Packet Switching
Path	Dedicated physical path	No fixed path
Setup	Required (long setup time)	None
Bandwidth reservation	Yes	No
Congestion	At setup time	During transmission
Out-of-order arrival	Impossible	Possible
Bandwidth waste	Yes (if idle)	No
Fault tolerance	Low (switch failure terminates circuits)	High (packets rerouted)
Charging basis	Distance and time	Traffic volume
Delay after setup	Propagation only	Store-and-forward + queuing

2. Data Link Layer Design Issues

The data link layer uses the physical layer's services to send and receive bits over communication channels. Its functions include: (1) providing a well-defined **service interface** to the network layer, (2) **dealing with transmission errors**, and (3) **regulating data flow** so slow receivers are not swamped by fast senders.

Encapsulation

Packets from the network layer are encapsulated into **frames**. Each frame contains: a **frame header**, a **payload field** (holds the packet), and a **frame trailer**.

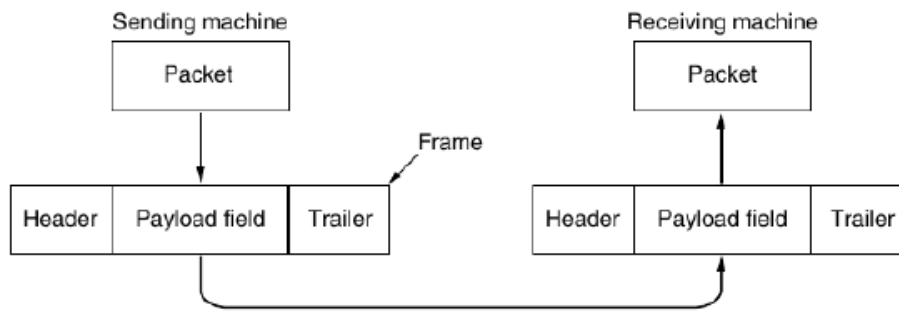


Figure 3-1. Relationship between packets and frames.

Figure 2-1: Frame structure: Header + Payload field + Trailer; packet encapsulated into frame.

2.1 Services Provided to the Network Layer

Communication Model

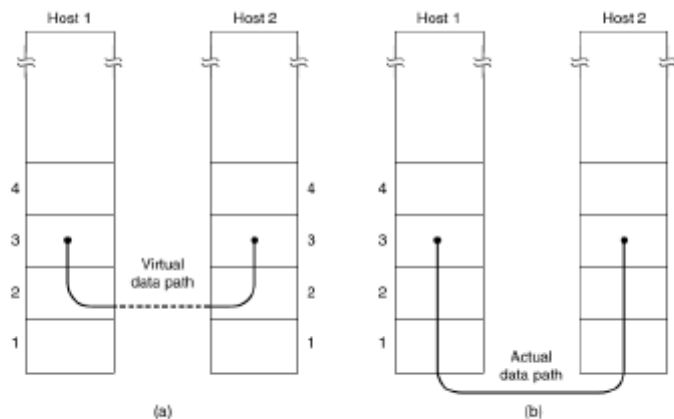


Figure 3-2. (a) Virtual communication. (b) Actual communication.

Figure 2-2: (a) Virtual communication: data link layer processes appear to communicate directly. (b) Actual communication: path goes down through layers, across physical link, up through layers.

- **Virtual communication** (Fig. 2-2a): Two data link layer processes appear to communicate directly using a data link protocol (conceptual model).
- **Actual communication** (Fig. 2-2b): The actual transmission follows a path down through the layers, across the physical medium, and up through the layers on the destination machine.

Service Type 1: Unacknowledged Connectionless Service

- Source machine sends independent frames without requiring acknowledgements.
- No logical connection established beforehand or released afterward.
- Lost frames due to noise are **not detected or recovered** at the data link layer.

- **Appropriate for:** Very low error rates (recovery left to higher layers), real-time traffic (e.g., voice) where late data is worse than bad data.
- **Example:** Ethernet.

Service Type 2: Acknowledged Connectionless Service

- No logical connections, but **each frame is individually acknowledged**.
- Sender knows whether a frame arrived correctly or was lost.
- Lost frames can be retransmitted if not acknowledged within a specified time.
- **Appropriate for:** unreliable channels (e.g., wireless systems).
- **Example:** 802.11 (WiFi).

Important Note on Acknowledgements

Providing acknowledgements in the data link layer is an **optimization**, never a requirement. The network layer could always implement its own acknowledgement and retransmission. However, the data link layer knows link-specific parameters (maximum frame length, propagation delays) that the network layer does not. If a large packet is broken into 10 frames and 2 are lost, network-layer retransmission would be very slow. Individual frame acknowledgement and retransmission correct errors more directly and quickly.

Service Type 3: Acknowledged Connection-Oriented Service

- Most sophisticated service.
- Source and destination establish a **connection** before data transfer.
- Each frame is **numbered**.
- **Guarantees:** Each frame sent is received; each frame received **exactly once**; all frames received in the **right order**.
- Provides the network layer with the equivalent of a **reliable bit stream**.
- **Appropriate for:** long, unreliable links (e.g., satellite channels, long-distance telephone circuits).

Connection-oriented transfer phases:

1. **Connection establishment:** both sides initialize variables and counters to track received frames.
2. **Data transmission:** one or more frames are transmitted.
3. **Connection release:** variables, buffers, and resources are freed.

2.2 Framing

The physical layer delivers a raw bit stream that is not guaranteed to be error-free. The data link layer must: break the bit stream into **discrete frames**, compute a **checksum** for each frame, include the checksum in the transmitted frame, and recompute and verify the checksum at the destination.

Challenge

Breaking the bit stream into frames is non-trivial. A good design must make it easy for the receiver to find the start of new frames while using minimal channel bandwidth.

Method 1: Byte Count

A field in the header specifies the number of bytes in the frame. The destination sees the byte count and knows how many bytes follow, locating the end of the frame.

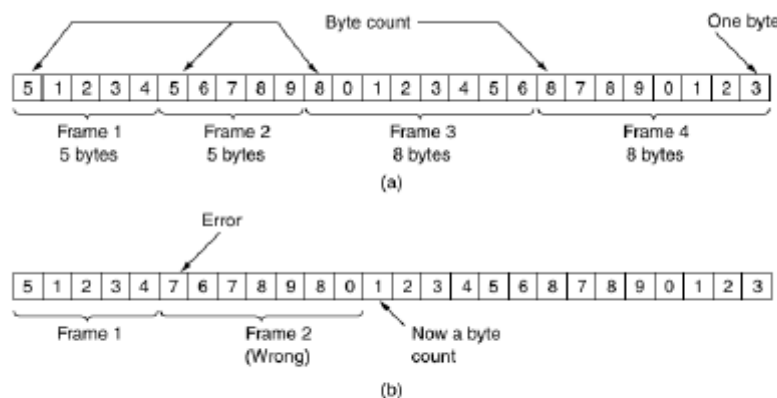


Figure 3-3. A byte stream. (a) Without errors. (b) With one error.

Figure 2-3: (a) Byte stream without errors: four frames of sizes 5, 5, 8, and 8 bytes. (b) With one error: the count can be garbled, causing loss of synchronization.

Vulnerability of Byte Count

The count can be garbled by transmission errors. If the byte count becomes corrupted, the destination loses synchronization and cannot locate the correct start of the next frame. Even if checksum detects the error, the destination cannot tell where the next frame starts. Retransmission requests are ineffective. **Rarely used by itself.**

Method 2: Flag Bytes with Byte Stuffing

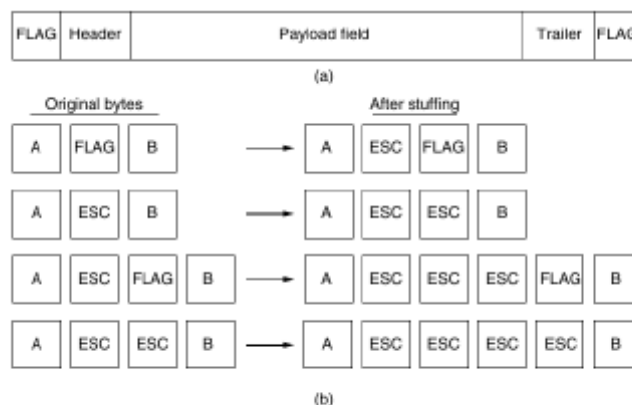


Figure 3-4. (a) A frame delimited by flag bytes. (b) Four examples of byte sequences before and after byte stuffing.

Figure 2-4: (a) Frame delimited by flag bytes. (b) Four examples of byte sequences before and after byte stuffing.

Mechanism: Each frame starts and ends with a special **flag byte**. Two consecutive flag bytes indicate the end of one frame and the start of the next. If the receiver loses synchronization, it searches for two flag bytes to resynchronize.

Problem: Flag bytes may occur in the data (e.g., binary data like photographs or songs).

Solution — Byte Stuffing: The sender's data link layer inserts a special **escape byte (ESC)** before each accidental flag byte in the data. A framing flag byte is distinguished from one in the data by the absence or presence of an escape byte before it. Receiver removes escape bytes before giving data to the network layer.

Table 2-1: Byte Stuffing Examples

Original bytes	After stuffing
A, FLAG, B	A, ESC, FLAG, B
A, ESC, B	A, ESC, ESC, B
A, ESC, FLAG, B	A, ESC, ESC, ESC, FLAG, B
A, ESC, ESC, B	A, ESC, ESC, ESC, ESC, B

Note: The byte-stuffing scheme shown is a slight simplification of the one used in **PPP (Point-to-Point Protocol)**.

Method 3: Flag Bits with Bit Stuffing

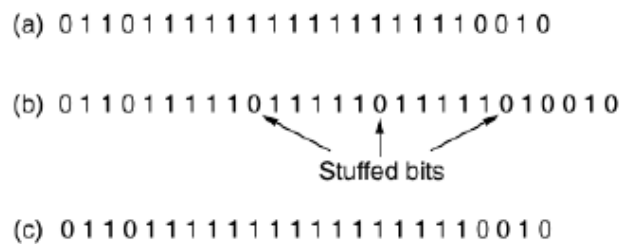


Figure 3-5. Bit stuffing. (a) The original data. (b) The data as they appear on the line. (c) The data as they are stored in the receiver's memory after destuffing.

Figure 2-5: Bit stuffing: (a) Original data. (b) The data as they appear on the line with stuffed bits. (c) The data as stored in receiver memory after destuffing.

Motivation: Byte stuffing is tied to 8-bit bytes. Framing can be done at the bit level, allowing frames with arbitrary bit counts.

Mechanism: Each frame begins and ends with the special bit pattern **01111110** (hexadecimal 0x7E). Whenever the sender encounters **five consecutive 1s** in the data, it automatically **stuffs a 0 bit** into the outgoing bit stream. This is analogous to byte stuffing but operates at the bit level. Also ensures a minimum density of transitions to help the physical layer maintain synchronization. **USB** uses bit stuffing for this reason.

Destuffing: When the receiver sees five consecutive incoming 1 bits followed by a 0 bit, it automatically **deletes** the 0 bit.

Transparency: Bit stuffing is completely transparent to the network layer. If user data contains the flag pattern 01111110, it is transmitted as 011111010 but stored in receiver memory as 01111110.

Method 4: Physical Layer Coding Violations

- Uses redundancy in physical layer encoding.
- Example: in **4B/5B** line code, 4 data bits map to 5 signal bits (16 of 32 signal possibilities are unused).
- Reserved signal patterns indicate start and end of frames.
- **Advantage:** no need to stuff data; reserved signals are easy to find.

Combined Methods

Many protocols use combinations for safety. Common pattern (Ethernet, 802.11): frame begins with a **preamble** (well-defined pattern, ~72 bits for 802.11) to allow receiver preparation, followed by a **length/count field** in the header to locate the end of the frame.

2.3 Error Control

Objective

Ensure all frames are eventually delivered to the network layer at the destination **exactly once** and in the **proper order**.

Mechanisms

1. **Acknowledgements** — Receiver sends back special control frames with **positive** or **negative** acknowledgements. Positive acknowledgement: frame arrived safely. Negative acknowledgement: something went wrong, frame must be retransmitted.
2. **Timers** — Problem: hardware troubles may cause a frame to vanish completely (noise burst), or the acknowledgement may be lost. If sender waits indefinitely for an acknowledgement, the protocol **hangs forever**. **Solution**: sender starts a timer when transmitting a frame. Timer set to expire after an interval long enough for: frame to reach destination, processing at destination, acknowledgement to propagate back. If timer expires before acknowledgement arrives, the frame is **retransmitted**.
3. **Sequence Numbers** — Danger: receiver may accept the same frame multiple times and pass it to the network layer more than once. **Solution**: assign **sequence numbers** to outgoing frames. Receiver can distinguish retransmissions from originals.

2.4 Flow Control

A sender on a fast, powerful computer may transmit frames faster than a slow receiver can accept them. Even with error-free transmission, the receiver may lose frames. *Example*: A smartphone requesting a Web page from a powerful server — the server may overwhelm the phone.

Approach 1: Feedback-Based Flow Control

- Receiver sends information back to the sender: granting permission to send more data, OR telling the sender how the receiver is doing.
- Protocol contains well-defined rules about when a sender may transmit the next frame.
- Often prohibits sending until receiver grants permission (implicitly or explicitly).

Approach 2: Rate-Based Flow Control

- Protocol has a built-in mechanism limiting sender transmission rate.
- Does **not** use feedback from the receiver.
- Primarily seen in the **transport layer**.

Prevalence

Feedback-based schemes appear at both link layer and higher layers. Higher layers are more common today. Link layer hardware (NICs) is often designed to run at "**wire speed**" — handling frames as fast as they arrive. Overruns are then handled by higher layers, not the link layer.

3. Error Detection and Correction

Communication channels exhibit a wide range of error characteristics: **optical fiber** in telecommunications has tiny error rates where transmission errors are rare; **wireless links** and aging local loops have error rates orders of magnitude larger where errors are the norm and cannot be avoided at reasonable cost. **Conclusion:** Transmission errors are here to stay. Network designers must learn to deal with them.

Two Basic Strategies

Table 3-1: Error Handling Strategies

Strategy	Mechanism	Term
Error-correcting codes	Include enough redundancy for receiver to deduce what the transmitted data must have been	FEC (Forward Error Correction)
Error-detecting codes	Include only enough redundancy to detect that an error occurred; receiver requests retransmission	—

Ecological niches: Highly reliable channels (e.g., fiber): cheaper to use error-detecting codes + occasional retransmission. Noisy channels (e.g., wireless): better to add redundancy so receiver can figure out the original block; FEC is preferred because retransmissions are just as likely to be in error as the first transmission.

Error Models

Model 1: Isolated Single-Bit Errors — Caused by extreme values of thermal noise overwhelming the signal briefly. Isolated, occasional single-bit flips.

Model 2: Burst Errors — Errors come in bursts rather than singly. Caused by physical processes: deep fade on wireless channel, transient electrical interference on wired channel.

Trade-offs of Burst Errors

Advantage: Data sent in blocks. With block size 1000 bits and error rate 0.001/bit: independent errors mean most blocks contain an error; bursts of 100 mean only one block in 100 affected on average.

Disadvantage: Much harder to correct than isolated errors when they do occur.

Model 3: Erasure Channels — Location of error is known (physical layer received analog signal far from expected value and declared bit lost). **Easier to correct:** even if the bit value is lost, we know which bit is in error. However, erasures are not always available.

3.1 Error-Correcting Codes

Code Fundamentals

A frame consists of: **m data (message) bits**, **r redundant (check) bits**, with total length $n = m + r$.

Table 3-2: Code Classifications

Property	Definition
Block code	r check bits computed solely as a function of the m data bits with which they are associated (like a lookup table)
Systematic code	m data bits sent directly, along with check bits (not encoded before sending)

Linear code r check bits computed as a linear function of m data bits (XOR or modulo 2 addition); encoding via matrix multiplication or simple logic circuits

Notation: An **(n, m) code** — n -bit unit containing data and check bits is an **n-bit codeword**.

Code rate: fraction of codeword carrying non-redundant information = m/n . Noisy channels: $\sim 1/2$ (half redundant). High-quality channels: close to 1 (small number of check bits).

Hamming Distance

Definition

Given two codewords, the **Hamming distance** is the number of bit positions in which they differ. **Computation:** XOR the two codewords and count the 1 bits in the result.

Example

```
10001001
10110001
-----
00111000 → 3 bits differ → Hamming distance = 3
```

Significance: If two codewords are Hamming distance d apart, it requires d **single-bit errors** to convert one into the other.

Hamming distance of a complete code: The smallest Hamming distance between any two legal codewords in the code.

Sparseness property: With r check bits, only fraction $2^m/2^n = 1/2^r$ of possible messages are legal codewords. This sparseness allows the receiver to detect and correct errors.

Error-Handling Properties by Hamming Distance

Table 3-3: Error Handling Capabilities

Capability	Required Distance	Reasoning
Detect d errors	$d + 1$	d single-bit errors cannot change a valid codeword into another valid codeword; receiver sees illegal codeword

Correct d errors	$2d + 1$	Legal codewords are so far apart that even with d changes, original codeword is still closer than any other; uniquely determinable
-----------------------	----------	--

Example

Code with four valid codewords: 0000000000, 0000011111, 1111100000, 1111111111.

Hamming distance = 5. Can correct double errors or detect quadruple errors.

Hamming Codes

Design Requirement

Code with m message bits and r check bits that allows all **single errors** to be corrected.

Counting argument: Each of the 2^m legal messages has n illegal codewords at distance 1 (formed by inverting each of the n bits). Each legal message requires $n + 1$ dedicated bit patterns.

Total bit patterns available: 2^n . Requirement:

$$(n + 1) \cdot 2^m \leq 2^n$$

Using $n = m + r$:

$$(m + r + 1) \leq 2^r \quad (\text{Equation 3-1})$$

This gives a **lower limit** on check bits needed to correct single errors.

Hamming code construction (Hamming, 1950):

- Bits numbered consecutively starting with bit 1 at left end.
- **Check bits:** positions that are powers of 2 (1, 2, 4, 8, 16, ...)
- **Data bits:** remaining positions (3, 5, 6, 7, 9, ...)

Parity computation: Each check bit forces the modulo 2 sum (parity) of some collection of bits (including itself) to be even (or odd). A bit may be included in several check bit computations.

Rule for check bit coverage: To see which check bits data bit in position k contributes to, rewrite k as a sum of powers of 2. Example: $11 = 1 + 2 + 8$; $29 = 1 + 4 + 8 + 16$. Bit is checked by just those check bits in its expansion.

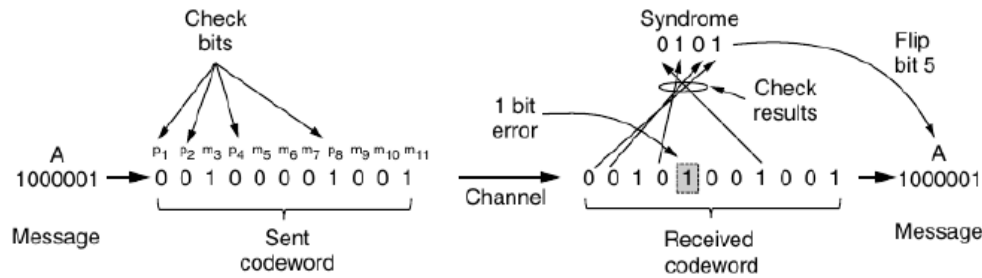


Figure 3-6. Example of an (11, 7) Hamming code correcting a single-bit error.

Figure 3-1: Example of an (11, 7) Hamming code correcting a single-bit error. Message A (ASCII 1000001) is encoded, sent, corrupted at bit 5, detected via syndrome 0101 = 5, and corrected.

(11, 7) Hamming Code Example

Message A (ASCII): 1000001

Position:	1	2	3	4	5	6	7	8	9	10	11
	p1	p2	m1	p4	m2	m3	m4	p8	m5	m6	m7
	0	0	1	0	0	0	0	1	0	0	1

Sent codeword: 00100001001. Hamming distance of 3 → can **correct single errors** or **detect double errors**.

Decoding process:

1. Receiver redoes check bit computations including received check bit values → **check results**.
2. If all check results are zero (for even parity): codeword accepted as valid.
3. If check results not all zero: error detected.

Error syndrome: The set of check results forms the error syndrome, used to pinpoint and correct the error. Example: Check results for $k = 8, 4, 2, 1$: 0, 1, 0, 1. Syndrome: $0101 = 4 + 1 = 5$. By design: **5th bit is in error**. Flip incorrect bit, discard check bits → correct message (ASCII "A").

Convolutional Codes

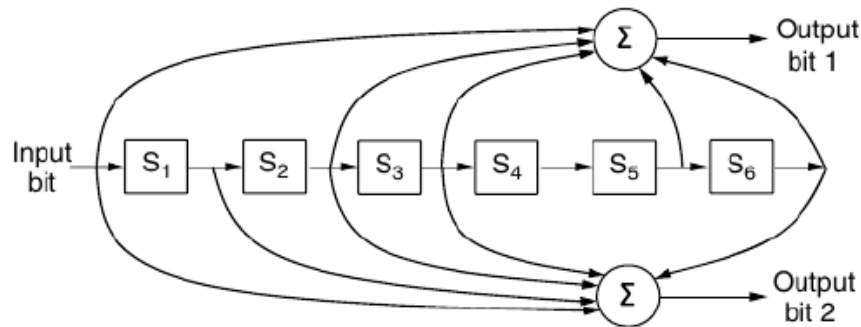


Figure 3-7. The NASA binary convolutional code used in 802.11.

Figure 3-2: The NASA binary convolutional code used in 802.11 (rate $r = 1/2$, constraint length $k = 7$).

Key difference from block codes: Not a block code. Encoder processes a sequence of input bits and generates a sequence of output bits. No natural message size or encoding boundary.

Memory property: Output depends on current and previous input bits. The number of previous bits on which output depends is the **constraint length**.

Specification: In terms of **rate** and **constraint length**. **Deployment:** Widely used in GSM mobile phone system, satellite communications, 802.11.

NASA binary convolutional code (Fig. 3-2): Rate $r = 1/2$, constraint length $k = 7$. First used for Voyager space missions (1977). Reused in 802.11.

- Each input bit produces **two output bits**.
- Output bits are XOR sums of the input and internal state.
- Binary, linear convolutional code.
- Code rate: $1/2$ (1 input $\rightarrow$ 2 output).
- **Not systematic** (none of output bits is simply the input bit).

3.2 Error-Detecting Codes

Usage context: Error-correcting codes are widely used on wireless links (notoriously noisy). Error-detecting codes are usually more efficient over fiber or high-quality copper for occasional errors.

Three Error-Detecting Codes

All are **linear, systematic block codes**:

1. **Parity**
2. **Checksums**
3. **Cyclic Redundancy Checks (CRCs)**

1. Parity

Mechanism: Single parity bit appended to data. Chosen so that the number of 1 bits in the codeword is even (or odd). **Computation:** Equivalent to modulo 2 sum or XOR of data bits.

Parity Example

Data: 1011010

Even parity: 10110100 (total 1s is even)

Odd parity: 10110101 (total 1s is odd)

Properties: Distance of 2 → can **detect single-bit errors**.

Efficiency Comparison

Channel: isolated errors, error rate 10^{-6} per bit. Block size: 1000 bits.

Table 3-4: Overhead Comparison per Megabit

Approach	Overhead per megabit
Hamming correction (Eq. 3-1: 10 check bits)	10,000 bits
Single parity + retransmission	~2,001 bits

Parity calculation: one parity bit per 1000-bit block. One block in 1000 has error → retransmit 1001 bits. Total: ~2001 bits overhead per megabit.

Burst Error Vulnerability

Single parity bit can only reliably detect single-bit errors. Long burst error: detection probability only **0.5** (unacceptable).

Interleaving

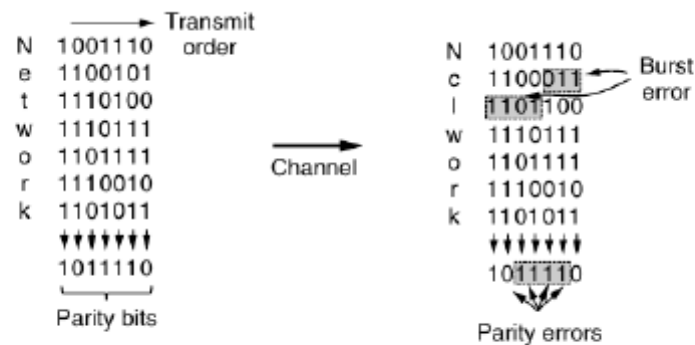


Figure 3-8. Interleaving of parity bits to detect a burst error.

Figure 3-3: Interleaving of parity bits to detect a burst error: 7×7 matrix, transmit order vs. burst error spread across columns causing parity errors.

Technique: Compute parity bits over data in different order than transmission order. Compute parity bit for each of n columns. Send all data bits as k rows (top to bottom, left to right). Last row: n parity bits.

Interleaving principle: Converts code detecting isolated errors into code detecting burst errors. Burst error of length n : bits in error spread across different columns. At most 1 bit in each of n columns affected. Parity bits on those columns detect the error.

Properties: Uses n parity bits on blocks of kn data bits. Detects single burst error of length $\leq n$. Burst of length $n + 1$: passes undetected if first and last bits inverted, others correct. Long burst or multiple shorter bursts: probability of bad block accepted = 2^{-n} .

2. Checksums

Definition: "Checksum" often means any group of check bits associated with a message. A group of parity bits is one example.

Stronger checksums: Based on running sum of data bits. Usually at end of message, as complement of sum function. **Verification:** Sum entire received codeword (data + checksum). If result is zero: no error detected.

Example: 16-bit Internet checksum (IP protocol): Sum of message bits divided into 16-bit words. Operates on words rather than bits. Detects errors that leave parity unchanged (e.g., flipping lowest-order bit in two different words).

One's complement arithmetic: Negative number = bitwise complement of positive counterpart. On two's complement computers: one's complement sum $\equiv$ sum modulo $2^{16} +$

overflow added back into low-order bits.

Weaknesses of Internet Checksum

- Assumption that frames contain random bits is wrong — real data patterns are non-random.
- Undetected errors more common than previously thought (Partridge et al., 1995).
- Does not detect deletion/addition of zero data.
- Does not detect swapping parts of message.
- Weak protection against message splices (parts of two packets combined).

Stronger alternative — Fletcher's checksum (Fletcher, 1982): Includes **positional component**: adds product of data and its position to running sum. Stronger detection of changes in data position.

3. Cyclic Redundancy Checks (CRCs)

Also known as **polynomial codes**. The strongest and most widely used error-detecting code at the link layer.

Polynomial Representation

- Bit string treated as polynomial with coefficients 0 and 1.
- k -bit frame = coefficient list for polynomial with k terms, from x^{k-1} to x^0 .
- High-order (leftmost) bit = coefficient of x^{k-1} .

Example

$$110001 \text{ (6 bits)} = 1 \cdot x^5 + 1 \cdot x^4 + 0 \cdot x^3 + 0 \cdot x^2 + 0 \cdot x^1 + 1 \cdot x^0$$

Polynomial arithmetic: Modulo 2 (no carries, no borrows). Addition and subtraction identical to XOR.

CRC Algorithm

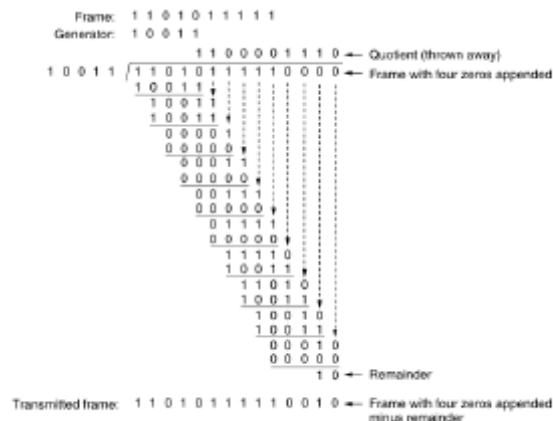


Figure 3-9. Example calculation of the CRC.

Figure 3-4: Example calculation of the CRC: long division of frame 1101011111 by generator 10011 ($x^4 + x + 1$), showing quotient, remainder, and transmitted frame.

Prerequisite: Sender and receiver agree on **generator polynomial** $G(x)$ in advance. Both high- and low-order bits of $G(x)$ must be 1. Frame to be checksummed must be longer than generator polynomial.

Goal: Append CRC such that polynomial of checksummed frame is **divisible by** $G(x)$.

Algorithm:

1. Let $r = \text{degree of } G(x)$. Append r **zero bits** to low-order end of frame. Frame now has $m + r$ bits, corresponding to polynomial $x^r \cdot M(x)$.
2. Divide bit string of $G(x)$ into bit string of $x^r \cdot M(x)$, using **modulo 2 division**.
3. **Subtract the remainder** (always r or fewer bits) from bit string of $x^r \cdot M(x)$ using modulo 2 subtraction.
4. Result = $T(x)$, the checksummed frame to be transmitted.

CRC Example (Fig. 3-4)

Frame: 1101011111

Generator $G(x) = x^4 + x + 1 \rightarrow$ bit string: 10011 (degree $r = 4$)

Append 4 zeros: 1101011111 0000

Divide by 10011 $\rightarrow$ Quotient: 1100001110 (discarded), Remainder: 10

Transmitted frame: 1101011111 **0010** (frame with 4 zeros minus remainder)

Why it works: $T(x)$ is divisible by $G(x)$. In any division, if you diminish dividend by remainder, result is divisible by divisor.

Error Detection Analysis

Received frame: $T(x) + E(x)$, where $E(x)$ has 1s at positions of inverted bits.

Receiver computes: $[T(x) + E(x)]/G(x) = E(x)/G(x)$ (since $T(x)/G(x) = 0$)

Errors slip by iff $E(x)$ contains $G(x)$ as a factor.

Table 3-5: CRC Error Detection Capabilities

Error Type	Detection
Single-bit error ($E(x) = x^i$)	Always detected if $G(x)$ has ≥ 2 terms
Two isolated single-bit errors	Detected if $G(x)$ not divisible by x and $G(x)$ does not divide $x^k + 1$ for k up to max frame length
Odd number of bit errors	Always detected if $x + 1$ is a factor of $G(x)$
Burst error of length $\leq r$	Always detected if $G(x)$ contains x^0 term
Burst error of length $r + 1$	Undetected iff burst identical to $G(x)$. Probability: $1/2^{r-1}$
Burst error $> r + 1$	Probability of undetected bad frame: $1/2^r$

Standard CRC Polynomials**IEEE 802 / Ethernet standard:**

$$G(x) = x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x^1 + 1$$

Properties: Detects all bursts of length ≤ 32 . Detects all bursts affecting odd number of bits. Used widely since 1980s.

Hardware implementation: Easy to compute and verify with simple **shift register circuits** (Peterson and Brown, 1961). Nearly always used in practice.

Deployment: Dozens of networking standards include various CRCs — virtually all LANs (Ethernet, 802.11) and point-to-point links (packets over SONET).

4. Carrier Sense Multiple Access Protocols

Motivation

With **slotted ALOHA**, the best channel utilization achievable is $1/e \approx 0.368$. This low result is expected because stations transmit at will without knowing what others are doing, leading to many collisions. In **LANs**, stations can often detect what other stations are doing and adapt behavior accordingly. These networks achieve **much better utilization** than $1/e$.

Carrier Sense Protocols

Protocols in which stations listen for a carrier (a transmission) and act accordingly.

4.1 CSMA Variants

1-Persistent CSMA

Mechanism:

1. Station has data to send → first **listens to the channel**.
2. If channel is **idle** → station sends its data immediately.
3. If channel is **busy** → station waits until it becomes idle, then transmits.
4. If a **collision occurs** → station waits a random amount of time and starts over.

Name origin: "1-persistent" because the station transmits with **probability 1** when it finds the channel idle.

Collision Scenarios (despite carrier sensing)

1. **Simultaneous ready stations:** Two stations become ready during a third station's transmission. Both wait politely until it ends, then begin transmitting simultaneously → collision.
2. **Propagation delay effect:** Just after a station begins sending, another station may become ready and sense the channel. If the first station's signal has not yet reached the second, the latter senses an idle channel and also begins sending → collision.

Propagation delay significance: Depends on the **bandwidth-delay product** of the channel. In most LANs: propagation delay is small, only a tiny fraction of a frame fits on the channel

→ small collision chance. Larger bandwidth-delay product → more important effect → worse protocol performance.

Nonpersistent CSMA

Design goal: Consciously less greedy than 1-persistent CSMA.

Mechanism:

1. Station senses channel when it wants to send.
2. If channel is **idle** → station begins transmitting.
3. If channel is **busy** → station **does not continually sense** it to seize immediately upon the end of previous transmission.
4. Instead, it **waits a random period of time** and repeats the algorithm.

Trade-off: Better channel utilization but **longer delays** than 1-persistent CSMA.

p-Persistent CSMA

Applicability: Slotted channels.

Mechanism:

1. Station becomes ready → senses channel.
2. If channel is **idle**: transmits with **probability** p , defers until next slot with **probability** $q = 1 - p$.
3. If next slot is also idle: repeat transmit/defer with probabilities p and q .
4. Process repeats until: frame has been transmitted, OR another station has begun transmitting (treat as collision: wait random time and start again).
5. If channel initially sensed as **busy** → wait until next slot and apply above algorithm.

Deployment: IEEE 802.11 uses a refinement of p-persistent CSMA.

Performance Comparison

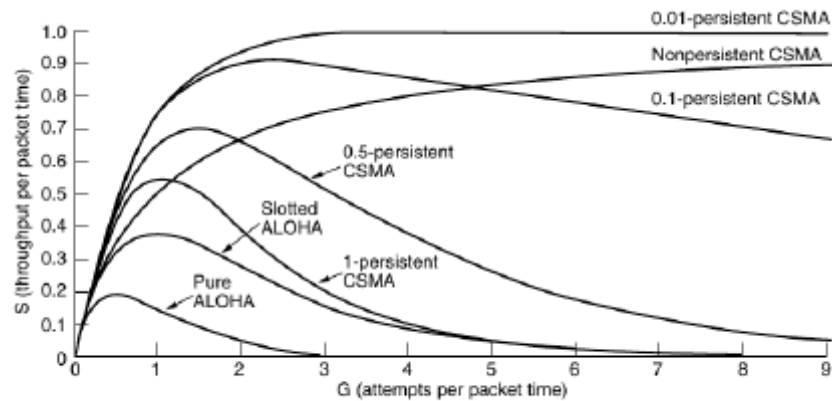


Figure 4-4. Comparison of the channel utilization versus load for various random access protocols.

Figure 4-1: Comparison of channel utilization S versus offered load G for various random access protocols.

Table 4-1: Throughput Comparison of Random Access Protocols

Protocol	Peak Throughput	Behavior at High Load
Pure ALOHA	~ 0.18	Collapses quickly
Slotted ALOHA	~ 0.37 ($1/e$)	Collapses after peak
1-persistent CSMA	~ 0.5	Degrades steadily
0.5-persistent CSMA	~ 0.7	Moderate degradation
0.1-persistent CSMA	~ 0.85	Slow degradation
0.01-persistent CSMA	~ 0.95	Very slow degradation
Nonpersistent CSMA	~ 0.9	Maintains high throughput

Key observations: All CSMA variants outperform ALOHA. Lower persistence (p) $\rightarrow$ higher peak throughput and better high-load behavior. Nonpersistent CSMA achieves near-optimal throughput. Trade-off: lower persistence $\rightarrow$ higher delays.

4.2 CSMA/CD (CSMA with Collision Detection)

Core Improvement

If two stations sense channel idle and begin transmitting simultaneously, their signals still collide. **CSMA/CD** adds the ability to **quickly detect collisions** and **abruptly stop transmitting** (rather than finishing the frame), since garbled frames are irretrievable anyway. This saves time and bandwidth. **Basis of classic Ethernet LAN.**

Collision detection as analog process: Station hardware must **listen to the channel while transmitting**. If received signal differs from transmitted signal → collision is occurring. **Implications:** Received signal must not be tiny compared to transmitted signal (difficult for wireless: received signals may be 1,000,000× weaker). Modulation must allow collision detection.

Conceptual Model

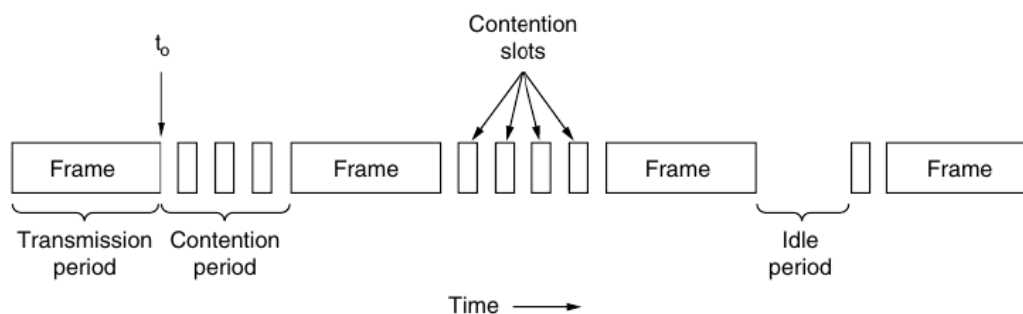


Figure 4-5. CSMA/CD can be in contention, transmission, or idle state.

Figure 4-2: CSMA/CD can be in contention, transmission, or idle state. Alternating transmission periods, contention periods (with slots), and idle periods.

CSMA/CD operates in three states:

- **Transmission periods:** Stations successfully transmitting frames.
- **Contention periods:** Multiple stations attempting to transmit, collisions may occur.
- **Idle periods:** All stations quiet (lack of work).

Contention Period Analysis

Critical question: How long until two colliding stations realize they have collided?

Minimum detection time: Time for signal to propagate from one station to the other.

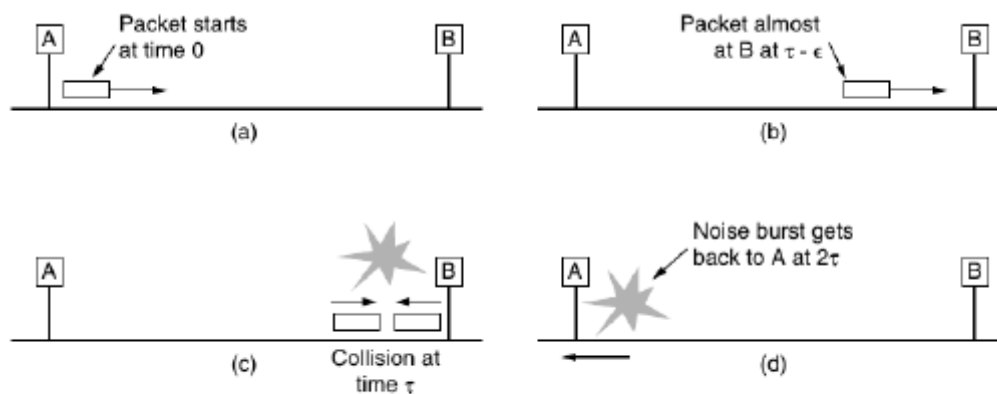


Figure 4-15. Collision detection can take as long as 2τ .

Figure 4-3: Collision detection can take as long as 2τ . (a) Packet starts at A at time 0. (b) Packet almost at B at $\tau - \epsilon$. (c) Collision at B at time τ . (d) Noise burst gets back to A at 2τ .

Table 4-2: Worst-Case Collision Detection Timing

Time	Event
0	Station A begins transmitting
$\tau - \epsilon$	Station B (farthest station) begins transmitting just before A's signal arrives
τ	Collision occurs at B
$2\tau - \epsilon$	Noise burst from collision reaches back to A

Worst-Case Conclusion

A station cannot be sure it has seized the channel until it has transmitted for 2τ without hearing a collision.

CSMA/CD as slotted ALOHA: Can think of CSMA/CD contention as slotted ALOHA with **slot width** = 2τ . **Practical example:** 1-km coaxial cable $\rightarrow \tau \approx 5 \mu\text{sec}$. **Performance improvement over slotted ALOHA:** Slots in which only one station transmits (channel seized) are followed by the **rest of a frame**. Greatly improves performance if frame time $\gg$ propagation time.

5. Classic Ethernet MAC Sublayer Protocol

5.1 Frame Format

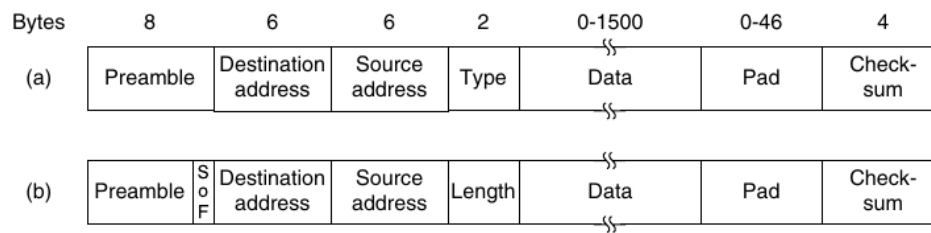


Figure 4-14. Frame formats. (a) Ethernet (DIX). (b) IEEE 802.3.

Figure 5-1: Frame formats: (a) Ethernet (DIX) with Type field. (b) IEEE 802.3 with Length field and SOF delimiter.

(a) Ethernet (DIX — Digital, Intel, Xerox)

Field	Size (bytes)	Description
Preamble	8	Synchronization pattern
Destination address	6	Recipient MAC address
Source address	6	Sender MAC address
Type	2	Indicates upper-layer protocol
Data	0–1500	Payload
Pad	0–46	Padding to meet minimum frame size
Checksum	4	32-bit CRC

(b) IEEE 802.3

Field	Size (bytes)	Description
Preamble	7	Synchronization
SOF (Start of Frame delimiter)	1	Marks frame start
Destination address	6	Recipient MAC address
Source address	6	Sender MAC address
Length	2	Frame length (≤ 1500)

Data	0–1500	Payload (includes LLC header)
Pad	0–46	Padding
Checksum	4	32-bit CRC

Field Details

Preamble

- **8 bytes** (DIX) or **7 bytes + SOF** (802.3).
- Each byte contains bit pattern 10101010.
- Exception: last byte (SOF delimiter) has last 2 bits set to 11.
- **Manchester encoding** produces 10-MHz square wave for 6.4 μ sec.
- Allows receiver's clock to synchronize with sender's.
- Last two 1 bits tell receiver that the rest of the frame is about to start.

Destination and Source Addresses

Size: 6 bytes (48 bits) each. **Destination address** — **first transmitted bit:** 0 → ordinary (unicast) address; 1 → group address.

Table 5-1: Address Types

Type	Description
Unicast	Ordinary address; frame delivered to single station
Multicast (group address)	Multiple stations listen to single address; all stations in group receive frame
Broadcast	Special address of all 1s; accepted by all stations on network

Source addresses — **globally unique:** Assigned centrally by **IEEE**. **OUI (Organizationally Unique Identifier):** First 3 bytes assigned by IEEE indicate manufacturer. Last 3 bytes assigned by manufacturer, programmed into NIC before sale.

Type vs. Length Field

Ethernet (DIX) — **Type field:** Tells receiver what to do with the frame. Multiple network-layer protocols may be in use simultaneously. Example: Type code 0x0800 = data contains IPv4 packet.

IEEE 802.3 — **Length field:** Carries frame length. No way for receiver to figure out what to do with incoming frame. **Solution:** Added **LLC (Logical Link Control)** header within data.

Resolution (1997)

By 1997, so much DIX Ethernet hardware/software in use that few wanted to repackage Type/Length fields. IEEE compromise: **both ways are fine. Rule:** Values $\leq 0x600$ (1536) → interpreted as **Length**; values $> 0x600$ → interpreted as **Type**.

Data Field

- **Maximum:** 1500 bytes.
- Chosen somewhat arbitrarily when Ethernet standard was finalized.
- Based on transceiver RAM requirements (RAM was expensive in 1978).

Minimum Frame Length and Pad Field

Minimum frame length: 64 bytes (from destination address to checksum, inclusive).

Why Minimum Length is Required

1. **Distinguish valid frames from garbage:** When transceiver detects collision, it truncates current frame. Stray bits and pieces of frames appear on cable constantly. Valid frames must be ≥ 64 bytes to distinguish from garbage.
2. **Prevent undetected collisions on short frames** (Fig. 5-2): If a station transmits a very short frame, transmission may complete **before** the noise burst gets back at 2τ . Sender incorrectly concludes frame was successfully sent.

Solution: All frames must take **more than 2τ** to send, so transmission is still in progress when noise burst returns.

Calculation for 10-Mbps LAN

Maximum length: 2500 meters + four repeaters. Worst-case round-trip time (including repeaters): nearly **50 μsec** . At 10 Mbps: 1 bit = 100 nsec. Minimum guaranteed frame: **500 bits**. With safety margin: rounded up to **512 bits = 64 bytes**.

Checksum

- **32-bit CRC.**
- Defined by the same generator polynomial from Section 3 (used in PPP, ADSL, and other links).
- Error-detecting code: frame **dropped** if error detected.
- No retransmission at data link layer; recovery left to higher layers.

5.2 CSMA/CD with Binary Exponential Backoff

Classic Ethernet uses **1-persistent CSMA/CD**:

- Stations sense medium when they have a frame to send.
- Send frame as soon as medium becomes idle.
- Monitor channel for collisions during transmission.
- If collision: abort with short **jam signal**, retransmit after random interval.

Binary Exponential Backoff Algorithm

Slot Definition

After collision, time divided into discrete slots of length = worst-case round-trip propagation time (2τ). For Ethernet: **512 bit times = 51.2 μ sec**.

Table 5-2: Binary Exponential Backoff

Collision Number (i)	Randomization Interval	Slots to Wait
1	[0, 1]	0 or 1
2	[0, 3]	0, 1, 2, or 3
3	[0, 7]	0 to 7
...	...	...
i	$[0, 2^i - 1]$	0 to $2^i - 1$
≥ 10	[0, 1023]	frozen at max 1023
16	—	Failure reported to computer

Design rationale:

- **Low delay when few stations collide:** Small randomization interval (0 or 1 slots) after first collision.
- **Reasonable resolution when many stations collide:** Interval grows exponentially, ensuring collisions resolve in reasonable time.
- **Truncation at 1023:** Prevents bound from growing too large.

No Acknowledgements in CSMA/CD

If no collision, sender assumes frame was probably successfully delivered. **No acknowledgements** in CSMA/CD or Ethernet. Appropriate for wired/optical fiber with low error rates. Errors detected by CRC, recovered by higher layers. Wireless channels (more errors) use acknowledgements.

6. Data Link Control Protocols

Traditionally, **four protocols** have been defined for the data-link layer to deal with flow and error control:

Table 6-1: Data-Link Layer Protocols

Protocol	Status at Data-Link Layer
Simple	Still used; discussed here
Stop-and-Wait	Still used; discussed here
Go-Back-N	Disappeared from data-link layer; full detail in transport layer
Selective-Repeat	Disappeared from data-link layer; full detail in transport layer

Finite State Machine (FSM) Representation

Figure 11.6 Connectionless and connection-oriented service represented as FSMs

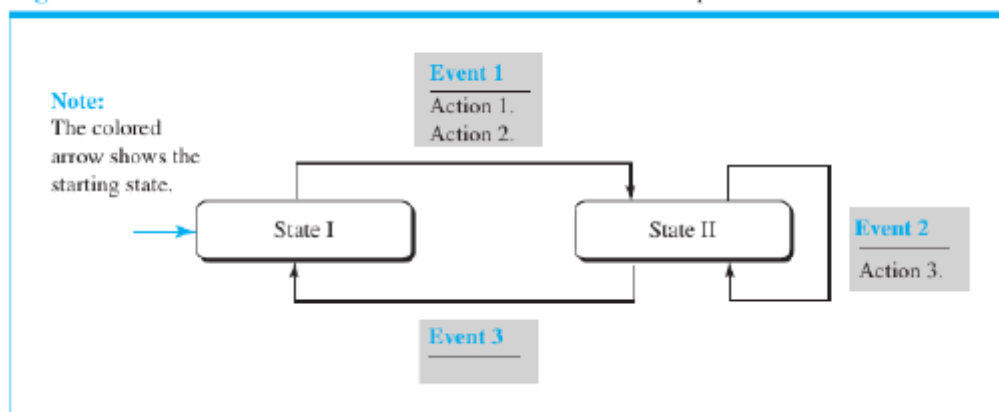


Figure 6-1: FSM notation example: Rounded-corner rectangles = states, colored text = events, regular black text = actions, horizontal line separates event from actions, arrow = movement to next state.

The behavior of data-link-layer protocols is best shown as a **finite state machine (FSM)**. A machine with a finite number of states. The machine is always in one state until an event occurs. **Each event is associated with two reactions:** (1) A list (possibly empty) of **actions** to be performed, and (2) Determination of the **next state** (which may be the same as the current state). **One state must be defined as the initial state.**

6.1 Simple Protocol

Characteristics

- **No flow control.**
- **No error control.**
- Assumption: receiver can immediately handle any frame it receives (receiver can never be overwhelmed).

Figure 11.7 Simple protocol

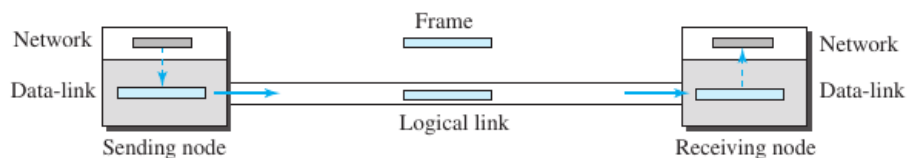


Figure 6-2: Simple protocol layout: sending node → logical link → receiving node.

Operation:

- **Sender data-link layer:** Gets packet from network layer → makes a frame → sends the frame.
- **Receiver data-link layer:** Receives frame from link → extracts packet → delivers packet to network layer.

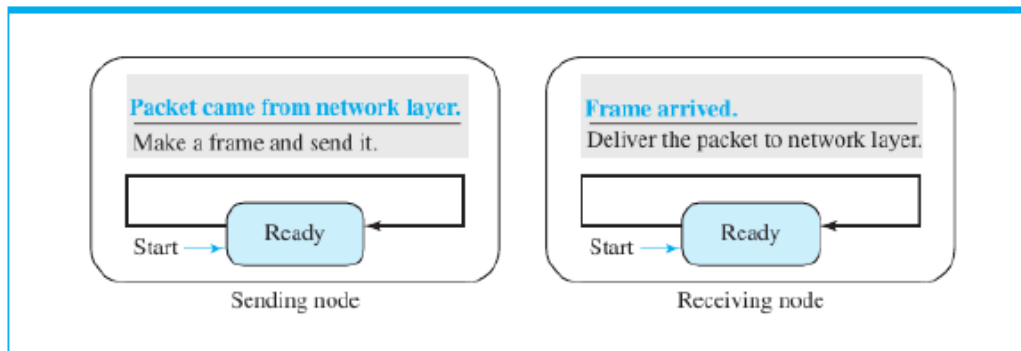
Figure 11.8 FSMs for the simple protocol

Figure 6-3: FSMs for simple protocol: sender (Ready → send frame) and receiver (Ready → deliver packet).

FSMs for Simple Protocol

Sending node: One state: Ready. Event: "Packet came from network layer." Action: "Make a frame and send it." Transition: Self-loop (remains in Ready).

Receiving node: One state: Ready. Event: "Frame arrived." Action: "Deliver the packet to network layer." Transition: Self-loop (remains in Ready).

Example 11.2

Figure 11.9 shows an example of communication using this protocol. It is very simple. The sender sends frames one after another without even thinking about the receiver.

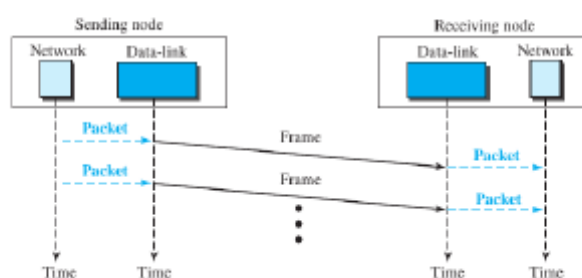
Figure 11.9 Flow diagram for Example 11.2

Figure 6-4: Flow diagram for Simple protocol: continuous unidirectional frame transmission without flow/error control.

6.2 Stop-and-Wait Protocol

Characteristics

- Uses **both flow control and error control**.

- Primitive version discussed here; sophisticated version with sliding windows in transport layer.

Figure 11.10 *Stop-and-Wait protocol*

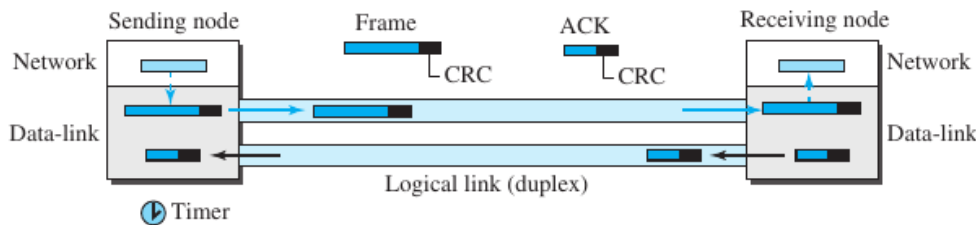


Figure 6-5: *Stop-and-Wait protocol layout: duplex logical link with Frame + CRC, ACK + CRC, and timer.*

Mechanism

- **Logical link is duplex.**
- Sender sends **one frame at a time** and waits for an **acknowledgment (ACK)** before sending the next.
- **CRC** added to each data frame for error detection.
- Only **one frame and one ACK** can be in the channels at any time.

Error detection: When a frame arrives at receiver, CRC is checked. If CRC is **incorrect**: frame is **corrupted** and **silently discarded**. The **silence of the receiver** signals the sender that a frame was either corrupted or lost.

Timer operation: Every time sender sends a frame, it **starts a timer**. If ACK arrives **before timer expires**: timer is stopped, sender sends next frame (if ready). If **timer expires**: sender **resends previous frame** (assuming frame was lost or corrupted). Sender must **keep a copy** of the frame until its ACK arrives.

Figure 11.11 *FSM for the Stop-and-Wait protocol*

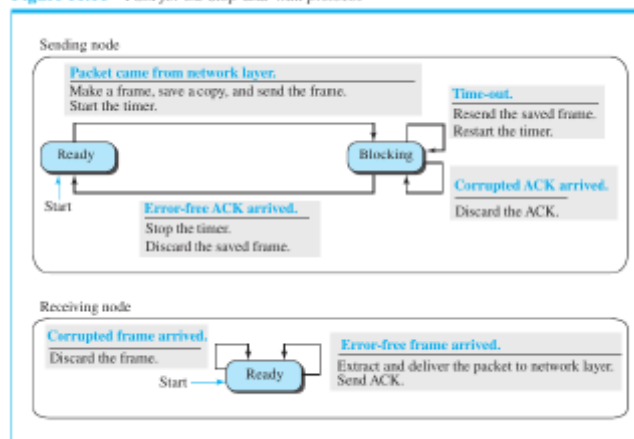


Figure 6-6: *FSMs for Stop-and-Wait: sender (Ready/Blocking states) and receiver (Ready state).*

FSMs for Stop-and-Wait

Sending Node States:

State	Description
Ready	Waiting for packet from network layer
Blocking	Waiting for ACK or time-out

Ready State: Event: "Packet came from network layer." Actions: (1) Make a frame, (2) Save a copy of the frame, (3) Start the timer, (4) Send the frame. Next state: Blocking.

Blocking State — three possible events:

Event	Actions	Next State
Time-out	Resend the saved frame; Restart the timer	Blocking (self-loop)
Corrupted ACK arrived	Discard the ACK	Blocking (self-loop)
Error-free ACK arrived	Stop the timer; Discard the saved frame	Ready

Receiving Node: Always in Ready state.

Event	Actions	Next State
Error-free frame arrived	Extract and deliver packet to network layer; Send ACK	Ready (self-loop)
Corrupted frame arrived	Discard the frame	Ready (self-loop)

Sequence and Acknowledgment Numbers

Figure 11.13 Flow diagram for Example 11.4

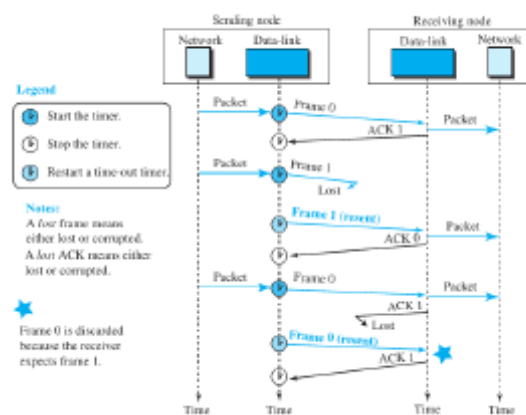


Figure 6-7: Flow diagram for Stop-and-Wait with sequence and acknowledgment numbers preventing duplicate delivery.

The Duplicate Problem

Scenario: First frame sent and acknowledged successfully. Second frame sent but **lost** → after time-out, resent. Third frame sent and acknowledged, but the **ACK is lost** → frame is resent. The network layer at the receiver receives **two copies of the third packet**. Duplicate packets are incorrect.

Solution: Add **sequence numbers** to data frames and **acknowledgment numbers** to ACK frames.

Numbering scheme (simple 1-bit alternating):

- **Sequence numbers:** 0, 1, 0, 1, 0, 1, ...
- **Acknowledgment numbers:** 1, 0, 1, 0, 1, 0, ...
- Sequence numbers start with **0**.
- Acknowledgment numbers start with **1**.
- An acknowledgment number always defines the **sequence number of the next frame to receive**.

Key insight (Fig. 6-7): The receiver uses sequence numbers to detect duplicates. When the resent Frame 0 arrives, the receiver was expecting Frame 1 (acknowledgment number 1 indicates next expected frame is 1), so it discards the duplicate Frame 0 and retransmits ACK 1.

Piggybacking

The Simple and Stop-and-Wait protocols are designed for **unidirectional communication**.

Piggybacking concept: Data in one direction is combined with (piggybacked on) the acknowledgment in the other direction. Piggybacking is **not a common practice** at the data-link layer because it makes communication significantly more complicated.

6.3 HDLC (High-level Data Link Control)

Overview

- **Bit-oriented protocol** for communication over **point-to-point** and **multipoint** links.
- Implements the **Stop-and-Wait protocol**.
- More of a **theoretical foundation** than a practical protocol today.
- **Basis for other practical protocols:** PPP, Ethernet protocol, Wireless LAN protocols.

Configurations and Transfer Modes

Figure 11.14 Normal response mode

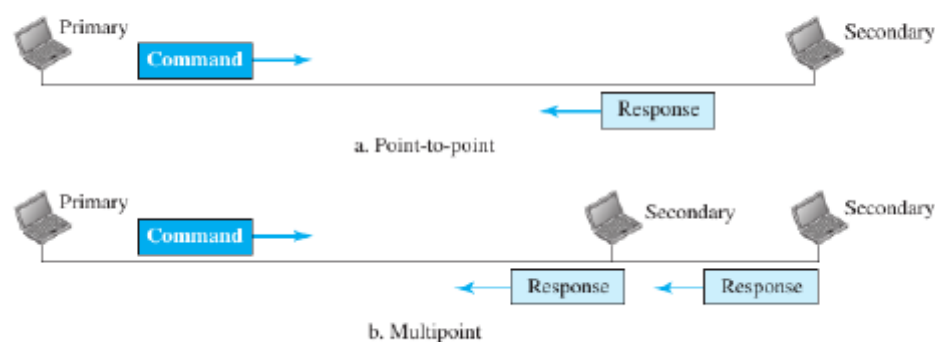


Figure 6-8: Normal Response Mode (NRM): (a) point-to-point and (b) multipoint configurations. One primary station and multiple secondary stations.

1. Normal Response Mode (NRM)

- **Configuration: Unbalanced** — one primary station and multiple secondary stations.
- Primary station: can send **commands**.
- Secondary stations: can only send **responses**.

- **Usage:** Both point-to-point and multipoint links.

Figure 11.15 Asynchronous balanced mode



Figure 6-9: Asynchronous Balanced Mode (ABM): peer-to-peer combined stations. This is the common mode used today.

2. Asynchronous Balanced Mode (ABM)

- **Configuration: Balanced** — link is point-to-point.
- Each station functions as both **primary and secondary** (peers).
- Either station can initiate communication.
- This is the **common mode used today**.

Framing

Figure 11.16 HDLC frames

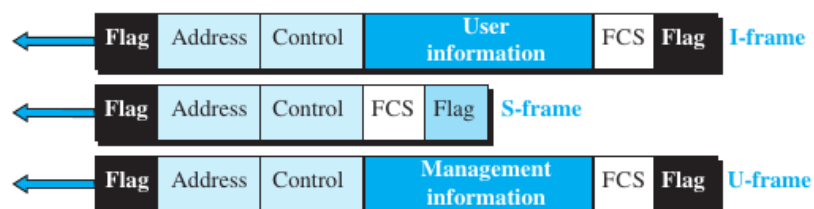


Figure 6-10: HDLC frame formats: I-frame (with user info), S-frame (no info), U-frame (with management info).

HDLC defines **three types of frames**:

Frame Type	Purpose	User Data	Control Info
I-frame (Information)	Carry user data and control info (piggybacking)	Yes	Yes
S-frame (Supervisory)	Transport control information only	No	Yes

U-frame (Unnumbered)	System management	No (management info only)	Yes
-----------------------------	-------------------	---------------------------	-----

Each frame contains up to **six fields**: Flag | Address | Control | Information | FCS | Flag

Table 6-2: HDLC Frame Fields

Field	Description
Flag	Synchronization pattern 01111110; identifies both beginning and end of frame
Address	Address of secondary station. Primary-created frame: to-address . Secondary-created frame: from-address
Control	1 or 2 bytes for flow and error control. Format depends on frame type
Information	User's data from network layer or management information
FCS	Frame Check Sequence; contains 2- or 4-byte CRC

Control Field Formats

Figure 11.17 Control field format for the different frame types

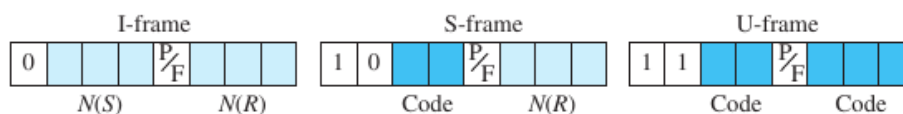


Figure 6-11: Control field formats for I-frame ($0 + N(S) + P/F + N(R)$), S-frame ($10 + \text{Code} + P/F + N(R)$), and U-frame ($11 + \text{Prefix} + P/F + \text{Suffix}$).

I-Frame Control Field

Subfield	Bits	Description
First bit	1	Always 0 → identifies I-frame
N(S)	3	Sequence number of frame (0–7)
P/F	1	Poll/Final bit (Poll when primary → secondary, Final when secondary → primary)
N(R)	3	Acknowledgment number (for piggybacking)

S-Frame Control Field

Code	Type	Description
00	RR — Receive Ready	Acknowledges receipt of safe frame(s)
10	RNR — Receive Not Ready	Acknowledges receipt + announces receiver is busy (congestion control)
01	REJ — Reject	NAK frame for Go-Back-N ARQ
11	SREJ — Selective Reject	NAK frame for Selective Repeat ARQ

Note: S-frames do **not** have information fields.

U-Frame Control Field

- First 2 bits: 11 → identifies U-frame.
- 5 bits total (2-bit prefix + 3-bit suffix) → up to **32 different types of U-frames**.
- Purpose: Exchange session management and control information.
- Contains information field for **system management information** (not user data).

Figure 11.18 Example of connection and disconnection

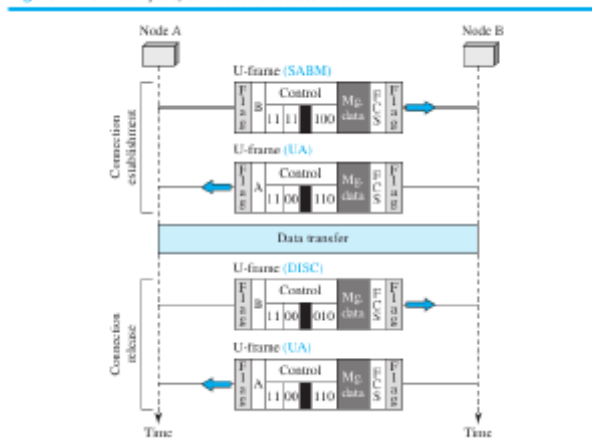


Figure 6-12: Example of connection establishment (SABM/UA) and connection release (DISC/UA) using U-frames.

Connection Establishment and Release

Table 6-3: Connection Phases

Phase	Direction	Frame	Details
-------	-----------	-------	---------

Establishment	A → B	U-frame (SABM)	Set Asynchronous Balanced Mode
	B → A	U-frame (UA)	Unnumbered Acknowledgment
Data Transfer	Data transfer occurs between the two nodes		
Release	A → B	U-frame (DISC)	Disconnect
	B → A	U-frame (UA)	Unnumbered Acknowledgment

Figure 11.19 Example of piggybacking with and without error

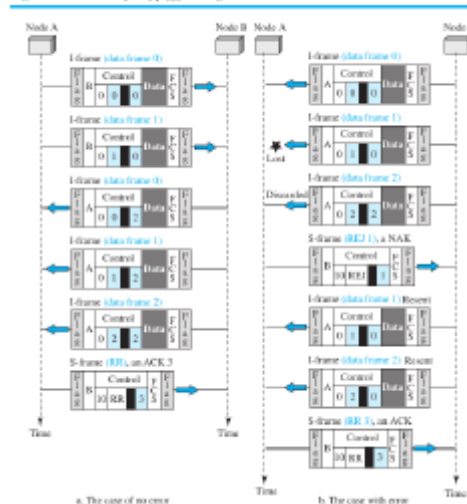


Figure 6-13: Example of piggybacking: (a) no error case with data frames carrying piggybacked acknowledgments. (b) error case with REJ/NAK recovery.

6.4 Point-to-Point Protocol (PPP)

Overview

- One of the most common protocols for **point-to-point access**.
- Millions of Internet users connect home computers to ISP servers via PPP.
- Typical setup: traditional modem over telephone line (physical layer) + PPP at data-link layer.
- **Most common** point-to-point data-link protocol.

Services Provided by PPP

Service	Description
Frame format definition	Defines format of frames exchanged between devices

Link negotiation	Defines how two devices negotiate link establishment and data exchange
Multi-network-layer support	Accepts payloads from several network layers (not only IP)
Authentication	Provided but optional
Network address configuration	Particularly useful for home users needing temporary network address

Services NOT Provided by PPP

Service	Status	Details
Flow control	Not provided	Sender can send several frames one after another
Error control	Minimal	CRC field used to detect errors. Corrupted frame silently discarded; upper-layer handles recovery
Multipoint addressing	Not provided	No sophisticated addressing mechanism

Framing

Figure 11.20 PPP frame format

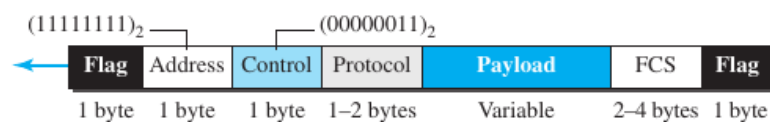


Figure 6-14: PPP frame format: Flag, Address, Control, Protocol, Payload, FCS, Flag.

PPP uses a **character-oriented (byte-oriented) frame**.

Table 6-4: PPP Frame Fields

Field	Size	Value / Description
Flag	1 byte	01111110 — marks start and end of frame
Address	1 byte	Constant 11111111 (broadcast address)
Control	1 byte	Constant 00000011 — imitates unnumbered frames in HDLC
Protocol	1-2 bytes	Defines what is carried in data field

Payload	Variable (max 1500 bytes)	Carries user data or other information. Byte-stuffed if flag pattern appears
FCS	2 or 4 bytes	Standard CRC

Byte Stuffing in PPP

Since PPP is **byte-oriented**, the flag byte must be escaped when it appears in the data section.

Aspect	Detail
Escape byte	01111101
Mechanism	Whenever flag-like pattern (01111110) appears in data, escape byte is stuffed before it
Escape byte itself	Must also be stuffed with another escape byte if it appears in data

Transition Phases

Figure 11.21 Transition phases

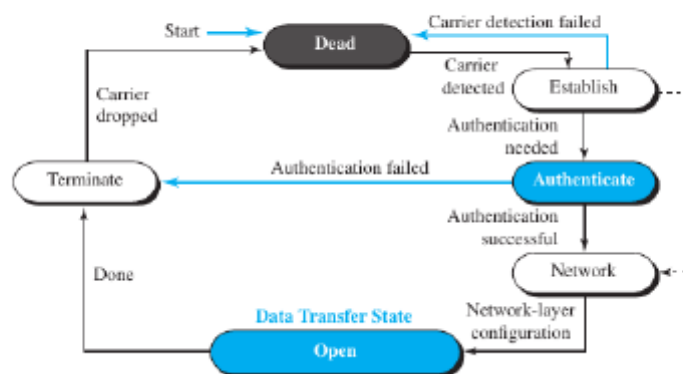


Figure 6-15: PPP transition phases: *Dead* → *Establish* → *Authenticate* → *Network* → *Open* → *Terminate* → *Dead*.

Table 6-5: PPP Phase Descriptions

Phase	Description
Dead	Initial state. No active carrier at physical layer; line is quiet
Establish	One node starts communication. Options are negotiated between parties
Authenticate	(Optional) If authentication is needed. If fails → goes to Terminate

Network	Network-layer configuration performed
Open	Data Transfer State. Exchange of data packets begins
Terminate	Connection termination requested. Returns to Dead when carrier is dropped

7. Practice Questions & Answers

How to Use This Section

This section contains exam-style questions covering both theoretical concepts and numerical problems. Review these after studying the notes to test your understanding. Numerical problems include step-by-step solutions.

7.1 Theoretical Questions

Q1. What are the two principal parts of the telephone system from an engineering perspective? Explain each.

Answer: The two principal parts are: (1) **Outside plant** — consists of local loops and trunks, physically located outside switching offices. (2) **Inside plant** — consists of switches located inside switching offices. The outside plant handles the physical connections to subscribers and between offices, while the inside plant handles the switching and routing of calls.

Q2. Explain the fundamental difference between circuit switching and packet switching with respect to path establishment.

Answer: In **circuit switching**, a dedicated physical path must be established from sender to receiver before any data can be transmitted. This setup takes time (easily 10+ seconds for long-distance calls). In **packet switching**, there is no need to set up a dedicated path in advance. Packets are sent as soon as they are available, and each packet is routed independently through the network.

Q3. Why is packet switching more fault-tolerant than circuit switching?

Answer: In circuit switching, if a switch fails, **all circuits using it are terminated** because each call depends on a specific dedicated path. In packet switching, packets can be **dynamically rerouted around dead switches** since there is no fixed path. Each packet can take a different route, so the failure of one switch does not prevent packets from reaching their destination.

Q4. What is the fundamental trade-off between circuit switching and packet switching?

Answer: The fundamental trade-off is: **guaranteed service + wasted resources** (circuit switching) versus **no guarantee + no waste** (packet switching). Circuit switching provides guaranteed bandwidth and constant delay but wastes resources when the connection is idle. Packet switching uses bandwidth efficiently but cannot guarantee delay or throughput.

Q5. List and briefly explain the three types of services provided by the data link layer to the network layer.

Answer: (1) **Unacknowledged Connectionless Service:** Frames sent independently without acknowledgements. No logical connection. Lost frames not recovered at data link layer. Used in Ethernet. (2) **Acknowledged Connectionless Service:** No logical connections, but each frame is individually acknowledged. Lost frames can be retransmitted. Used in 802.11 (WiFi). (3) **Acknowledged Connection-Oriented Service:** Connection established before data transfer. Each frame numbered. Guarantees exactly-once, in-order delivery. Used for satellite channels and long-distance circuits.

Q6. Why is providing acknowledgements at the data link layer considered an "optimization" rather than a requirement?

Answer: The network layer could always implement its own acknowledgement and retransmission mechanism. However, the data link layer knows **link-specific parameters** (maximum frame length, propagation delays) that the network layer does not. If a large packet is broken into 10 frames and 2 are lost, network-layer retransmission of the entire packet would be very slow. Data-link-layer frame acknowledgement and retransmission correct errors more directly and quickly, making it an optimization for efficiency.

Q7. Explain the four methods of framing used in the data link layer.

Answer: (1) **Byte Count:** A field in the header specifies the number of bytes in the frame. Vulnerable to errors in the count field. (2) **Flag Bytes with Byte Stuffing:** Special flag bytes delimit frames. If flag pattern appears in data, an escape byte (ESC) is inserted before it. (3) **Flag Bits with Bit Stuffing:** Flag pattern 01111110 delimits frames. After five consecutive 1s in data, a 0 is stuffed. Used in HDLC, USB. (4) **Physical Layer Coding Violations:** Uses reserved signal patterns in physical encoding (e.g., 4B/5B) to mark frame boundaries. No stuffing needed.

Q8. Why is byte count framing rarely used by itself?

Answer: The byte count field can be **garbled by transmission errors**. If the count becomes corrupted (e.g., due to a bit flip), the destination loses synchronization and cannot locate the correct start of the next frame. Even if the checksum detects the error, the receiver doesn't know how many bytes to skip to find the next frame. Retransmission requests are ineffective because the destination cannot tell where the next frame starts.

Q9. Explain the difference between error-correcting codes (FEC) and error-detecting codes.

Answer: Error-correcting codes (FEC — Forward Error Correction) include enough redundant information for the receiver to deduce what the original transmitted data must have been. The receiver can correct errors without requesting retransmission. Preferred on noisy channels (e.g., wireless) where retransmissions are likely to also be corrupted. **Error-detecting codes** include only enough redundancy to detect that an error occurred. The receiver then requests retransmission. More efficient on reliable channels (e.g., fiber) where errors are rare.

Q10. What is Hamming distance and why is it important?

Answer: The **Hamming distance** between two codewords is the number of bit positions in which they differ. It is computed by XORing the two codewords and counting the 1 bits. It is important because: (1) To **detect d errors**, a code needs Hamming distance of at least $d + 1$. (2) To **correct d errors**, a code needs Hamming distance of at least $2d + 1$. The minimum Hamming distance between any two valid codewords determines the error-handling capability of the entire code.

Q11. Explain the relationship between the number of check bits r and message bits m in Hamming codes (Equation 3-1).

Answer: The Hamming code requirement is: $(m + r + 1) \leq 2^r$. This ensures there are enough bit patterns to represent all 2^m legal messages plus their $n = m + r$ single-error variants (distance-1 codewords), plus the valid codeword itself. Each of the 2^m messages requires $n + 1 = m + r + 1$ dedicated patterns, and the total available patterns is $2^n = 2^{m+r}$.

Q12. Compare parity, checksum, and CRC as error-detecting mechanisms.

Answer: Parity: Single bit appended to make total 1s even/odd. Distance of 2. Detects single-bit errors only. Vulnerable to burst errors (50% detection probability). **Checksum:** Based on running sum of data bits (e.g., 16-bit Internet checksum). Detects errors that leave parity unchanged. Weaker than CRC; doesn't detect zero additions or message swaps. **CRC (Cyclic Redundancy Check):** Strongest and most widely used. Based on polynomial division. Can detect all burst errors up to length r , all odd-numbered bit errors (if $x + 1$ is a factor), and has controllable undetected error probability of $1/2^r$.

Q13. What is the "hidden terminal problem" and how does CSMA/CD address it differently from wireless networks?

Answer: The hidden terminal problem occurs when two stations cannot sense each other's transmissions but can both reach a common receiver, causing collisions. **CSMA/CD (wired)** addresses this by having stations **listen while transmitting** — if the received signal differs from the transmitted signal, a collision is detected and transmission is aborted. This works because wired signals have predictable strength. **In wireless**, received signals may be 1,000,000× weaker than transmitted signals, making collision detection impractical. Wireless uses CSMA/CA (Collision Avoidance) instead, with RTS/CTS handshaking.

Q14. Explain binary exponential backoff in CSMA/CD. Why is it designed this way?

Answer: After a collision, the randomization interval grows exponentially: $[0, 2^i - 1]$ slots after the i -th collision, capped at $[0, 1023]$ after 10 collisions, and failure reported after 16 collisions. **Rationale:** (1) **Low delay when few stations collide:** Small interval (0 or 1 slots) after first collision. (2) **Reasonable resolution when many collide:** Exponential growth ensures collisions resolve in reasonable time. (3) **Truncation at 1023** prevents the bound from growing too large and causing excessive delays.

Q15. Why is there a minimum frame size of 64 bytes in Ethernet?

Answer: Two reasons: (1) **Distinguish valid frames from garbage:** When transceiver detects collision, it truncates current frame. Stray bits appear on cable constantly. Valid frames must be ≥ 64 bytes to be distinguishable from garbage. (2) **Prevent undetected collisions on short frames:** A station must still be transmitting when the collision noise burst returns at time 2τ . If a frame is too short, transmission completes before the noise burst returns, and the sender incorrectly assumes success. At 10 Mbps with worst-case round-trip of $\sim 50 \mu\text{sec}$, minimum frame = 512 bits = 64 bytes.

Q16. Explain the difference between Ethernet (DIX) and IEEE 802.3 frame formats.

Answer: Both have the same overall structure but differ in: (1) **Preamble:** DIX uses 8 bytes; 802.3 uses 7 bytes + 1 SOF delimiter byte. (2) **Type/Length field:** DIX uses a **Type** field (values > 1536) indicating the upper-layer protocol (e.g., $0x0800 = \text{IPv4}$). IEEE 802.3 uses a **Length** field (≤ 1500) carrying frame length. This required adding an LLC header inside the data field to identify the protocol. (3) The 1997 compromise: values $\leq 0x600 = \text{Length}$; values $> 0x600 = \text{Type}$.

Q17. What is piggybacking and why is it not commonly used at the data link layer?

Answer: **Piggybacking** is the technique of combining data in one direction with acknowledgements in the other direction within the same frame (e.g., when node A sends data to B, it also includes ACK for data received from B). It is **not commonly used at the data link layer** because it makes communication significantly more complicated. The data link layer typically handles unidirectional traffic efficiently, and the added complexity of managing bidirectional data+ack combinations outweighs the benefit for most link-layer scenarios.

Q18. Compare NRM and ABM in HDLC.

Answer: **NRM (Normal Response Mode):** Unbalanced configuration with one primary station and multiple secondary stations. Primary sends commands; secondaries can only send responses. Used for both point-to-point and multipoint links. **ABM (Asynchronous Balanced Mode):** Balanced configuration for point-to-point links. Each station functions as both primary and secondary (peers). Either station can initiate communication. ABM is the **common mode used today**.

Q19. What services does PPP NOT provide compared to HDLC?

Answer: PPP does **not provide**: (1) **Flow control** — sender can send several frames one after another without concern about overwhelming the receiver. (2) **Error control** (beyond detection) — CRC is used to detect errors, but corrupted frames are silently discarded; upper-layer protocol handles recovery. (3) **Multipoint addressing** — no sophisticated addressing for multipoint configurations. The Address field is set to constant 11111111 (broadcast), and the Control field is constant 00000011.

Q20. List the four data-link layer protocols for flow and error control and their current status.

Answer: (1) **Simple Protocol**: Still used; no flow or error control. (2) **Stop-and-Wait Protocol**: Still used; implements both flow and error control. (3) **Go-Back-N**: Disappeared from data-link layer; full detail in transport layer. (4) **Selective-Repeat**: Disappeared from data-link layer; full detail in transport layer. The first two are discussed at the data link layer because they are needed to understand wired and wireless LANs.

7.2 Numerical Problems

N1. A code has four valid codewords: 0000000000, 0000011111, 1111100000, and 1111111111. What is the Hamming distance of this code? How many errors can it correct? How many can it detect?

Solution:

Step 1: Find Hamming distance between pairs:

- $d(0000000000, 0000011111) = 5$
- $d(0000000000, 1111100000) = 5$
- $d(0000000000, 1111111111) = 10$
- $d(0000011111, 1111100000) = 10$
- $d(0000011111, 1111111111) = 5$
- $d(1111100000, 1111111111) = 5$

Step 2: Minimum Hamming distance = 5.

Step 3: For correction: need $d \geq 2t + 1$, so $5 \geq 2t + 1 \rightarrow t \leq 2$. Can correct **2 errors**.

Step 4: For detection: need $d \geq e + 1$, so $5 \geq e + 1 \rightarrow e \leq 4$. Can detect **4 errors**.

N2. Given a message with $m = 8$ data bits, what is the minimum number of check bits r required for a Hamming code that can correct all single-bit errors?

Solution:

Using the Hamming code inequality: $(m + r + 1) \leq 2^r$

Substitute $m = 8$: $(8 + r + 1) \leq 2^r \rightarrow (9 + r) \leq 2^r$

Test values of r :

- $r = 3$: $9 + 3 = 12 \leq 8$? **No**
- $r = 4$: $9 + 4 = 13 \leq 16$? **Yes**

Minimum check bits required: $r = 4$. This gives a (12, 8) Hamming code.

N3. A channel has an error rate of 10^{-6} per bit. Blocks are 1000 bits. Compare the overhead per megabit for: (a) Hamming code correction (using 10 check bits per block), and (b) single parity bit plus retransmission.

Solution:

(a) Hamming correction:

Overhead = 10 check bits per 1000-bit block.

Blocks per megabit = $10^6 / 1000 = 1000$ blocks.

Total overhead = $10 \times 1000 = \mathbf{10,000}$ bits.

(b) Single parity + retransmission:

Error probability per block = $1000 \times 10^{-6} = 10^{-3}$.

Expected errors per megabit = $1000 \times 10^{-3} = 1$ block in error.

Overhead per error = 1 parity bit + 1000-bit retransmission = 1001 bits.

Plus parity bits for all blocks = $1 \times 1000 = 1000$ bits.

Total overhead = $1000 + 1001 \approx \mathbf{2,001}$ bits.

Conclusion: Parity + retransmission is about 5× **more efficient** for this reliable channel.

N4. In a 10-Mbps CSMA/CD network, the maximum distance between any two stations is 2.5 km and there are four repeaters. The signal propagation speed is 2×10^8 m/s. Calculate the minimum frame size if the worst-case round-trip time (including repeaters) is 50 μ sec.

Solution:

Step 1: Bit time at 10 Mbps = $1/10^7 = 100$ nsec = 0.1 μ sec.

Step 2: Minimum frame bits = Worst-case RTT / Bit time
 $= 50 \mu\text{sec} / 0.1 \mu\text{sec} = 500$ bits.

Step 3: With safety margin, round up to nearest power of 2: **512 bits = 64 bytes.**

Step 4: Verify propagation time:

$$\tau = 2500 \text{ m} / (2 \times 10^8 \text{ m/s}) = 12.5 \mu\text{sec one way.}$$

Worst-case collision detection = $2\tau = 25 \mu\text{sec}$ (plus repeater delays = 50 μ sec total).

N5. A CSMA/CD network uses binary exponential backoff. After the 5th collision, what is the range of slots a station must wait? What is the maximum wait after 12 collisions?

Solution:

Step 1: For the i -th collision, randomization interval is $[0, 2^i - 1]$.

Step 2: After **5th collision**:

$$\text{Interval} = [0, 2^5 - 1] = [0, 31].$$

Station waits a random number of slots from **0 to 31**.

Step 3: After **12th collision**:

Since $i \geq 10$, the interval is **frozen at** $[0, 1023]$.

Station waits a random number of slots from **0 to 1023**.

Step 4: After **16 collisions**, failure is reported to the computer.

N6. An Ethernet frame carries 1000 bytes of data. Calculate the total frame size including all overhead (use DIX Ethernet format). What is the efficiency?

Solution:

Step 1: DIX Ethernet overhead per frame:

- Preamble: 8 bytes
- Destination address: 6 bytes
- Source address: 6 bytes
- Type: 2 bytes
- Pad: 0 bytes (data $\geq$ 46 bytes, so no padding needed)
- Checksum (CRC): 4 bytes

Step 2: Total overhead = $8 + 6 + 6 + 2 + 0 + 4 = 26$ bytes.

Step 3: Total frame size = $1000 + 26 = \mathbf{1026}$ bytes.

Step 4: Efficiency = Data / Total = $1000/1026 = \mathbf{97.5\%}$.

N7. A CRC generator polynomial is $G(x) = x^3 + x + 1$ (bit string 1011). The data frame is 1101. Compute the transmitted frame.

Solution:

Step 1: Degree of $G(x) = r = 3$. Append 3 zeros to data: **1101000**.

Step 2: Divide 1101000 by 1011 using modulo 2 division:

```

      1110
1011 | 1101000
      1011
      ----
        1100
        1011
        ----
          1110
          1011
          ----
            1010
            1011
            ----
              0010 ← Remainder = 010

```

Step 3: Transmitted frame = 1101000 + 010 = **1101010** (data + CRC).

Step 4: Verify: 1101010 / 1011 should have remainder 0.

N8. Using interleaving with a 5×5 matrix, compute the parity bits for the data "HELLO" (ASCII: H=1001000, E=1000101, L=1001100, L=1001100, O=1001111). Show how a burst error of 5 consecutive bits is detected.

Solution:

Step 1: Arrange data in 5×5 matrix (using last 5 bits of each 7-bit ASCII):

```

Row 0 (H): 0 0 1 0 0
Row 1 (E): 1 0 1 0 1
Row 2 (L): 1 0 0 1 1
Row 3 (L): 1 0 0 1 1
Row 4 (O): 1 1 1 1 1

```

Step 2: Compute column parity bits (even parity):

Col 0: $0+1+1+1+1 = 4 \rightarrow \text{parity} = 0$

Col 1: $0+0+0+0+1 = 1 \rightarrow \text{parity} = 1$

Col 2: $1+1+0+0+1 = 3 \rightarrow \text{parity} = 1$

Col 3: $0+0+1+1+1 = 3 \rightarrow \text{parity} = 1$

Col 4: $0+1+1+1+1 = 4 \rightarrow \text{parity} = 0$

Parity row: **0 1 1 1 0**

Step 3: Transmit column by column. A burst of 5 bits affects at most 1 bit per column when spread across the 5 rows. Since each column has its own parity bit, the error in each affected column will be detected.

N9. In a Stop-and-Wait protocol, the frame size is 1000 bits, the channel rate is 1 Mbps, and the one-way propagation delay is 10 msec. Calculate the channel utilization (efficiency).

Solution:

Step 1: Transmission time per frame (T_t) = Frame size / Bandwidth

$$T_t = 1000 \text{ bits} / 10^6 \text{ bps} = 1 \text{ msec.}$$

Step 2: Total time per frame cycle = $T_t + 2 \times T_p + T_{ACK}$

(Assume ACK is small, so $T_{ACK} \approx 0$)

$$\text{Total} = 1 + 2(10) + 0 = 21 \text{ msec.}$$

Step 3: Utilization = $T_t / \text{Total cycle time}$

$$= 1/21 = \mathbf{4.76\%}.$$

Note: This low efficiency is the main limitation of Stop-and-Wait protocols on high-delay channels.

N10. In p-persistent CSMA, a station senses the channel idle in a slotted system with $p = 0.3$. What is the probability that the station transmits in the first slot? What is the probability it defers for exactly 3 slots before transmitting?

Solution:

Given: $p = 0.3$, so $q = 1 - p = 0.7$.

Step 1: Probability of transmitting in first slot = $p = 0.3$.

Step 2: Probability of deferring exactly 3 slots then transmitting:

This means: defer (slot 1) $\times$ defer (slot 2) $\times$ defer (slot 3) $\times$ transmit (slot 4).

$$P = q \times q \times q \times p = q^3 \times p = (0.7)^3 \times 0.3$$

$$= 0.343 \times 0.3 = \mathbf{0.1029 \text{ or } 10.29\%}.$$

N11. An HDLC I-frame has sequence number $N(S) = 5$ and acknowledgment number $N(R) = 3$. The P/F bit is set to 1 and the frame is sent from a secondary to a primary station. (a) What does $N(S) = 5$ mean? (b) What does $N(R) = 3$ mean? (c) What does $P/F = 1$ indicate in this context?

Solution:

(a) $N(S) = 5$: This is the **sequence number of the current frame being sent**. It indicates this is the 5th I-frame in the current sequence.

(b) $N(R) = 3$: This is the **acknowledgment number**. It means the sender has received all I-frames up to sequence number 2 and expects to receive frame 3 next from the other station.

(c) $P/F = 1$ (sent by secondary to primary): Since the frame goes from secondary to primary, the P/F bit set to 1 indicates **Final**. This means it is the final frame in the current response from the secondary station.

N12. A Stop-and-Wait protocol uses 1-bit sequence numbers. Show what happens when: (a) Frame 0 is sent and ACK 1 is received, (b) Frame 1 is lost, (c) Frame 0 (resent after timeout) arrives when receiver expects Frame 1.

Solution:

(a) Frame 0 sent, ACK 1 received:

- Sender transmits Frame (seq=0).
- Receiver accepts Frame 0, delivers packet to network layer, sends ACK (ack=1).
- Sender receives ACK 1, stops timer, discards copy of Frame 0.
- Sender is now ready to send Frame (seq=1).

(b) Frame 1 is lost:

- Sender transmits Frame (seq=1), starts timer.
- Frame 1 is lost in transit.
- Receiver does not receive Frame 1, sends nothing.
- Timer expires. Sender resends Frame (seq=1), restarts timer.
- Receiver gets Frame 1, delivers packet, sends ACK (ack=0).

(c) Duplicate Frame 0 arrives:

- Suppose Frame 0 was sent, ACK 1 was lost.
- Timer expires, sender resends Frame (seq=0).
- Receiver has already accepted Frame 0 and is expecting Frame 1 (ack=1 indicates next expected = 1).
- Receiver sees seq=0 but expects seq=1 → **discards as duplicate**, retransmits ACK 1.
- Sender receives ACK 1, proceeds to send Frame (seq=1).

N13. In a CSMA/CD network with slot time = 51.2 μ sec (512 bit times at 10 Mbps), calculate the throughput efficiency if the average frame size is 64 bytes and the average contention period is 2 slots.

Solution:

Step 1: Frame transmission time = $64 \times 8 \text{ bits} / 10 \text{ Mbps} = 512 \text{ bits} / 10^7 = 51.2 \mu\text{sec}$.

Step 2: Average contention period = $2 \times 51.2 = 102.4 \mu\text{sec}$.

Step 3: Total cycle time = Contention + Transmission = $102.4 + 51.2 = 153.6 \mu\text{sec}$.

Step 4: Efficiency = Transmission time / Total cycle time
 $= 51.2 / 153.6 = 1/3 = \mathbf{33.3\%}$.

Note: This is a simplified model. Actual efficiency depends on number of stations, traffic load, and frame size distribution.

N14. For an (11, 7) Hamming code, a received codeword is 00101001001. The check bits are at positions 1, 2, 4, 8. Check bits cover: p1 covers positions 1,3,5,7,9,11; p2 covers 2,3,6,7,10,11; p4 covers 4,5,6,7; p8 covers 8,9,10,11. Determine if an error occurred and correct it.

Solution:

Received: Position 1 2 3 4 5 6 7 8 9 10 11

Bits: 0 0 1 0 1 0 0 1 0 0 1

Step 1: Check p1 (positions 1,3,5,7,9,11): $0+1+1+0+0+1 = 3$ (odd) → **Error in group** → $c1 = 1$

Step 2: Check p2 (positions 2,3,6,7,10,11): $0+1+0+0+0+1 = 2$ (even) → OK → $c2 = 0$

Step 3: Check p4 (positions 4,5,6,7): $0+1+0+0 = 1$ (odd) → **Error in group** → $c4 = 1$

Step 4: Check p8 (positions 8,9,10,11): $1+0+0+1 = 2$ (even) → OK → $c8 = 0$

Step 5: Syndrome = $c8\ c4\ c2\ c1 = 0101$ (binary) = **5 (decimal)**.

Step 6: **Bit 5 is in error.** Flip it: bit 5 changes from 1 to 0.

Step 7: Corrected codeword: 00100001001

Data bits (positions 3,5,6,7,9,10,11): 1 0 0 0 0 0 1 = **1000001** (ASCII "A").

N15. A Pure ALOHA system operates at 100 kbps. The average frame is 1000 bits. Calculate the maximum achievable throughput in kbps.

Solution:

Step 1: Frame transmission time $T_f = 1000 \text{ bits} / 100 \text{ kbps} = 10 \text{ msec}$.

Step 2: For Pure ALOHA, maximum throughput occurs at $G = 0.5$:

$$S_{max} = G \times e^{-2G} = 0.5 \times e^{-1} = 0.5 / 2.718 = 0.1839.$$

Step 3: Maximum throughput = $0.1839 \times 100 \text{ kbps} = \mathbf{18.39 \text{ kbps}}$.

Note: This is the theoretical maximum (~18.4% of channel capacity). For Slotted ALOHA, $S_{max} = 1/e \approx 36.8\%$ at $G = 1$.

N16. In a byte-stuffing scheme using flag byte 01111110 and escape byte 01111101, show the transmitted byte sequence for the data: FLAG, A, ESC, B, FLAG, C.

Solution:

Original data: | FLAG | A | ESC | B | FLAG | C |

Step 1: Flag at start of frame → delimiter (not stuffed).

Step 2: FLAG in data → stuff with ESC: **ESC, FLAG**

Step 3: A → A (no change)

Step 4: ESC in data → stuff with ESC: **ESC, ESC**

Step 5: B → B (no change)

Step 6: FLAG in data → stuff with ESC: **ESC, FLAG**

Step 7: C → C (no change)

Step 8: Flag at end of frame → delimiter (not stuffed).

Transmitted frame:

| FLAG | ESC | FLAG | A | ESC | ESC | B | ESC | FLAG | C | FLAG |



UNIT 5

The Network Layer

Complete Exam Notes with Diagrams, Q&A

Computer Networks

Comprehensive Study Material

May 2026



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Exam Questions & Answers

5.1 Network Layer Design Issues

This section introduces the fundamental issues that network layer designers must address, including the service model provided to the transport layer and the internal organization of the network itself.

Core Responsibilities of the Network Layer

- Routing packets from source to destination across multiple hops
- Providing a service interface to the transport layer
- Managing the internal network organization
- Handling addressing, fragmentation, and congestion

5.1.1 Store-and-Forward Packet Switching

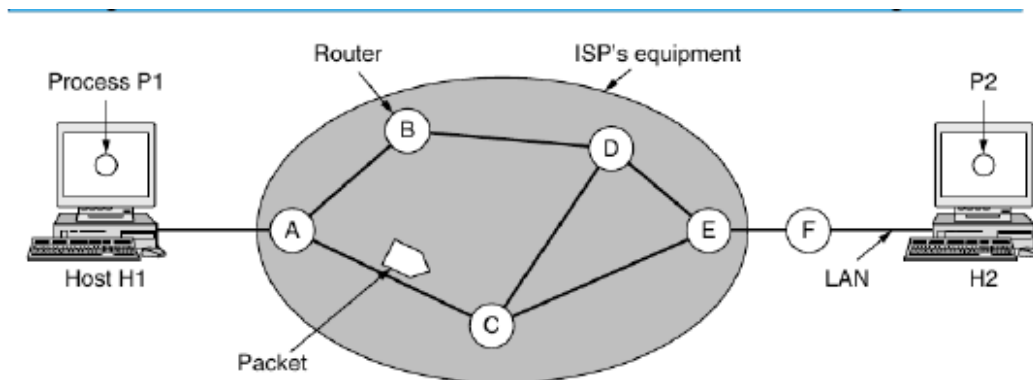


Figure 5-1. The environment of the network layer protocols.

Figure 5-1: The environment of the network layer protocols. ISP equipment (shaded oval) consists of routers connected by transmission lines. Customer equipment (hosts and customer-owned routers) is outside the oval.

The network layer protocols operate within an environment composed of two major components:

Table 5-1 Network Layer Environment Components

Component	Description
ISP's Equipment	Routers connected by transmission lines, shown inside the shaded oval.
Customers' Equipment	Hosts and customer-owned routers, shown outside the oval.

Example Topology:

- **Host H1** is directly connected to ISP router **A** (e.g., a home computer plugged into a DSL modem).
- **Host H2** resides on a LAN (e.g., an office Ethernet) with a customer-owned router **F**, which connects to the ISP via a leased line.
- **Router F** is shown outside the oval because it does not belong to the ISP. However, for routing algorithm analysis, customer-premise routers are considered part of the ISP network because they run the same routing algorithms.

Store-and-Forward Packet Switching Mechanism

1. A host with a packet to send transmits it to the nearest router (either on its own LAN or over a point-to-point link to the ISP).
2. The packet is **stored** at the router until it has fully arrived and the link has finished processing it (including checksum verification).
3. The packet is then **forwarded** to the next router along the path.
4. This process repeats until the packet reaches the destination host, where it is delivered.

Key Point

This is the classic **store-and-forward packet switching** mechanism. Each router must receive the entire packet before forwarding it, introducing a store-and-forward delay at each hop.

5.1.2 Services Provided to the Transport Layer

The network layer provides services to the transport layer at the **network layer/transport layer interface**. The design of these services must satisfy three critical goals:

Table 5-2 Network Layer Service Design Goals

Goal	Description
Router Technology Independence	The services should be independent of the underlying router technology.
Topology Shielding	The transport layer should be shielded from the number, type, and topology of the routers present.
Uniform Addressing	The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

The Fundamental Debate: Connection-Oriented vs. Connectionless Service

These goals give designers considerable freedom, which has historically led to a deep division between two competing philosophies:

Camp 1: The Internet Community (Connectionless)

- Philosophy: routers move packets, nothing else
- Premise: network is inherently unreliable
- Hosts must do error control and flow control
- Service: **connectionless** (SEND PACKET, RECEIVE PACKET)
- Each packet carries **full destination address**
- Exemplifies the **end-to-end argument**

Camp 2: Telephone Companies (Connection-Oriented)

- Philosophy: network should provide reliable service
- Premise: 100 years of telephone system success
- QoS is paramount for voice/video
- Service: **connection-oriented**
- Without connections, QoS is very difficult
- Requires setup before data transfer

Historical Evolution

Table 5-3 Historical Network Technologies

Era	Technology	Orientation	Status
1970s	X.25	Connection-oriented	Early data network
1980s	Frame Relay	Connection-oriented	Successor to X.25
ARPANET/Early Internet	IP	Connectionless	Grew tremendously
1980s-1990s	ATM	Connection-oriented	Now niche uses
Present	IP	Connectionless (dominant)	Taking over telephony

Important Nuance

Despite IP's connectionless dominance, the Internet is evolving connection-oriented features under the covers:

- **MPLS (MultiProtocol Label Switching):** Connection-oriented technology used within ISP networks.
- **VLANs:** Widely used (seen in Chapter 4).

5.1.3 Implementation of Connectionless Service (Datagram Networks)

Organization Principle

- Packets are injected into the network **individually** and routed **independently** of each other.
- **No advance setup** is required.
- Packets are called **datagrams** (by analogy with telegrams).
- The network is called a **datagram network**.

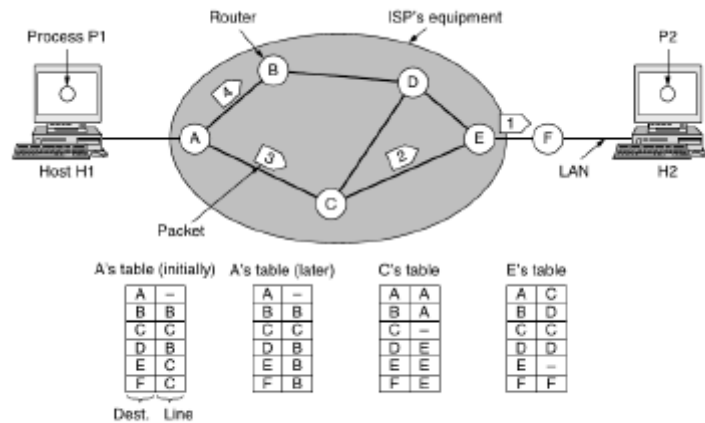
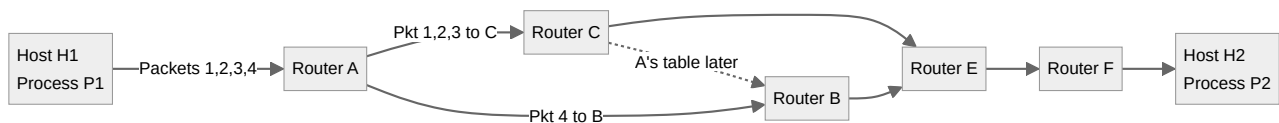


Figure 5-2. Routing within a datagram network.

Figure 5-2: Routing within a datagram network. Router A's routing table maps destinations to outgoing lines. Packets are routed independently.

How Datagram Routing Works: Detailed Example

Scenario: Process **P1** on host **H1** has a long message for process **P2** on host **H2**.



Process:

1. **Transport Layer (H1):** Prepends transport header, hands to network layer.
2. **Packet Fragmentation:** Message is four times max packet size, broken into packets **1, 2, 3, 4**.
3. **Router Operation:** Every router maintains a **routing table** mapping destinations to outgoing lines. Each entry: (Destination, Outgoing Line).

Table 5-4 Router A's Routing Tables

Destination	Line (Initially)	Line (Later)
A	--	--
B	B	B
C	C	C
D	B	B
E	C	B
F	C	B

Packet Journey:

- **Packets 1, 2, 3:** A → C → E → F → H2
- **Packet 4:** A → B (because A updated its routing table due to traffic jam along ACE path)
- The algorithm managing these tables is the **routing algorithm**.

IP: The Dominant Datagram Protocol

- **IP (Internet Protocol)** is the dominant connectionless network service.
- Each packet carries a **destination IP address** for individual forwarding.
- **IPv4:** 32-bit addresses; **IPv6:** 128-bit addresses.

5.1.4 Implementation of Connection-Oriented Service (Virtual-Circuit Networks)

Organization Principle

- A path must be **established before** any data packets can be sent.
- This connection is called a **VC (virtual circuit)**, by analogy with telephone circuits.
- The network is called a **virtual-circuit network**.
- Each packet carries an **identifier** telling which VC it belongs to.

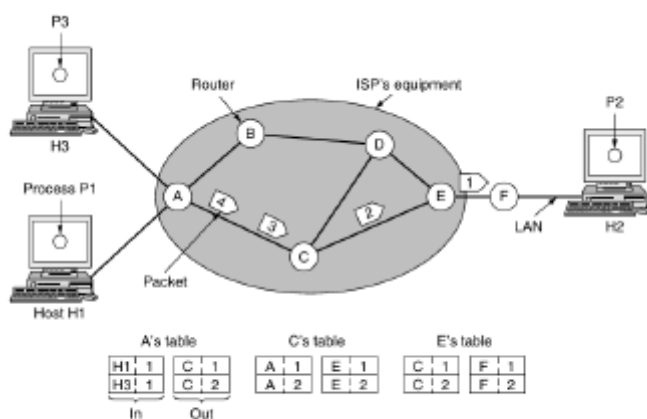


Figure 5-3. Routing within a virtual-circuit network.

Figure 5-3: Routing within a virtual-circuit network. Router tables are indexed by connection identifiers. Routes are fixed at setup time.

How Virtual-Circuit Routing Works: Detailed Example

Scenario: Host **H1** has established **connection 1** with host **H2**.

Connection 1: H1 → H2

Table 5-5 VC Routing Tables for Connection 1

Router	In	Out
A	From H1, connection ID 1	To C, connection ID 1
C	From A, connection ID 1	To E, connection ID 1
E	From C, connection ID 1	To F, connection ID 1

Second Connection: H3 → H2

- Host **H3** chooses **connection identifier 1** (as initiator, its only connection).
- This creates a **connection identifier conflict** at router C.

Table 5-6 Connection Identifier Conflict Resolution

Router	Can Distinguish?	Resolution
A	Yes - easily distinguishes packets from H1 vs. H3	None needed at input
C	No - cannot tell which connection a packet belongs to	A assigns a different connection identifier for the second connection

Label Switching

Routers must have the ability to **replace connection identifiers** in outgoing packets to avoid conflicts. This process is called **label switching**.

MPLS: A Connection-Oriented Network Service

- **MPLS (MultiProtocol Label Switching)** is a prominent connection-oriented service used within ISP networks.
- IP packets are wrapped in an MPLS header containing a **20-bit connection identifier (label)**.
- Often hidden from customers; ISPs establish long-term connections for large traffic volumes.

5.1.5 Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Figure 5-4. Comparison of datagram and virtual-circuit networks.

Figure 5-4: Comparison of datagram and virtual-circuit networks across key dimensions.

Table 5-7 Detailed Comparison: Datagram vs. Virtual-Circuit Networks

Issue	Datagram Network	Virtual-Circuit Network
Circuit setup	Not needed	Required
Addressing	Full source and destination address in each packet	Short VC number only
State information	Routers hold no connection state	Each VC requires router table space
Routing	Each packet routed independently	Route chosen at setup; all packets follow it
Router failure effect	Only packets lost during crash	All VCs through failed router terminated
Quality of service	Difficult	Easy if resources allocated in advance
Congestion control	Difficult	Easy if resources allocated in advance

Detailed Trade-off Analysis

1. Setup Time vs. Address Parsing Time

Aspect	Virtual Circuit	Datagram
Setup	Requires setup phase (takes time, consumes resources)	No setup needed
Forwarding	Easy: circuit number indexes table	More complicated lookup

2. Address Overhead

Datagram networks use **longer destination addresses** because addresses have global meaning. If packets are short, full destination address in every packet = significant overhead.

3. Router Table Space

Network Type	Table Requirement
Datagram	Entry for every possible destination
Virtual Circuit	Entry only for each virtual circuit

4. Quality of Service and Congestion Control

- **Virtual Circuits:** Resources can be **reserved in advance** during connection establishment. Clear advantage for QoS.
- **Datagrams:** No advance reservation; congestion avoidance is more difficult.

5. Suitability for Different Traffic Patterns

Traffic Type	Best Fit	Rationale
Short transactions (credit card verification)	Datagram	Setup/clear overhead dwarfs actual use
Long-running flows (VPN between offices)	Virtual Circuit (Permanent VC)	Permanent VCs lasting months/years useful

6. Fault Tolerance

Failure Scenario	Datagram	Virtual Circuit
Router crash	Only queued packets lost	All VCs through router must be aborted
Communication line loss	Route around failure easily	Fatal to all VCs using that line
Load balancing	Can change routes mid-transmission	Route fixed at setup time

Exam Tip: Key Takeaways for 5.1

1. The network layer provides either **connectionless (datagram)** or **connection-oriented (virtual circuit)** service.
2. **Store-and-forward packet switching** is the fundamental mechanism for packet traversal.
3. The Internet is predominantly connectionless (IP), but connection-oriented tech (MPLS, VLANs) is increasingly used for QoS.
4. The choice involves trade-offs in setup overhead, addressing, state management, fault tolerance, and QoS.

5.2 Routing Algorithms

Overview

The **primary function** of the network layer is routing packets from the source machine to the destination machine. In most networks, packets require **multiple hops** to complete their journey. The only notable exception is broadcast networks.

The **routing algorithm** is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on.

Routing vs. Forwarding: Critical Distinction

Process	Description
Routing	Making the decision about which routes to use. The planning phase.
Forwarding	What happens when a packet actually arrives. Looking up outgoing line in routing tables and sending the packet.

A router has **two internal processes**:

1. **Forwarding process:** Handles each arriving packet, looks up the outgoing line, and transmits it.
2. **Routing algorithm process:** Fills in and updates the routing tables.

When Routing Decisions Are Made

Table 5-8 Timing of Routing Decisions

Network Type	When Decisions Are Made
Datagram networks	Anew for every arriving data packet
Virtual-circuit networks	Only when a new VC is being set up ; data packets follow established route

Desirable Properties of Routing Algorithms

Table 5-9 Essential Properties of Routing Algorithms

Property	Description
Correctness	Must route packets to their intended destinations.
Simplicity	Should not be unnecessarily complex.
Robustness	Must cope with hardware/software failures and topology changes without network-wide reboots.
Stability	Must converge to fixed paths and stay there; converge quickly.
Fairness	Should treat connections equitably.
Efficiency	Should make effective use of network resources.

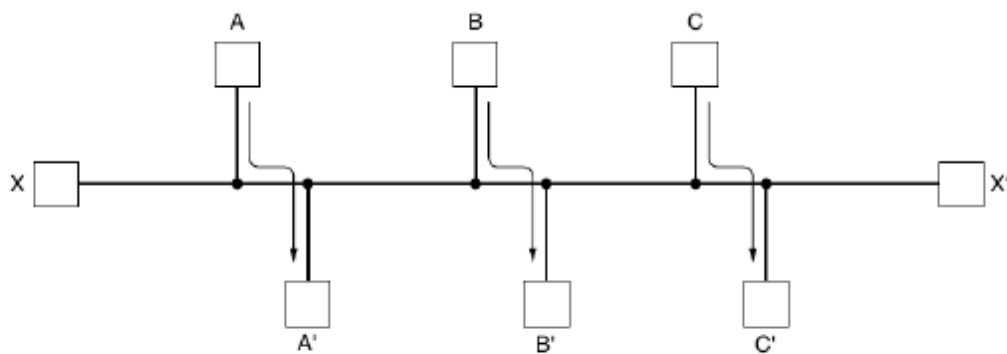


Figure 5-5. Network with a conflict between fairness and efficiency.

Figure 5-5: Network with a conflict between fairness and efficiency. Traffic between A-A', B-B', C-C' saturates horizontal links.

The Fairness-Efficiency Conflict

These goals are often contradictory:

- **Efficiency-maximizing:** Shut off all X-to-X' traffic to maximize total flow.
- **Fairness:** X and X' would not accept being completely denied service.
- **Conclusion:** A compromise between global efficiency and fairness is necessary.

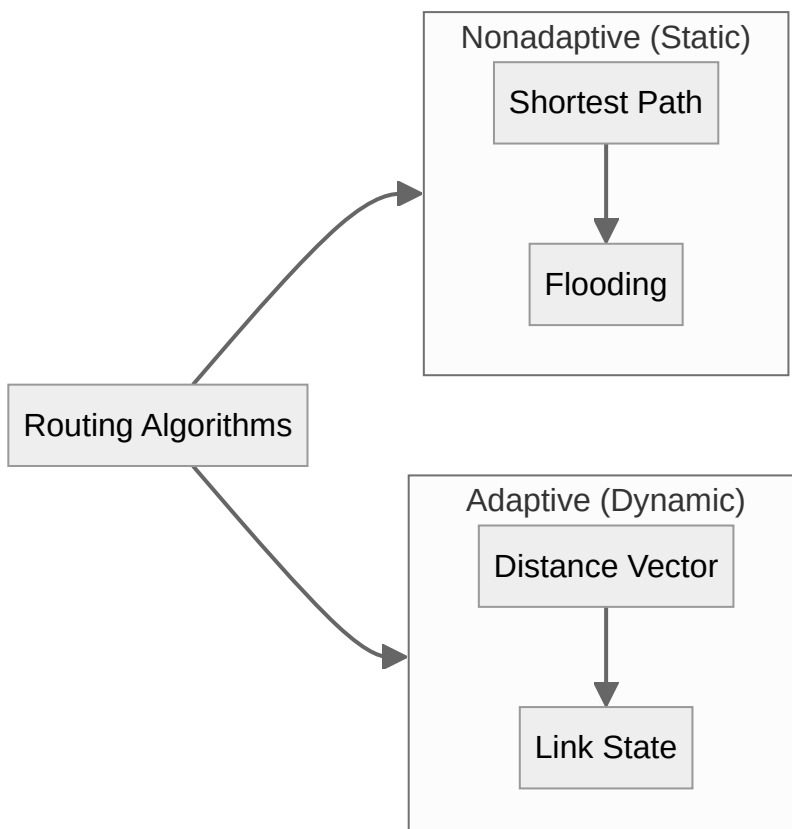
Classification of Routing Algorithms

1. Nonadaptive (Static) Routing

- Do not use measurements of current topology/traffic
- Routes computed **in advance, off-line**
- Downloaded when network boots
- Do **not respond to failures**
- Best for clear routing choices

2. Adaptive (Dynamic) Routing

- Change decisions to reflect topology/traffic changes
- Information from: local, adjacent, or all routers
- Timing: when topology changes or every ΔT seconds
- Metrics: distance, hops, estimated transit time



5.2.1 The Optimality Principle

Statement of the Principle (Bellman, 1957)

If router **J** is on the optimal path from router **I** to router **K**, then the optimal path from **J** to **K** also falls along the same route.

Proof by Contradiction: Let the optimal route from **I** to **K** be decomposed into r_1 (**I** to **J**) and r_2 (**J** to **K**). If a better route than r_2 existed from **J** to **K**, it could be concatenated with r_1 to produce a better route from **I** to **K**. This contradicts the assumption that r_1r_2 is optimal.

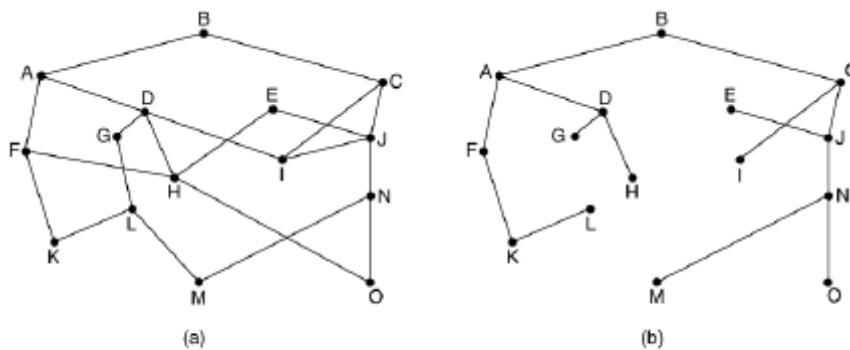


Figure 5-6. (a) A network. (b) A sink tree for router **B**.

Figure 5-6: (a) A general network topology. (b) The sink tree for router **B** - the set of optimal paths from all other routers to **B**.

Consequence: The Sink Tree

The set of optimal routes from **all sources** to a **given destination** forms a **tree rooted at the destination**. This is called a **sink tree**.

- **Tree structure:** Rooted at destination; no loops.
- **Finite delivery:** Packet delivered within bounded number of hops.
- **Generalization (DAG):** If all optimal paths are allowed, becomes a Directed Acyclic Graph.

Practical Considerations

- Links and routers go down and come back up dynamically.
- Different routers may have different views of the topology.
- Despite complications, the sink tree provides a **benchmark** for measuring routing algorithms.

5.2.2 Shortest Path Algorithm (Dijkstra's Algorithm)

Fundamental Approach

The shortest path algorithm computes optimal paths given a complete picture of the network. These are the paths that distributed routing algorithms aim to find.

Table 5-10 Graph-to-Network Mapping

Graph Element	Network Element
Node	Router
Edge	Communication line (link)

Metrics for Path Length

Table 5-11 Path Length Metrics

Metric	Description	Example
Number of hops	Count of edges traversed	Paths ABC and ABE equally long
Geographic distance	Physical distance in km	ABC much longer than ABE if drawn to scale
Mean delay	Measured delay of test packet	Fastest path, not fewest edges
General composite	Function of distance, bandwidth, traffic, cost	Computed by changing weighting function

Dijkstra's Algorithm (1959)

Goal: Find shortest paths from a **source** to **all destinations**.

Key Properties:

- Distances must be **non-negative**.
- Each node labeled with distance from source along best known path.

Label Types:

- **Tentative:** Initial state; may improve.
- **Permanent:** Represents shortest possible path; never changed.

Algorithm Steps:

1. Initialize all labels as **tentative**, set to **infinity**.
2. Make source node **permanent** with distance 0.
3. Examine neighbors of the newest permanent node; update tentative labels.
4. Select the tentative node with **smallest label**; make it permanent.
5. Repeat steps 3-4 until all nodes are permanent.

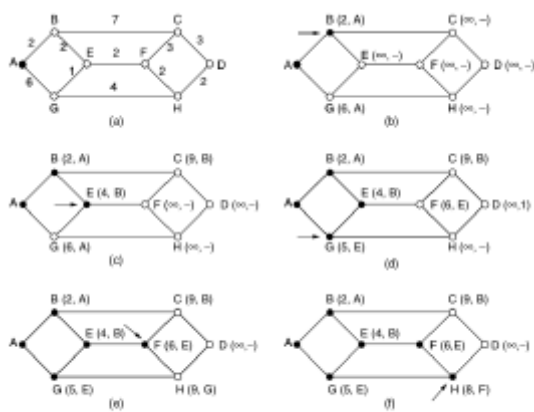


Figure 5-7. The first six steps used in computing the shortest path from A to D. The arrows indicate the working node.

Figure 5-7: The first six steps used in computing the shortest path from A to D using Dijkstra's algorithm. Arrows indicate the working node.

Network Topology for Figure 5-7

Table 5-12 Edge Weights for Dijkstra Example

Edge	Weight	Edge	Weight
A-B	2	A-E	2
A-G	6	B-C	7
B-E	2	E-F	2
E-G	1	G-H	4
C-F	3	C-D	3
F-H	2	H-D	2

Algorithm Execution Trace

Table 5-13 Dijkstra's Algorithm Step-by-Step

Step	Working Node	Action	Result
(b)	A (perm)	Examine B, E, G	B(2,A), E(2,A), G(6,A); C,D,F,H= ∞
(c)	B (perm)	Examine C, E	C(9,B); E stays (2,A)
(d)	E (perm)	Examine F, G	F(4,E); G(3,E)
(e)	F (perm)	Examine C, H	C(7,F); H(6,E)
(f)	G (perm)	Examine H	H stays (6,E)

How to Answer Dijkstra's Algorithm Questions

1. Draw the network graph with all edge weights.
2. Create a table with columns for each node: (distance, predecessor).
3. Mark tentative distances as ∞ initially, except source = 0.
4. At each step: pick the unvisited node with smallest tentative distance, make it permanent.
5. Update neighbors: if new path is shorter, update their tentative distance.
6. Continue until destination is permanent or all nodes are processed.
7. Trace back predecessors to find the actual path.

5.2.4 Distance Vector Routing

Computer networks generally use **dynamic routing algorithms**. The two most popular are:

1. **Distance Vector Routing** (this section)
2. **Link State Routing** (next section - not detailed in notes)

Alternative Names: Distributed Bellman-Ford routing algorithm; **RIP (Routing Information Protocol)**.

Core Mechanism

Each router maintains a **routing table (vector)** containing:

- **Best known distance** to each destination
- **Preferred outgoing line** to reach that destination

Tables updated by **exchanging information with neighbors**.

Table Structure

Table 5-14 Distance Vector Routing Table

Field	Description
Destination	One entry for each router in network
Preferred outgoing line	Best next hop
Distance estimate	Estimated cost to reach destination

Update Process

Once every **T msec**, each router:

1. **Sends** to each neighbor a list of estimated delays to each destination.
2. **Receives** a similar list from each neighbor.
3. For neighbor **X** with delay estimate **X_i** to router **i**, and local delay **m** to **X**:
Delay to i via X = $X_i + m$
4. Selects the **best estimate** and updates table.

Critical Point

The **old routing table** is **NOT used** in the calculation. Each update is computed entirely from neighbor vectors.

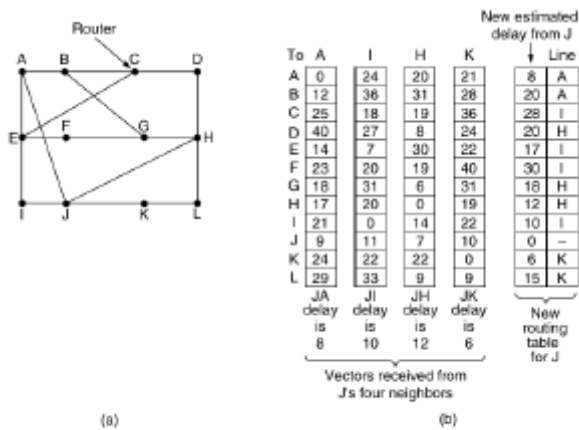


Figure 5-9: (a) A network. (b) Input from A, I, H, K, and the new routing table for J.

Figure 5-9: (a) A network. (b) Input from A, I, H, K and the new routing table for J.

Detailed Example: Router J

J's Delay to Neighbors: A = 8, I = 10, H = 12, K = 6

Computing New Route to Router G

Table 5-15 Distance Calculation to G via Each Neighbor

Via	Calculation	Total Delay
A	$18 (A \rightarrow G) + 8 (J \rightarrow A)$	26
I	$31 (I \rightarrow G) + 10 (J \rightarrow I)$	41
H	$6 (H \rightarrow G) + 12 (J \rightarrow H)$	18 (Best)
K	$31 (K \rightarrow G) + 6 (J \rightarrow K)$	37

Result: J's routing table entry for G: **Delay = 18 msec, Route = via H**

The Count-to-Infinity Problem

Convergence Characteristics

News Type	Reaction Speed
Good news	Rapid - one vector exchange per hop
Bad news	Slow - converges leisurely (count-to-infinity)

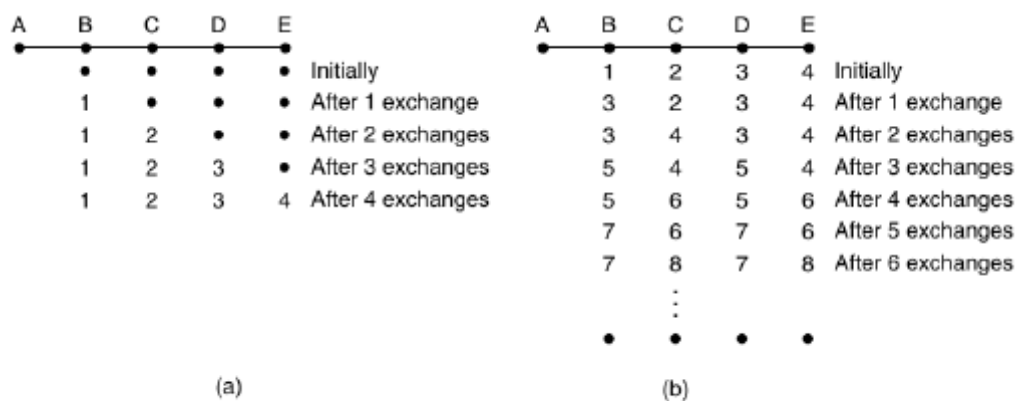


Figure 5-10. The count-to-infinity problem.

Figure 5-10: (a) Good news propagation - spreads at one hop per exchange. (b) Bad news propagation - the count-to-infinity problem.

Good News Propagation (Figure 5-10a)

Scenario: Five-node linear network (A-B-C-D-E), A initially down.

Table 5-16 Good News Propagation Speed

Exchange	B	C	D	E
Initially	∞	∞	∞	∞
After 1	1	∞	∞	∞
After 2	1	2	∞	∞
After 3	1	2	3	∞
After 4	1	2	3	4

Rate: Good news spreads at **one hop per exchange**. In a network with longest path of N hops, all routers know within **N exchanges**.

Bad News Propagation - The Problem (Figure 5-10b)

Scenario: All links up initially. A goes down.

Table 5-17 Count-to-Infinity Problem Trace

Exchange	B	C	D	E	Explanation
Initially	1	2	3	4	Normal state
After 1	3	2	3	4	B doesn't hear from A; C says path=2, B thinks via C = 3
After 2	3	4	3	4	C sees neighbors claim 3, updates to 4
After 3	5	4	5	4	D updates to 5
...	...	...	...	...	Gradually increases toward infinity

Why Bad News Travels Slowly

- No router ever has a value more than **one higher** than the minimum of all neighbors.
- Gradually all routers work up to infinity.
- **Core Problem:** When X tells Y it has a path somewhere, **Y has no way of knowing if it itself is on that path.**
- **Recommendation:** Set infinity to **longest path + 1**.

Attempted Solutions

Table 5-18 Solutions to Count-to-Infinity

Technique	Description	Effectiveness
Split horizon with poisoned reverse	Prevent routers from advertising best paths back to neighbors they heard them from (RFC 1058)	Does not work well in practice

Fundamental Issue

Despite many heuristic attempts, **none work well in practice** because the fundamental information asymmetry remains: a router cannot determine if it lies on a neighbor's advertised path.

Exam Tip: Key Takeaways for 5.2

1. **Routing** = decision-making; **Forwarding** = packet handling.
2. Algorithms must balance correctness, simplicity, robustness, stability, fairness, and efficiency.
3. **Nonadaptive (static)** algorithms compute routes in advance; **Adaptive (dynamic)** respond to changes.
4. **Optimality principle**: optimal routes form a **sink tree**.
5. **Dijkstra's algorithm**: computes shortest paths using non-negative weights with tentative/permanent labels.
6. **Distance vector routing**: routers exchange delay vectors with neighbors.
7. **Count-to-infinity**: good news fast (1 hop/exchange), bad news slow.
8. **Split horizon with poisoned reverse** attempts to mitigate but doesn't solve the core problem.

5.6 The Network Layer in the Internet

5.6.0 Design Principles of the Internet

The foundational principles driving the Internet's design are enumerated in **RFC 1958**, drawing from Clark (1988) and Saltzer et al. (1984).

Table 5-19 Top 10 Internet Design Principles

Rank	Principle	Explanation
1	Make sure it works.	Multiple prototypes before finalizing standards.
2	Keep it simple.	Use simplest solution; fight features (Occam's razor).
3	Make clear choices.	If several ways exist, choose one .
4	Exploit modularity.	Protocol stacks with independent layers.
5	Expect heterogeneity.	Different hardware, facilities, applications.
6	Avoid static options.	Have sender and receiver negotiate values.
7	Good design > perfect.	Go with good design; burden edge cases.
8	Strict sending, tolerant receiving.	Send compliant packets; accept non-conformant.
9	Think scalability.	No centralized databases; spread load evenly.
10	Consider performance and cost.	Poor performance = nobody uses it.

Internet Architecture Overview

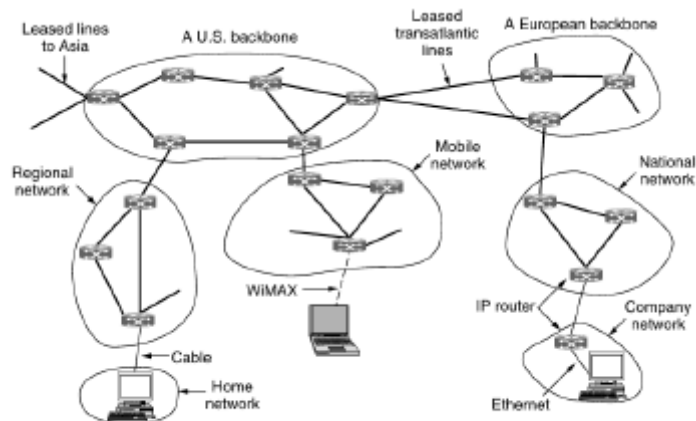


Figure 5-45. The Internet is an interconnected collection of many networks.

Figure 5-45: *The Internet as an interconnected collection of many networks: backbones, regional networks, ISPs, home networks, mobile networks.*

Table 5-20 Internet Components

Component	Description
Tier 1 networks	Biggest backbones; high-bandwidth lines and fast routers.
Backbones	High-bandwidth infrastructure (U.S., European, etc.)
ISPs	Provide Internet access to homes and businesses.
Data centers	Full of server machines serving content.
Regional networks	Connect to backbones and smaller ISPs.
Edge networks	LANs at universities, companies, home networks, mobile.

Typical Packet Journey

A packet from a home network may traverse **four networks** and numerous IP routers before reaching a destination on a company network. Redundant connectivity exists throughout, creating many possible paths.

The IP Protocol's Core Function

The network layer provides a **best-effort** (i.e., **not guaranteed**) way to transport packets from source to destination.



IP Packet Size

- **In theory:** Up to **64 KB** each.
- **In practice:** Usually not more than **1500 bytes** (to fit in one Ethernet frame).

5.6.1 The IP Version 4 Protocol

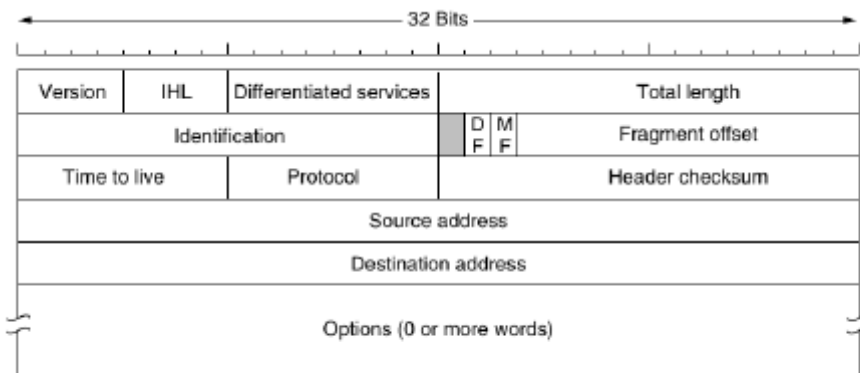


Figure 5-46. The IPv4 (Internet Protocol) header.

Figure 5-46: The IPv4 (Internet Protocol) header format. 32 bits wide, transmitted left-to-right, top-to-bottom, big-endian.

An IPv4 datagram consists of:

- **Header:** 20-byte fixed part + variable-length optional part
- **Body/Payload:** The actual data being transported

Field-by-Field Analysis of IPv4 Header

Table 5-21 IPv4 Header Fields

Field	Size	Description
Version	4 bits	4 = IPv4, 6 = IPv6. Allows version transition.
IHL	4 bits	Internet Header Length in 32-bit words. Min=5 (20 bytes), Max=15 (60 bytes).
Differentiated Services	8 bits	Originally Type of Service (TOS). Top 6 bits = service class; Bottom 2 bits = ECN.
Total Length	16 bits	Header + data. Maximum 65,535 bytes.
Identification	16 bits	Allows destination to determine which fragment a packet belongs to.
Flags (DF, MF)	3 bits	DF = Don't Fragment; MF = More Fragments.
Fragment Offset	13 bits	Position of fragment in original packet. Unit = 8 bytes.
Time to Live (TTL)	8 bits	Counts hops (decremented each hop). Packet discarded at 0. Prevents infinite loops.
Protocol	8 bits	Which transport process to give packet to (TCP=6, UDP=17).
Header Checksum	16 bits	One's complement sum of 16-bit halfwords. Must be recomputed at each hop.
Source Address	32 bits	IP address of source network interface.
Destination Address	32 bits	IP address of destination network interface.
Options	Variable	Variable length, padded to multiple of 4 bytes.

Fragmentation Fields Working Together

Identification + **MF** + **Fragment offset** implement fragmentation:

- All fragments of a packet contain the same **Identification** value.
- **MF** set on all fragments except the last one.
- **Fragment offset** tells where the fragment belongs in the original packet.
- Elementary fragment unit = **8 bytes**.

IP Options

Table 5-22 IPv4 Options

Option	Description	Status
Security	How secret the information is.	Ignored by all routers.
Strict source routing	Complete path as sequence of IP addresses.	Useful for emergency/timing.
Loose source routing	Must traverse listed routers, may pass through others.	For forcing particular paths.
Record route	Each router appends its IP address.	40 bytes too small now.
Timestamp	Like record route plus 32-bit timestamp.	Network measurement.

Current Status of IP Options

IP options have **fallen out of favor**. Many routers ignore them or don't process them efficiently. They are **only partly supported and rarely used**.

5.6.2 IP Addresses

Fundamental Characteristics

Table 5-23 IP Address Fundamentals

Aspect	Detail
Size	32-bit addresses
Scope	Every host and router has an IP address
Usage	Source and Destination address fields of IP packets
Important	IP address refers to a network interface , not a host

Multi-homed Hosts and Routers

- Host on one network: **One IP address** (typical)
- Host on two networks: **Two IP addresses** (one per interface)
- Routers: **Multiple IP addresses** (one per interface)

Prefixes and Hierarchical Structure

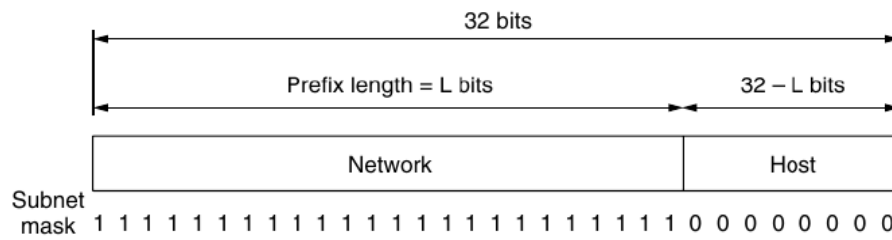


Figure 5-48. An IP prefix and a subnet mask.

Figure 5-48: An IP prefix and a subnet mask. The prefix length = L bits for network, $32-L$ bits for host.

Each 32-bit address consists of:

Portion	Location	Description
Network portion	Top bits (variable length)	Same for all hosts on a single network
Host portion	Bottom bits (remaining)	Unique within the network

Prefix Notation (CIDR)

A prefix is written as: `lowest_IP_address/prefix_length`

Example: $128.208.0.0/24 = 2^8 = 256$ addresses (24 network bits, 8 host bits)

Subnet Mask

Binary mask of 1s in the network portion. ANDed with IP address to extract network portion.

Example for /24: $255.255.255.0 = 11111111\ 11111111\ 11111111\ 00000000$

Advantages and Disadvantages of Hierarchical Addresses

Table 5-24 Hierarchical Addressing Pros and Cons

Aspect	Detail
Key advantage	Routers forward based only on network portion . Routing tables: ~300,000 prefixes vs. ~1 billion hosts.
Disadvantage 1	Location-dependent: IP address depends on network location. Mobile IP needed for moving hosts.
Disadvantage 2	Address waste: Hierarchy is wasteful unless carefully managed.

Subnets

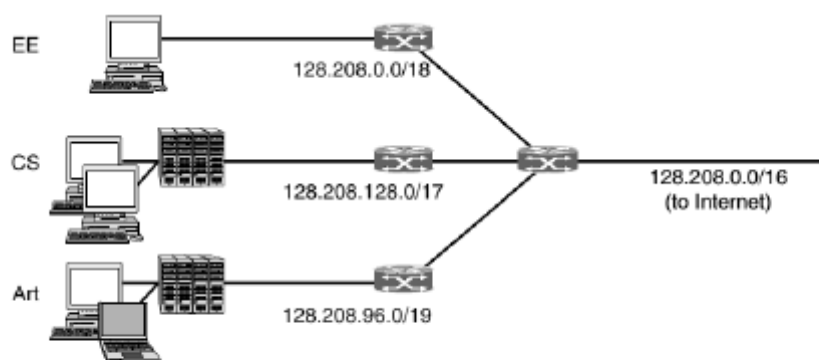


Figure 5-49. Splitting an IP prefix into separate networks with subnetting.

Figure 5-49: Splitting an IP prefix into separate networks with subnetting: CS (/17), EE (/18), Art (/19).

Problem: University has /16 prefix (65,536 addresses) for CS. Later EE and Art want connectivity.

Subnetting

- **Subnetting:** Splitting a block of addresses into several parts for internal use while acting like a single network externally.
- **Subnets:** The resulting networks from dividing a larger network.

Table 5-25 Subnetting Example for 128.208.0.0/16

Department	Prefix	Size	Fraction
Computer Science	128.208.128.0/17	32,768	1/2
Electrical Eng.	128.208.0.0/18	16,384	1/4
Art	128.208.96.0/19	8,192	1/8
Unallocated	--	8,192	1/8

Router Forwarding with Subnets

When a packet arrives at the main router:

1. AND the destination address with each subnet mask
2. Check if result matches the corresponding prefix

Example: Packet for 128.208.2.151

- AND with CS mask (255.255.128.0): 128.208.0.0 $\neq$ 128.208.128.0 $\rightarrow$ **No match**
- AND with EE mask (255.255.192.0): 128.208.0.0 = 128.208.0.0 $\rightarrow$ **Match!** $\rightarrow$ Forward to EE

Key Point

Subnet divisions can be changed later by updating subnet masks at internal routers. No need to contact ICANN or change external databases.

CIDR - Classless InterDomain Routing

Problem: Routing Table Explosion

- **Edge routers:** Can use default route to ISP for external destinations.
- **Core routers:** In the **default-free zone** - must know routes to every network. Tables potentially millions of entries.

Solution: Route Aggregation

- **Route aggregation:** Combining multiple small prefixes into one larger prefix.
- **Supernet:** The resulting larger prefix (opposite of subnet).
- **CIDR:** Works with **variable-length prefixes**.

University	First address	Last address	How many	Prefix
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

Figure 5-50. A set of IP address assignments.

Figure 5-50: A set of IP address assignments for a block of 8,192 addresses (/19).

Table 5-26 University Address Block Example (194.24.0.0/19)

University	Needed	Assigned Range	Mask	Prefix
Cambridge	2,048	194.24.0.0 - 194.24.7.255	255.255.248.0	194.24.0.0/21
Edinburgh	1,024	194.24.8.0 - 194.24.11.255	255.255.252.0	194.24.8.0/22
(Available)	1,024	194.24.12.0 - 194.24.15.255	--	194.24.12.0/22
Oxford	4,096	194.24.16.0 - 194.24.31.255	255.255.240.0	194.24.16.0/20

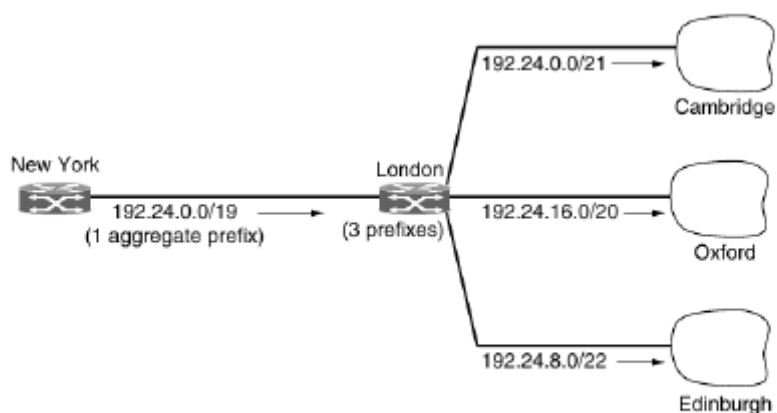
**Figure 5-51.** Aggregation of IP prefixes.

Figure 5-51: Aggregation of IP prefixes: London router has 3 entries; New York router receives 1 aggregate prefix (194.24.0.0/19).

How Aggregation Works

- **London router:** Needs entries for all three university prefixes (different outgoing lines).
- **New York router:** Receives **one aggregate prefix** 194.24.0.0/19 from London (covers all 8K addresses).
- **Result:** Three prefixes reduced to one, reducing New York's routing table.

Longest Matching Prefix Routing

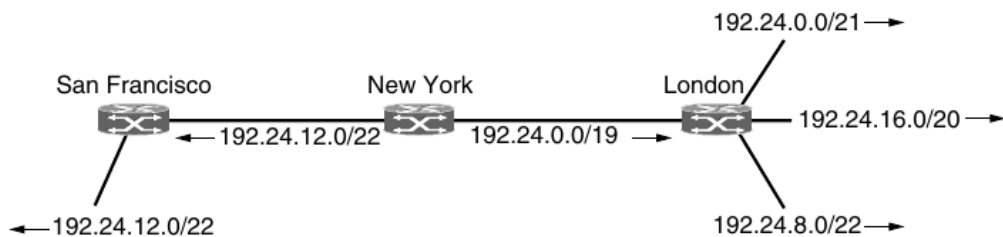


Figure 5-52. Longest matching prefix routing at the New York router.

Figure 5-52: Longest matching prefix routing at the New York router. The /22 to San Francisco is more specific than the /19 to London.

Longest Matching Prefix Rule

Packets are sent in the direction of the most specific route - the longest matching prefix (fewest IP addresses).

Example: 194.24.12.0/22 allocated to San Francisco within 194.24.0.0/19:

Prefix	Destination
194.24.0.0/19	London (aggregate)
194.24.12.0/22	San Francisco (more specific)

At New York router: addresses in 194.24.12.0/22 → **San Francisco** (/22 > /19).

Classful Addressing (Historical, Pre-1993)

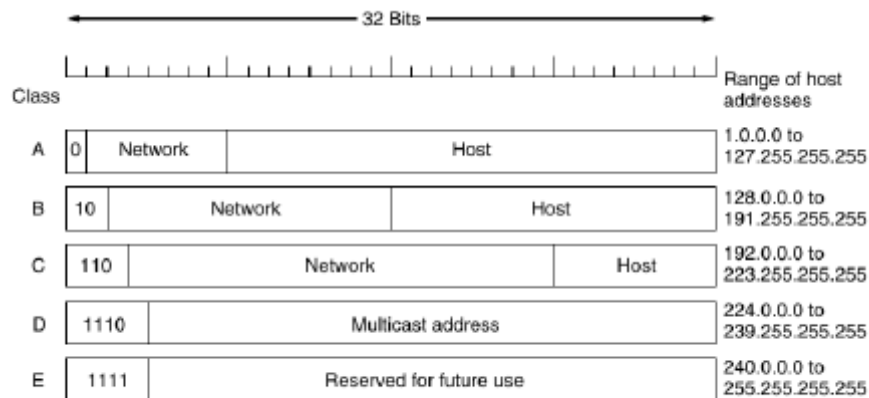


Figure 5-53. IP address formats.

Figure 5-53: IP address formats for Class A, B, C, D, and E networks.

Table 5-27 Classful Addressing

Class	Leading Bits	Network Bits	Host Bits	Address Range	Hosts/Network
A	0	7	24	1.0.0.0 - 127.255.255.255	16,777,216
B	10	14	16	128.0.0.0 - 191.255.255.255	65,536
C	110	21	8	192.0.0.0 - 223.255.255.255	256
D	1110	--	--	224.0.0.0 - 239.255.255.255	Multicast
E	1111	--	--	240.0.0.0 - 255.255.255.255	Reserved

The "Three Bears" Problem (Class B Crisis)

- **Class A** (16M hosts): Too large for most organizations.
- **Class B** (65K hosts): "Just right" but actually far too large for most.
- **Class C** (256 hosts): Too small.
- **Reality:** More than half of Class B networks had **fewer than 50 hosts**.
- **Waste:** Class C would have sufficed, but organizations feared future growth.

Transition to CIDR

- Subnets introduced for flexible internal assignment.
- CIDR added to reduce global routing table size.
- **Today:** Class bits are no longer used, though references remain common.

Special IP Addresses

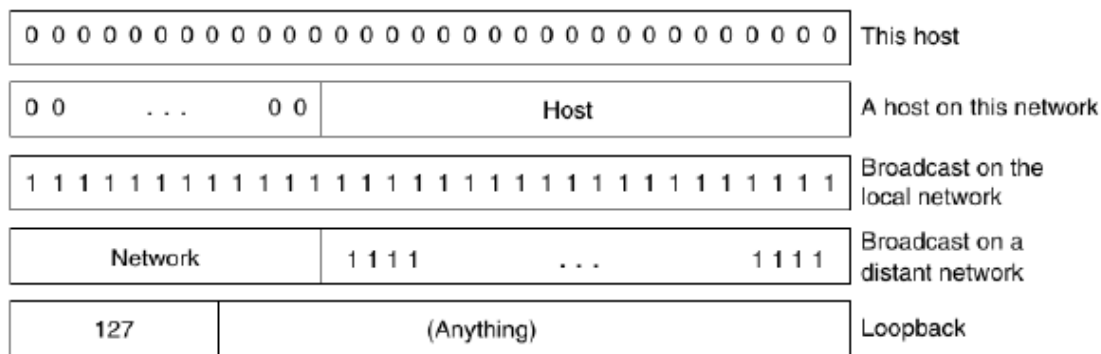


Figure 5-54. Special IP addresses.

Figure 5-54: Special IP addresses: this host, broadcast, loopback, and network-specific addresses.

Table 5-28 Special IP Addresses

Address	Binary Form	Meaning
0.0.0.0	00000000...	"This host" - used during booting
0.0.x.x	Network = 0	"A host on this network"
255.255.255.255	All 1s	Broadcast on the local network
network.255...	Host = all 1s	Broadcast on a distant network
127.x.x.x	01111111...	Loopback - processed locally

NAT - Network Address Translation

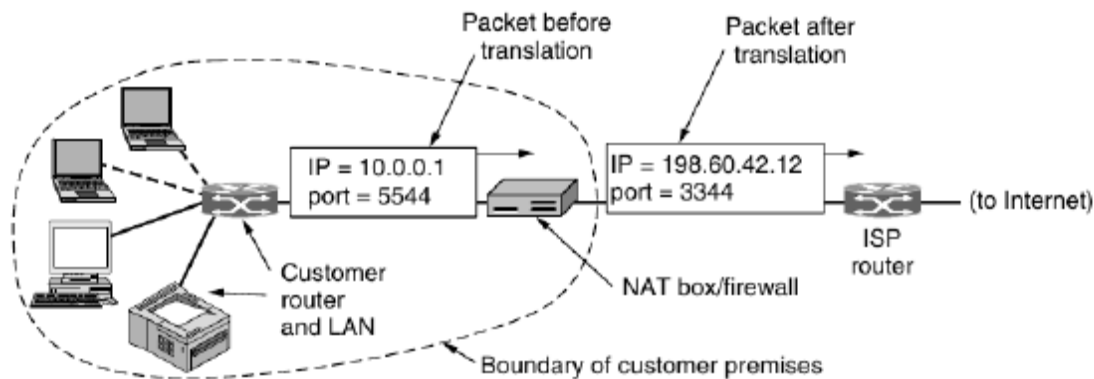


Figure 5-55. Placement and operation of a NAT box.

Figure 5-55: Placement and operation of a NAT box. Internal addresses (10.x.y.z) are translated to a public IP (198.60.42.12).

The IP Address Scarcity Problem

- ISP with /16: 65,534 usable host numbers.
- More customers than addresses - running out is happening **now**.

NAT: The Quick Fix (RFC 3022)

- **ISP assignment:** Single IP address per home/business for Internet traffic.
- **Internal addressing:** Every computer gets a unique **private IP address**.
- **Translation point:** Just before exiting to ISP, internal address → shared public address.

Private Address Ranges (RFC 1918)

Table 5-29 Private IP Address Ranges

Range	Prefix	Hosts Available
10.0.0.0 - 10.255.255.255	/8	16,777,216
172.16.0.0 - 172.31.255.255	/12	1,048,576
192.168.0.0 - 192.168.255.255	/16	65,536

Rule

No packets containing these private addresses may appear on the public Internet.

How NAT Works: TCP/UDP Port Mapping**Table 5-30 NAT Translation Process**

Direction	Steps
Outgoing packet	1. Replace 10.x.y.z source with public IP 2. Replace TCP Source port with index into NAT's 65,536-entry table 3. Table entry contains: original IP + original port 4. Recompute checksums
Incoming packet	1. Extract Source port from TCP header 2. Use as index into mapping table 3. Retrieve internal IP and original port 4. Insert into packet; recompute checksums 5. Forward for normal delivery

Why Replace the Source Port?

Connections from 10.0.0.1 and 10.0.0.2 may both use port 5000. Source port alone is insufficient to identify the sending process. **Analogy:** Company with one telephone number; callers reach an operator who asks for an extension.

Objections to NAT

Table 5-31 Seven Objections to NAT

#	Objection	Explanation
1	Violates IP architecture	IP says each address uniquely identifies one machine. NAT: thousands use 10.0.0.1.
2	Breaks end-to-end connectivity	NAT mappings set up by outgoing packets; incoming cannot be accepted until after outgoing.
3	Makes Internet connection-oriented	NAT box must maintain mapping state. If it crashes, all TCP connections destroyed.
4	Violates protocol layering	NAT inspects TCP/UDP headers (layer 4) from layer 3. If TCP upgraded, NAT fails.
5	Breaks non-TCP/UDP protocols	New transport protocols fail because NAT cannot locate TCP Source port.
6	Breaks embedded addresses	FTP inserts IP addresses in body; NAT knows nothing about these.
7	Limited by port space	16-bit port = at most ~61,440 machines per IP (first 4,096 reserved).

Why NAT Persists

- IP address shortage - only expedient technique during IPv6 transition
- Firewall integration - blocks unsolicited incoming packets
- Privacy - hides internal network structure

5.6.3 IP Version 6

Background and Motivation

IPv4's exponential growth success created a critical problem: **IPv4 is close to running out of addresses.**

Table 5-32 Why IPv6 Was Needed

Issue	Detail
Address exhaustion	Even with CIDR and NAT, last IPv4 addresses expected to be assigned before end of 2012
Smartphone proliferation	Millions of mobile devices requiring constant connectivity
Industry convergence	Computer, communication, entertainment industries merging
Multimedia demand	Billions of machines for audio/video on demand

IETF Goals for IPv6

Table 5-33 IPv6 Design Goals

#	Goal	Description
1	Scale	Support billions of hosts
2	Routing efficiency	Reduce routing table sizes
3	Simplicity	Faster router processing
4	Security	Better authentication and privacy
5	QoS	Real-time data support
6	Multicasting	Specify scopes
7	Mobility	Host roaming without address change
8	Evolvability	Allow future evolution
9	Coexistence	Old and new protocols coexist for years

Major Improvements of IPv6

Table 5-34 IPv6 Improvements over IPv4

#	Improvement	Description
1	Longer addresses	128-bit addresses - effectively unlimited supply
2	Simplified header	Only 7 fields (vs. 13 in IPv4) - faster processing
3	Better option support	Previously required fields now optional
4	Security	Authentication and privacy as key features
5	Quality of service	Greater urgency due to multimedia growth

The IPv6 Fixed Header

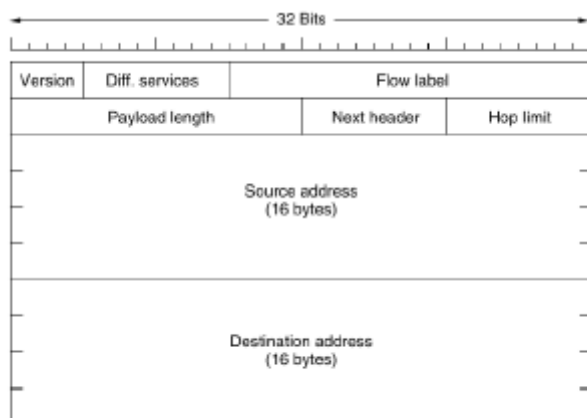


Figure 5-56. The IPv6 fixed header (required).

Figure 5-56: The IPv6 fixed header (required). Total size: 40 bytes. Version, Diff. services, Flow label, Payload length, Next header, Hop limit, Source (128 bits), Destination (128 bits).

Table 5-35 IPv6 Header Fields

Field	Size	Description
Version	4 bits	6 = IPv6, 4 = IPv4
Differentiated Services	8 bits	Distinguishes class of service (same as IPv4 DS + ECN)
Flow Label	20 bits	Marks groups of packets needing identical treatment (pseudo-connection)
Payload Length	16 bits	Bytes following 40-byte header (max 65,535 bytes)
Next Header	8 bits	Identifies extension header or transport protocol (TCP/UDP)
Hop Limit	8 bits	Decrement each hop; packet discarded at 0 (same as TTL)
Source Address	128 bits	16 bytes
Destination Address	128 bits	16 bytes

IPv6 Fixed Header: 40 bytes total

The IPv6 header is **twice as large** as IPv4's minimum (20 bytes) but requires **less processing** because options are handled via extension headers, and the header length is fixed.

IPv6 Address Notation

Standard format: 8000:0000:0000:0000:0123:4567:89AB:CDEF

Table 5-36 IPv6 Address Notation Rules

Rule	Example
1. Omit leading zeros	0123 → 123
2. Replace 0000 groups with ::	8000:0000:0000:0000:0123... → 8000::123:4567:89AB:CDEF
3. IPv4 as :: + dotted decimal	::192.31.20.46

Address Space Magnitude

- Total addresses: $2^{128} \approx 3 \times 10^{38}$
- If entire Earth covered with computers: 7×10^{23} addresses per m^2
- This exceeds **Avogadro's number** (6.022×10^{23})

What Was Removed from IPv4

Table 5-37 IPv4 Fields Removed in IPv6

IPv4 Field	IPv6 Status	Rationale
IHL	Removed	IPv6 header has fixed length (40 bytes)
Protocol	Replaced by Next header	Tells what follows last IP header
Fragmentation fields	Removed	Different approach to fragmentation
Header checksum	Removed	Performance penalty not worth value; other layers have checksums

IPv6 Fragmentation

Table 5-38 IPv6 Fragmentation Approach

Aspect	Detail
Host responsibility	All hosts determine packet size using path MTU discovery
Router behavior	If packet too large, router drops it and sends error to source
Minimum forwardable	Raised from 576 to 1280 bytes
Key difference	Only source host can fragment ; routers may NOT fragment

Extension Headers

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

Figure 5-57. IPv6 extension headers.

Figure 5-57: IPv6 extension headers: hop-by-hop, destination, routing, fragmentation, authentication, encrypted security payload.

Table 5-39 IPv6 Extension Headers

Extension Header	Description
Hop-by-hop options	Miscellaneous info for routers
Destination options	Additional info for destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of sender's identity
Encrypted security payload	Info about encrypted contents

Extension Header Properties

- Each is **optional**.
- If multiple present, must appear in listed order after fixed header.
- Variable-length option encoding: **(Type, Length, Value)** tuple.
- Type field first 2 bits tell routers what to do if they don't understand the option:
 - 00 = Skip; 01 = Discard; 10 = Discard + ICMP error; 11 = Discard, no ICMP for multicast

Hop-by-Hop Header (Figure 5-58)

Next header	0	194	4
Jumbo payload length			

Figure 5-58. The hop-by-hop extension header for large datagrams (jumbograms).

Figure 5-58: The hop-by-hop extension header for large datagrams (jumbograms). Option code 194 defines datagram size.

- **Purpose:** Information that **all routers must examine**.
- **Jumbograms:** Datagrams exceeding 64 KB using option code 194.
- **Critical for supercomputers** transferring gigabytes.

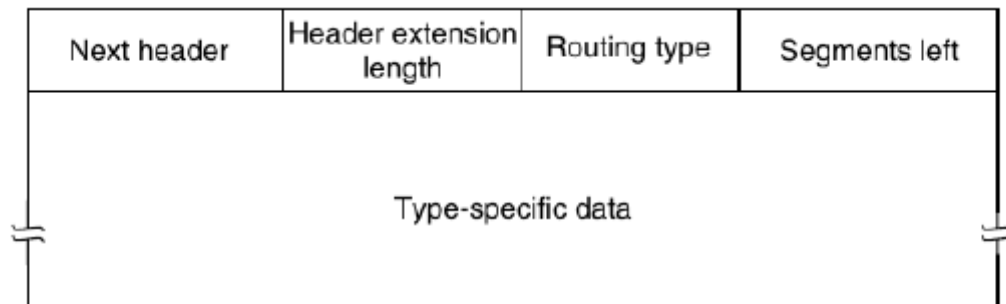
Routing Header (Figure 5-59)**Figure 5-59.** The extension header for routing.

Figure 5-59: The extension header for routing. Lists routers that must be visited en route to destination.

- **Function:** Lists routers that **must be visited** en route.
- **IPv4 equivalent:** Loose source routing.
- **Segments left field:** Tracks unvisited addresses; decremented at each visit.

IPv6 Deployment Reality**Table 5-40 IPv6 Deployment Status**

Aspect	Status
Transition plan	Isolated IPv6 islands communicating via tunnels; grow and merge
Actual deployment	Achilles heel - remains little used despite full OS support
Primary use cases	New situations needing large numbers of addresses (mobile operators)
Transition strategies	Automatic tunnel config; dual-stack hosts (IPv4 + IPv6)
Future	Substantial deployment inevitable when IPv4 addresses exhausted

5.6.4 Internet Control Protocols

In addition to IP for data transfer, the Internet has several **control protocols** in the network layer:

Table 5-41 Internet Control Protocols

Protocol	IPv4	IPv6	Function
ICMP	ICMPv4	ICMPv6	Error reporting and network testing
ARP	ARP	NDP	Address resolution
DHCP	DHCPv4	DHCPv6	Host configuration

ICMP - The Internet Control Message Protocol

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

Figure 5-60. The principal ICMP message types.

Figure 5-60: *The principal ICMP message types: destination unreachable, time exceeded, echo reply, etc.*

Purpose: When something unexpected occurs during packet processing, the event is reported to the sender via ICMP. Each ICMP message type is carried **encapsulated in an IP packet**.

Table 5-42 Principal ICMP Message Types

Message Type	Description	Key Details
Destination unreachable	Packet could not be delivered	Router cannot locate destination; or DF bit packet too big
Time exceeded	TTL field hit 0	Packet looping or counter too low
Parameter problem	Invalid header field	Bug in sending host's IP software
Source quench	Choke packet	Rarely used now ; congestion at transport layer
Redirect	Teach router about geography	Tells host to update to better route
Echo / Echo reply	Check if machine is alive	Used in ping utility
Timestamp request/reply	Same as Echo + timestamp	Network performance measurement
Router advertisement/solicitation	Find a nearby router	Hosts learn router IP addresses

TTL and Traceroute

Traceroute (Van Jacobson, 1987)

- Send sequence of packets with **TTL = 1, 2, 3, ...** successively.
- Counters reach zero at successive routers; each sends **TIME EXCEEDED** back.
- Result: IP addresses of all routers along the path, with timings.
- Perhaps the most useful network debugging tool of all time.

Source Quench (Deprecated)

Deprecated

Originally used to throttle hosts sending too many packets. Problem: during congestion, these packets add more fuel to the fire. Now rarely used; congestion control handled in **transport layer**.

ARP - The Address Resolution Protocol

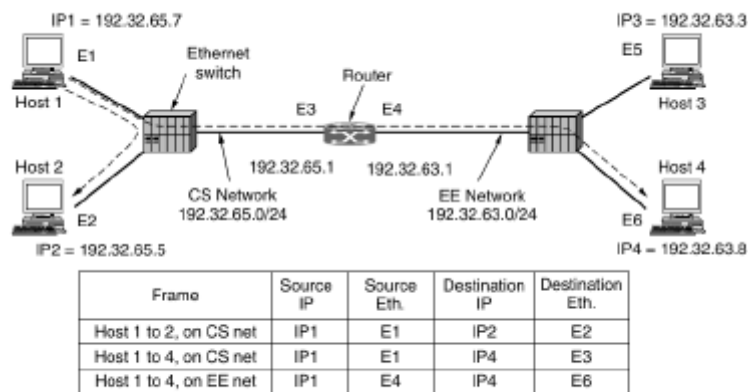


Figure 5-61. Two switched Ethernet LANs joined by a router.

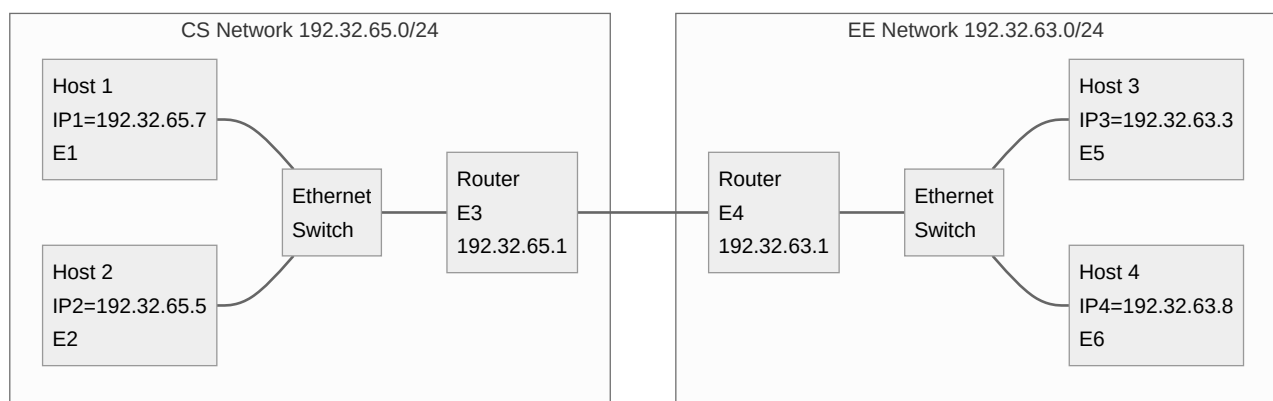
Figure 5-61: Two switched Ethernet LANs joined by a router. IP addresses remain constant; Ethernet addresses change each hop.

The Core Problem

Table 5-43 Layer Addressing Mismatch

Layer	Address Type	Size	Understanding
Network (IP)	IP address	32 bits	Routers and hosts understand
Data link (Ethernet)	Ethernet address	48 bits	NICs understand; unique per device

Ethernet NICs know nothing about 32-bit IP addresses. ARP bridges this gap.



ARP Operation: Intra-Network (Host 1 → Host 2)

Table 5-44 ARP Steps for Same Network

Step	Action
1	Sender knows receiver name; DNS returns IP2
2	IP software checks: destination is on same network
3	Needs Ethernet address E2 to send frame
4	Host 1 broadcasts ARP : "Who owns IP 192.32.65.5?"
5	All machines check their IP address
6	Only Host 2 responds with Ethernet address E2
7	IP software builds frame to E2, puts IP packet in payload
8	Host 2 receives, extracts IP packet, passes to IP software

ARP Optimizations

Table 5-45 ARP Optimization Techniques

Technique	Mechanism	Benefit
Caching	Cache ARP result after first use	Eliminates second broadcast
Include sender mapping	Sender includes its (IP, Ethernet) mapping	Receiver caches it too
Timeout	Cache entries time out after minutes	Allows mappings to change
Gratuitous ARP	Broadcast mapping when configured	Updates everyone's cache
Duplicate detection	If response to gratuitous ARP arrives	Two machines have same IP - error

ARP Operation: Inter-Network (Host 1 → Host 4)

Table 5-46 ARP Steps for Different Networks

Step	Action
1	IP software sees IP4 = 192.32.63.8 is NOT on CS network
2	Off-network traffic sent to default gateway (192.32.65.1)
3	Host 1 ARP broadcasts for 192.32.65.1, learns E3
4	Host 1 sends frame to E3 with IP packet still destined for IP4
5	Router receives; sees destination is on EE network
6	Router ARPs for 192.32.63.8 on EE network, learns E6
7	Router sends frame to E6; IP packet still destined for IP4

Key Observation

Ethernet addresses change with each network hop; IP addresses remain constant across the entire path.

Proxy ARP

Proxy ARP

Router answers ARPs on one network for a host on another network, giving its own Ethernet address. Used when a host doesn't know the destination is on a different network.

DHCP - The Dynamic Host Configuration Protocol

The Configuration Problem

Manual IP configuration is tedious and error-prone for thousands of machines.

DHCP Process

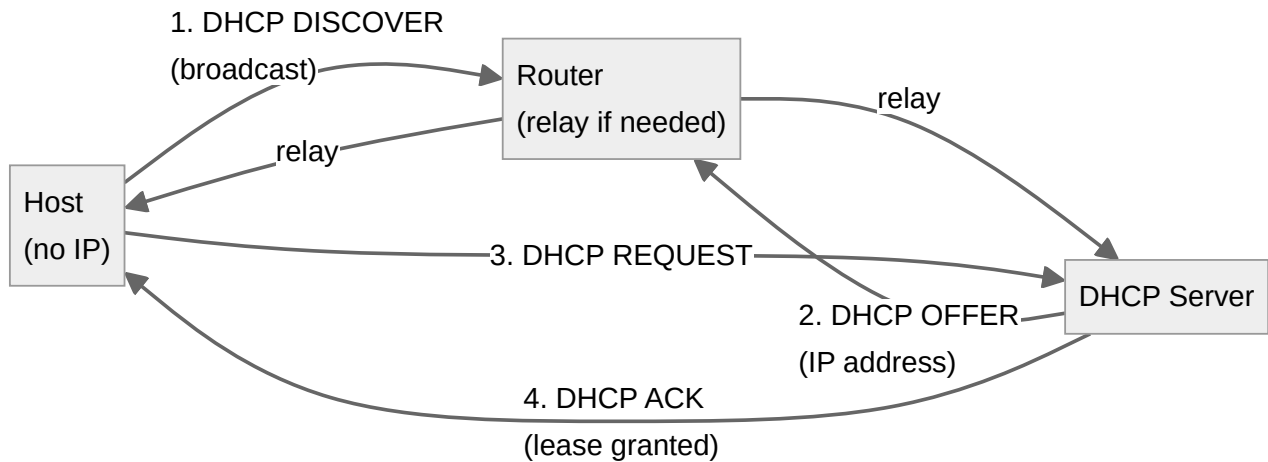


Table 5-47 DHCP Steps

Step	Action
1. DHCP DISCOVER	Computer broadcasts request for IP address
2. DHCP OFFER	Server allocates free IP, sends to host
3. DHCP REQUEST	Host requests offered IP
4. DHCP ACK	Server confirms; lease granted

Leasing

Table 5-48 DHCP Leasing

Aspect	Detail
Problem	If host leaves without returning IP, address is permanently lost
Solution	Leasing - IP assignment for fixed period
Renewal	Host must request renewal before lease expires
Configured params	Network mask, default gateway, DNS server, time server

DHCP Predecessors

DHCP largely replaced **RARP** and **BOOTP** (more limited functionality).

Exam Tip: Key Takeaways for 5.6

1. **Internet design:** 10 core principles emphasizing simplicity, modularity, scalability.
2. **Internet architecture:** Collection of interconnected autonomous systems (Tier 1, ISPs, regional, edge).
3. **IPv4:** Best-effort, connectionless; 20-byte header + options; fragmentation via ID/DF/MF/offset.
4. **IP addressing:** Hierarchical 32-bit with CIDR notation; subnetting splits prefixes internally.
5. **CIDR:** Route aggregation reduces core router tables; longest matching prefix for overlapping routes.
6. **Classful addressing (pre-1993):** A/B/C/D/E classes led to "three bears" address waste problem.
7. **NAT:** Quick fix for IPv4 exhaustion using private ranges + port translation; violates IP architecture.
8. **IPv6:** 128-bit addresses, 40-byte fixed header (7 fields), extension headers, no header checksum.
9. **ICMP:** Error reporting (destination unreachable, time exceeded); ping (echo), traceroute (TTL).
10. **ARP:** Resolves IP to Ethernet via broadcast; DHCP automates host configuration via leasing.

Exam Questions & Answers

How to Use This Section

This section contains both **theoretical** and **numerical** questions commonly asked in exams. Practice answering them before checking the solutions. Numerical questions are marked with an orange bar.

Section A: Theoretical Questions

Q1. What is store-and-forward packet switching? Explain with a diagram.

Answer: Store-and-forward packet switching is the fundamental mechanism where a packet is **fully received and stored** at each router before being **forwarded** to the next hop. The process: (1) Host sends packet to nearest router; (2) Router stores the packet until fully arrived and verified (checksum); (3) Packet is forwarded to the next router; (4) Process repeats until destination. This introduces a store-and-forward delay at each hop equal to the packet size divided by the link bandwidth.

Q2. Compare connection-oriented and connectionless services in the network layer. List two examples of each.

Answer:

Feature	Connectionless	Connection-Oriented
Setup	Not needed	Required before data transfer
Addressing	Full address per packet	Short VC number
Routing	Each packet independent	Fixed route for all packets
State	No connection state in routers	Router table space per VC
QoS	Difficult	Easy with resource reservation
Examples	IP (Internet Protocol)	MPLS, ATM, X.25

Q3. What is the end-to-end argument? How does it relate to the Internet's design philosophy?

Answer: The **end-to-end argument** (Saltzer et al., 1984) states that functions placed at lower layers of a system may be redundant or of little value when compared to the cost of providing them there. The Internet community applied this by keeping the network layer simple (connectionless, no error control) and pushing reliability to the end hosts (transport layer). This makes routers simpler and faster, and hosts can implement whatever error control they need.

Q4. Explain the optimality principle in routing. What is a sink tree?

Answer: The **optimality principle** (Bellman, 1957) states: If router J is on the optimal path from I to K, then the optimal path from J to K also falls along the same route. Proof by contradiction: if a better path from J to K existed, it could be combined with I-to-J to create a better I-to-K path. A **sink tree** is the set of optimal routes from all sources to a given destination, forming a tree rooted at that destination. It has no loops and guarantees finite delivery.

Q5. Differentiate between routing and forwarding.

Answer:

- **Routing:** The process of making decisions about which routes to use. It's the **planning phase** - computing paths and populating routing tables. Happens in background.
- **Forwarding:** The process of handling an arriving packet - looking up the outgoing line in the routing table and sending it on its way. It's the **action phase** - happens for every packet.

A router has two processes: one for forwarding packets and one for running the routing algorithm to update tables.

Q6. What are the desirable properties of a routing algorithm? Explain any two in detail.

Answer: Desirable properties: correctness, simplicity, robustness, stability, fairness, efficiency.

- **Robustness:** Must cope with hardware/software failures, topology changes, and line failures without requiring network-wide reboots. The algorithm should adapt to changes automatically.
- **Fairness vs. Efficiency:** Fairness means treating connections equitably. Efficiency means maximizing total throughput. These are often in conflict - maximizing total flow might

require shutting off some connections entirely, which fairness would not accept. A compromise is necessary.

Q7. Compare static and dynamic routing algorithms.

Answer:

Feature	Static (Nonadaptive)	Dynamic (Adaptive)
Information used	No current measurements	Current topology and traffic
Computation	In advance, offline	Continuously during operation
Response to failures	None	Adapts to new topology
Complexity	Simple	More complex
Best for	Stable, small networks	Large, changing networks

Q8. What is the count-to-infinity problem in distance vector routing? Why does it occur?

Answer: The **count-to-infinity problem** is the slow convergence of distance vector routing when bad news (link failure) occurs. Good news spreads at one hop per exchange, but bad news propagates slowly because routers gradually increment their distance estimates toward infinity. The core problem: **when X tells Y it has a path somewhere, Y has no way of knowing if it itself is on that path.** So Y may choose a path that actually goes through itself. Split horizon with poisoned reverse is an attempted solution but doesn't work well in practice.

Q9. Explain Dijkstra's shortest path algorithm. What are tentative and permanent labels?

Answer: Dijkstra's algorithm (1959) finds shortest paths from a source to all destinations using non-negative edge weights. Each node has a label (distance from source):

- **Tentative label:** Initial state, may be improved as algorithm proceeds. Initially set to infinity.
- **Permanent label:** Confirmed shortest path from source; never changed.

Steps: (1) Make source permanent with distance 0; (2) Examine neighbors of newest permanent node, update tentative labels; (3) Select tentative node with smallest label, make permanent; (4) Repeat until all nodes permanent.

Q10. List the 10 design principles of the Internet. Which is the most important?

Answer: Top 10 (RFC 1958): 1. Make sure it works; 2. Keep it simple; 3. Make clear choices; 4. Exploit modularity; 5. Expect heterogeneity; 6. Avoid static options; 7. Good design need not be perfect; 8. Be strict sending, tolerant receiving; 9. Think scalability; 10. Consider performance and cost. The most important is "**Make sure it works**" - don't finalize until multiple prototypes have successfully communicated.

Q11. Describe the IPv4 header format with a diagram. What is the purpose of the TTL field?

Answer: The IPv4 header has a 20-byte fixed part + variable options. Key fields: Version (4 bits), IHL (4 bits), Differentiated Services (8 bits), Total Length (16 bits), Identification (16 bits), Flags + Fragment Offset (16 bits), TTL (8 bits), Protocol (8 bits), Header Checksum (16 bits), Source Address (32 bits), Destination Address (32 bits), Options (variable). **TTL (Time to Live):** Originally meant time in seconds, now counts hops. Decrement at each router; packet discarded when reaches 0. Prevents packets from wandering forever if routing tables are corrupted. Maximum: 255 hops.

Q12. What is CIDR? How does it help reduce routing table size?

Answer: **CIDR (Classless InterDomain Routing)** allows variable-length network prefixes, eliminating fixed class boundaries (A/B/C). It enables **route aggregation**: combining multiple small prefixes into a single larger prefix (supernet). For example, three university prefixes (/21, /22, /20) can be advertised as one /19 aggregate to distant routers. This dramatically reduces core routing table entries. Routers use **longest matching prefix** to handle overlapping prefixes.

Q13. Explain subnetting with an example. Why is it needed?

Answer: **Subnetting** splits a block of addresses into multiple internal networks while appearing as one network externally. **Need:** A university with a /16 for CS later needs to add EE and Art

departments. Instead of requesting new blocks (expensive, wasteful), they subnet internally. **Example:** 128.208.0.0/16 can be split into CS (/17 = 32,768 addresses), EE (/18 = 16,384), Art (/19 = 8,192), with 8,192 unallocated. The main router uses subnet masks to forward packets to the correct department.

Q14. What is NAT? How does it work and what are its main objections?

Answer: NAT (Network Address Translation, RFC 3022) maps private internal addresses to a public IP address for Internet traffic. It uses TCP/UDP port numbers to distinguish connections from different internal hosts. **Key objections:** (1) Violates IP architecture - thousands of machines share one IP; (2) Breaks end-to-end connectivity - incoming connections need prior outgoing traffic; (3) Makes Internet connection-oriented - NAT box maintains state; (4) Violates protocol layering - inspects layer 4 headers from layer 3; (5) Breaks non-TCP/UDP protocols; (6) Breaks applications with embedded IP addresses (FTP, H.323); (7) Limited by 16-bit port space (~61,440 machines per IP).

Q15. Compare IPv4 and IPv6 headers. What fields were removed and why?

Answer:

Feature	IPv4	IPv6
Header size	20-60 bytes (variable)	40 bytes (fixed)
Address size	32 bits	128 bits
Fields	13 fields	7 fields
Checksum	Present (recomputed each hop)	Removed (other layers handle it)
Fragmentation	Routers can fragment	Only source can fragment
Options	In header (variable)	Extension headers (separate)
Flow label	Not present	20-bit flow label for QoS

Removed: IHL (fixed length), Protocol (replaced by Next Header), Fragmentation fields (different approach), Header Checksum (performance not worth value).

Q16. What are ICMP and ARP? Explain their functions with examples.

Answer: **ICMP (Internet Control Message Protocol):** Reports unexpected events during packet processing. Encapsulated in IP packets. Examples: Destination Unreachable (can't deliver), Time Exceeded (TTL=0), Echo Request/Reply (used by ping). **ARP (Address Resolution Protocol):** Resolves IP addresses to Ethernet (MAC) addresses. Host broadcasts "Who has IP X?" and the target responds with its MAC address. Result is cached. Used for intra-network communication. For inter-network, ARP is used to find the router's MAC address.

Q17. How does DHCP work? What is a lease?

Answer: **DHCP (Dynamic Host Configuration Protocol)** automates host IP configuration. Process: (1) Host broadcasts DHCP DISCOVER; (2) Server responds with DHCP OFFER (IP address); (3) Host sends DHCP REQUEST; (4) Server sends DHCP ACK. A **lease** is a time-limited IP assignment. Hosts must renew before expiry or lose the address. DHCP also provides network mask, default gateway, and DNS server addresses.

Section B: Numerical Problems

N1. A university has the IP block 192.168.0.0/16. They need to create subnets for 4 departments: Engineering (500 hosts), Science (200 hosts), Arts (100 hosts), and Admin (50 hosts). Design an efficient subnetting scheme.

Answer:

- **Engineering (500 hosts):** Need 9 host bits ($2^9 - 2 = 510$). Prefix: **192.168.0.0/23**, Mask: 255.255.254.0
- **Science (200 hosts):** Need 8 host bits ($2^8 - 2 = 254$). Prefix: **192.168.2.0/24**, Mask: 255.255.255.0
- **Arts (100 hosts):** Need 7 host bits ($2^7 - 2 = 126$). Prefix: **192.168.3.0/25**, Mask: 255.255.255.128

- **Admin (50 hosts):** Need 6 host bits ($2^6 - 2 = 62$). Prefix: **192.168.3.128/26**, Mask: 255.255.255.192

N2. Given IP address 172.16.5.130/22, find: (a) Network address, (b) Broadcast address, (c) First usable host, (d) Last usable host, (e) Number of usable hosts.

Answer:

- /22 means 22 network bits, 10 host bits. Subnet mask: 255.255.252.0
- Block size in 3rd octet: $256 - 252 = 4$. Network blocks: 172.16.0.0, 172.16.4.0, 172.16.8.0...
- 172.16.5.130 falls in **172.16.4.0/22**
- (a) **Network:** 172.16.4.0
- (b) **Broadcast:** 172.16.7.255 (next network - 1)
- (c) **First usable:** 172.16.4.1
- (d) **Last usable:** 172.16.7.254
- (e) **Usable hosts:** $2^{10} - 2 = 1022$

N3. In distance vector routing, router J has neighbors A (delay 8), I (delay 10), H (delay 12), K (delay 6). Neighbor A advertises: G=18, B=12, C=25. Neighbor H advertises: G=6, B=31, C=19. Calculate J's delay to G and B, and which neighbor to use.

Answer: To G:

- Via A: $18 + 8 = 26$
- Via I: (not given, assume no route)
- Via H: $6 + 12 = 18$ (**Best**)
- Via K: (not given, assume no route)

J's route to G: Delay = 18, via H

To B:

- Via A: $12 + 8 = 20$ (**Best**)
- Via H: $31 + 12 = 43$

J's route to B: Delay = 20, via A

N4. Using Dijkstra's algorithm, find the shortest path from A to all other nodes in the following network: A-B(2), A-C(4), B-C(1), B-D(7), C-D(3), C-E(5), D-E(1), D-F(2), E-F(3).

Answer:

Step	Permanent Node	B	C	D	E	F
0	A(0)	∞	∞	∞	∞	∞
1	A(0)	2(A)	4(A)	∞	∞	∞
2	B(2)	2(A)	3(B)	9(B)	∞	∞
3	C(3)	2(A)	3(B)	6(C)	8(C)	∞
4	D(6)	2(A)	3(B)	6(C)	7(D)	8(D)
5	E(7)	2(A)	3(B)	6(C)	7(D)	8(D)
6	F(8)	2(A)	3(B)	6(C)	7(D)	8(D)

Shortest paths from A: B=2 (A-B), C=3 (A-B-C), D=6 (A-B-C-D), E=7 (A-B-C-D-E), F=8 (A-B-C-D-F)

N5. An IPv4 packet has Total Length = 4000 bytes, Identification = 0x1A2B, DF=0, MF=0, Fragment Offset = 0. It needs to traverse a link with MTU = 1500 bytes. Show all fragments with their header field values.

Answer:

- IP header = 20 bytes. Max data per fragment = 1500 - 20 = 1480 bytes.
- Fragment offset unit = 8 bytes. 1480/8 = 185 (exact division).
- Original data = 4000 - 20 = 3980 bytes.

Fragment	Data Bytes	Offset	MF	Total Length
1	1480 (bytes 0-1479)	0	1	1500
2	1480 (bytes 1480-2959)	185	1	1500
3	1020 (bytes 2960-3979)	370	0	1040

All fragments: Identification = 0x1A2B, DF = 0.

N6. A router has these entries: 192.168.0.0/16 (port 1), 192.168.1.0/24 (port 2), 192.168.2.0/24 (port 3), 0.0.0.0/0 (port 4). Determine the outgoing port for: (a) 192.168.1.5, (b) 192.168.3.10, (c) 192.168.0.50, (d) 10.0.0.1.

Answer: Using **longest matching prefix**:

- (a) 192.168.1.5: Matches /16 (port 1) and /24 (port 2). Longest: /24 → **Port 2**
- (b) 192.168.3.10: Matches /16 (port 1) only. **Port 1**
- (c) 192.168.0.50: Matches /16 (port 1) only. **Port 1**
- (d) 10.0.0.1: No specific match. **Default** → **Port 4**

N7. A network has 5 routers in a line: A-B-C-D-E. Initially A is down. Show the count-to-infinity problem if the A-B link fails after all routers have converged. Assume infinity = 16.

Answer:

Initial converged state (distances to A): B=1, C=2, D=3, E=4

After A-B link fails:

Exchange	B	C	D	E	Explanation
Initial	1	2	3	4	Normal state
1	3	2	3	4	B hears C has path=2, so B via C = 3
2	3	4	3	4	C hears B=3 or D=3, picks B, updates to 4
3	5	4	5	4	D updates to 5
4	5	6	5	6	E updates to 6
5	7	6	7	6	Continuing...
...	...	...	...	...	Gradually approaches 16 (infinity)

This is the count-to-infinity problem. It takes many exchanges to converge to infinity.

N8. A company has 2000 employees and receives the block 203.10.20.0/21. They need subnets for: HQ (800 hosts), Branch1 (400 hosts), Branch2 (200 hosts), Branch3 (100 hosts), and a server farm (50 hosts). Design the subnetting.

Answer:

- /21 = 2048 addresses total. Available: 203.10.16.0 - 203.10.23.255
- **HQ (800):** Need 10 host bits ($2^{10}-2=1022$). **203.10.16.0/22** (203.10.16.0-203.10.19.255)
- **Branch1 (400):** Need 9 host bits ($2^9-2=510$). **203.10.20.0/23** (203.10.20.0-203.10.21.255)
- **Branch2 (200):** Need 8 host bits ($2^8-2=254$). **203.10.22.0/24** (203.10.22.0-203.10.22.255)
- **Branch3 (100):** Need 7 host bits ($2^7-2=126$). **203.10.23.0/25** (203.10.23.0-203.10.23.127)
- **Server farm (50):** Need 6 host bits ($2^6-2=62$). **203.10.23.128/26** (203.10.23.128-203.10.23.191)

N9. In a classful addressing system, determine the class, default mask, network address, and broadcast address for: (a) 10.5.3.2, (b) 172.16.20.5, (c) 192.168.1.1, (d) 224.0.0.1.

Answer:

Address	Class	Default Mask	Network	Broadcast
10.5.3.2	A (1-126)	255.0.0.0 (/8)	10.0.0.0	10.255.255.255
172.16.20.5	B (128-191)	255.255.0.0 (/16)	172.16.0.0	172.16.255.255
192.168.1.1	C (192-223)	255.255.255.0 (/24)	192.168.1.0	192.168.1.255
224.0.0.1	D (224-239)	N/A	N/A	Multicast

N10. A NAT box translates internal network 192.168.1.0/24 to public IP 203.0.113.5. Two hosts (192.168.1.10 and 192.168.1.20) both want to connect to a web server at 198.51.100.10:80. Host 1 uses source port 5000, Host 2 uses source port 5000. Show the NAT translation table after both connections are established.

Answer:

NAT Port	Internal IP	Internal Port	Remote IP	Remote Port
40001	192.168.1.10	5000	198.51.100.10	80
40002	192.168.1.20	5000	198.51.100.10	80

- **Outgoing Host 1:** Source 192.168.1.10:5000 → 203.0.113.5:40001
- **Outgoing Host 2:** Source 192.168.1.20:5000 → 203.0.113.5:40002
- The NAT box **must replace the source port** because both hosts use the same port 5000. The NAT port (40001, 40002) uniquely identifies each connection.
- **Return traffic:** Packets to 203.0.113.5:40001 forwarded to 192.168.1.10:5000; packets to 203.0.113.5:40002 forwarded to 192.168.1.20:5000.

N11. For the IPv6 address 2001:0DB8:0000:0000:0000:FF00:0042:8329, write: (a) The compressed form, (b) The :: notation form.

Answer:

- (a) **Compressed (omit leading zeros):** 2001:DB8:0:0:0:FF00:42:8329
- (b) **:: notation (replace longest zero run):** 2001:DB8::FF00:42:8329
- Note: Only one :: allowed per address to avoid ambiguity.

N12. Using Figure 5-9 from the notes, router J receives vectors from neighbors A (delay 8), I (delay 10), H (delay 12), K (delay 6). Complete J's new routing table for destinations B, C, D, E, F, G, L using the neighbor data below.

Answer:

Dest	From A (delay 8)	From I (delay 10)	From H (delay 12)	From K (delay 6)	J's Delay	J's Line
B	12	36	31	28	20 (12+8)	A
C	25	18	19	36	28 (18+10)	I
D	40	27	8	24	20 (8+12)	H
E	14	7	30	22	17 (7+10)	I
F	23	20	19	40	30 (20+10)	I
G	18	31	6	31	18 (6+12)	H
L	29	33	9	9	15 (9+6)	K

Section C: Quick Reference Tables

Table R-1 Quick Reference: Key Protocols

Protocol	Layer	Purpose	Key Feature
IP (IPv4/IPv6)	Network	Best-effort packet delivery	Connectionless, 32/128-bit addresses
ICMP	Network	Error reporting, testing	Ping, traceroute, TTL exceeded
ARP	Network	IP to MAC resolution	Broadcast query, cached replies
DHCP	Application	Auto IP configuration	Lease-based address assignment
NAT	Network edge	Private to public translation	Port-based multiplexing
MPLS	Network (ISP)	Fast packet forwarding	Label switching, connection-oriented

Table R-2 Quick Reference: Important RFCs

RFC	Topic
RFC 791	IPv4 Protocol
RFC 1918	Private Address Space
RFC 1958	Architectural Principles of the Internet
RFC 2131	DHCP
RFC 2460	IPv6 Specification
RFC 3022	Traditional NAT

Table R-3 Quick Reference: Common Port Numbers

Protocol	Port	Purpose
DHCP	67/68	Dynamic host configuration
DNS	53	Domain name resolution
HTTP	80	Web traffic
HTTPS	443	Secure web traffic
SSH	22	Secure shell

Final Exam Checklist

- Store-and-forward packet switching mechanism
- Connection-oriented vs. connectionless services and their trade-offs
- Routing vs. forwarding distinction
- Dijkstra's algorithm - be able to execute step-by-step
- Distance vector routing and count-to-infinity problem
- Sink tree concept and optimality principle
- IPv4 header fields - memorize key fields and their purposes
- IP addressing - CIDR notation, subnetting calculations
- Longest matching prefix routing
- NAT operation and objections
- IPv6 header - what changed from IPv4
- ICMP message types and their uses
- ARP process - intra-network and inter-network
- DHCP four-step process and leasing



COMPLETE EXAM NOTES

Unit 5

The Transport Layer, DNS, SMTP & HTTP

Transport Layer Protocols · Domain Name System

Email Protocols · World Wide Web Architecture

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Chapter 6: The Transport Layer

6.1 The Transport Service

6.1.1 Services Provided to the Upper Layers

Primary Objective

The transport layer provides **efficient, reliable, and cost-effective data transmission service** to application-layer processes. It uses the services of the network layer to achieve this goal.

The Transport Entity is the software and/or hardware within the transport layer that performs data transmission. It can be located in:

Table 6.1: Transport Entity Locations

Location	Description
Operating system kernel	Most common on the Internet
Library package bound into network applications	Most common on the Internet
Separate user process	Less common
Network interface card	Less common

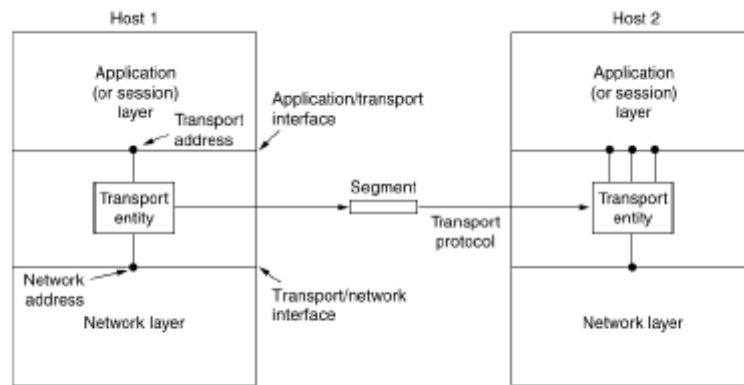


Figure 6-1. The network, transport, and application layers.

Figure 6.1: *The network, transport, and application layers. The transport entity resides in the transport layer and interfaces with the application layer above (via transport address) and the network layer below (via network address). Communication between transport entities occurs via the transport protocol, exchanging segments.*

Exam Tip

The transport entity is the key boundary: it uses **network addresses** below and **transport addresses (ports)** above. Segments are the unit of communication between transport entities.

Types of Transport Service

Table 6.2: Transport Service Types

Type	Characteristics
Connection-oriented	Three phases: establishment, data transfer, and release. Addressing and flow control similar to connection-oriented network service.
Connectionless	Very similar to connectionless network service.

Important

It is difficult to provide a connectionless transport service on top of a connection-oriented network service, because setting up a connection to send a single packet and tearing it down is inefficient.

Why a Distinct Transport Layer?

Central Question: If transport and network layer services are so similar, why have two layers?

Table 6.3: Transport vs Network Layer

Aspect	Transport Layer	Network Layer
Execution location	Runs entirely on users' machines	Mostly runs on routers (carrier-operated)
User control	Users have full control	Users have no real control
Service improvement	Can compensate for poor network service	Cannot be modified by end users

Key Insight

The transport layer makes it possible for the transport service to be **more reliable than the underlying network**. If packets are lost, the transport entity can detect and retransmit. If a network connection is terminated mid-transmission, it can set up a new connection and pick up from where it left off.

Network Independence: Transport primitives hide network service differences. Changing the network merely requires replacing one set of library procedures with another. Application programmers write code to standard primitives, and it works across Ethernet, WiMAX, etc.

Provider-User Distinction

- **Layers 1-4:** The transport service **provider**
- **Layers 5+:** The transport service **user**

The transport layer forms the major boundary. **It is the level that applications see.**

6.4 The Internet Transport Protocols: UDP

Table 6.4: Internet Transport Protocols

Protocol	Type	Characteristics
----------	------	-----------------

UDP (User Datagram Protocol)	Connectionless	Does almost nothing beyond sending packets between applications
TCP (Transmission Control Protocol)	Connection-oriented	Connections, reliability, retransmissions, flow control, congestion control

6.4.1 Introduction to UDP

UDP Definition

UDP is a **connectionless transport protocol** (RFC 768) that lets applications send encapsulated IP datagrams **without establishing a connection**.

UDP Segment Structure

Each UDP segment has an **8-byte header** followed by the payload data.

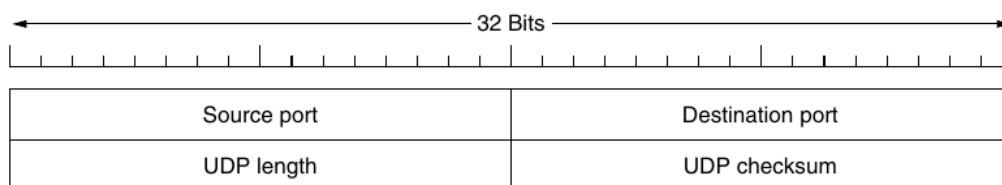


Figure 6-27. The UDP header.

Figure 6.27: The UDP header format (8 bytes). Fields: Source port (16 bits), Destination port (16 bits), UDP length (16 bits), UDP checksum (16 bits).

Table 6.5: UDP Header Fields

Field	Size	Description
Source port	16 bits	Identifies sending endpoint; needed for replies
Destination port	16 bits	Identifies receiving endpoint
UDP length	16 bits	Includes 8-byte header + data. Min=8, Max=65,515 bytes
UDP checksum	16 bits	Optional checksum for extra reliability

The Port Mechanism

When a UDP packet arrives, its payload is handed to the process attached to the destination port via the **BIND primitive**. Think of ports as **mailboxes** that applications rent to receive packets.

Main value of UDP over raw IP: The addition of source and destination ports enables **demultiplexing** -- delivering packets to the correct application.

UDP Checksum

- **Optional** but provided for extra reliability
- Covers: UDP header + UDP data + **conceptual IP pseudoheader**
- Stored as 0 means "checksum not computed"; a true computed 0 is stored as all 1s

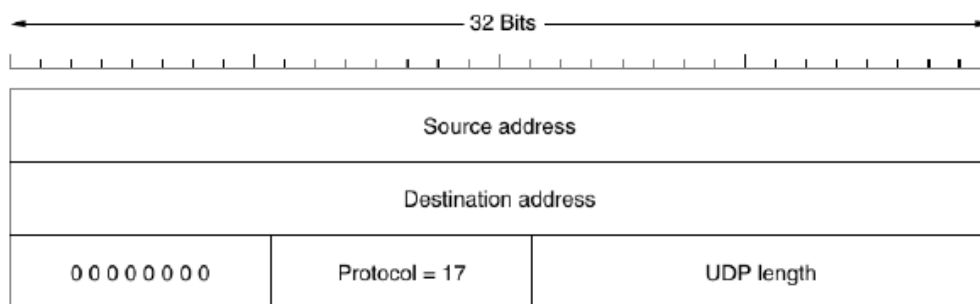


Figure 6-28. The IPv4 pseudoheader included in the UDP checksum.

Figure 6.28: The IPv4 pseudoheader included in the UDP checksum. Contains: Source address (32 bits), Destination address (32 bits), Padding (8 bits = 0), Protocol (8 bits = 17 for UDP), UDP length (16 bits).

Protocol Hierarchy Violation

Including the pseudoheader violates protocol hierarchy since IP addresses belong to the IP layer, not UDP. But it helps **detect misdelivered packets**. TCP uses the **same pseudoheader**.

What UDP Does NOT Do

Table 6.6: Features UDP Lacks

Missing Feature	Consequence
Flow control	Up to user processes

Congestion control	Up to user processes
Retransmission of bad segments	Up to user processes

What UDP Does

Table 6.7: What UDP Provides

Feature	Description
Interface to IP	Provides access to IP protocol
Demultiplexing	Uses ports to direct packets to correct process
Optional error detection	Via checksum mechanism

UDP Summary

UDP provides an interface to IP with demultiplexing via ports and optional end-to-end error detection. That is all it does -- and for many applications (DNS, streaming), that is exactly what is needed.

6.5 The TCP Protocol

6.5.3 The TCP Protocol Overview

Key Feature

Every byte on a TCP connection has its own **32-bit sequence number**. This dominates the entire protocol design.

Segment Structure: TCP entities exchange data in **segments**:

- Fixed **20-byte header** (plus optional part)
- Followed by **zero or more data bytes**

Table 6.8: Limits on Segment Size

Limit	Constraint
IP payload limit	Each segment (including TCP header) must fit in 65,515-byte IP payload
MTU	Must fit in MTU at sender and receiver for single unfragmented packet

Practical MTU: Generally **1,500 bytes** (Ethernet payload size), defining the upper bound on segment size.

Path MTU Discovery

Modern TCP uses **Path MTU discovery** (RFC 1191) with ICMP error messages to find the smallest MTU on the path, then adjusts segment size downward to avoid fragmentation.

Basic TCP Protocol: Sliding Window

1. Sender transmits a segment and **starts a timer**
2. Receiver sends back a segment with: **acknowledgement number** (next expected seq) + **remaining window size**
3. If timer expires before ACK, sender **retransmits**

Complexity Warning

Segments can arrive **out of order**, be **delayed** causing unnecessary retransmissions, or have **retransmission ranges** that differ from originals. TCP must track each byte with its unique offset to handle all cases.

6.5.4 The TCP Segment Header

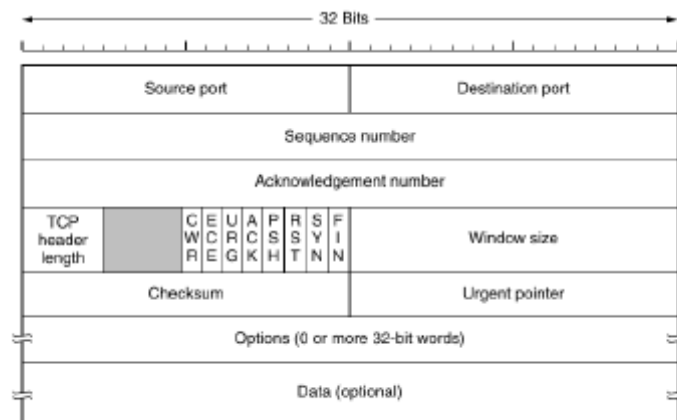


Figure 6-36. The TCP header.

Figure 6.36: The TCP header. Fixed 20-byte header + optional options. Max data = 65,535 - 20 (IP) - 20 (TCP) = 65,495 bytes.

Field-by-Field Dissection

Table 6.9: TCP Header Fields

Field	Size	Description
Source port	16 bits	Local endpoint on sending host
Destination port	16 bits	Local endpoint on receiving host
Sequence number	32 bits	Sequence number of first data byte
Acknowledgement number	32 bits	Next expected in-order byte (cumulative ack)
TCP header length	4 bits	Number of 32-bit words in TCP header
Reserved/Unused	4 bits	Unused for 30+ years (testament to good design)
Flags (8 bits)	8 x 1 bit	CWR, ECE, URG, ACK, PSH, RST, SYN, FIN
Window size	16 bits	Bytes allowed to be sent (flow control)

Checksum	16 bits	Covers header, data, pseudoheader (mandatory)
Urgent pointer	16 bits	Byte offset of urgent data
Options	Variable	TLV-encoded; up to 40 bytes
Data	Variable	Optional payload (up to 65,495 bytes)

Memory Trick

Remember the 8 flags with: **United A**Peaceful **S**ociety **R**equires **S**ome **F**riends **C**alm --
URG, ACK, PSH, RST, SYN, FIN, ECE, CWR

The 5-Tuple Connection Identifier

Table 6.10: 5-Tuple Components

Component	Value
Protocol	TCP
Source IP address	Host A's IP
Source port	Port on Host A
Destination IP address	Host B's IP
Destination port	Port on Host B

Flag Details

Table 6.11: TCP Flag Functions

Flag	Name	Function
CWR	Congestion Window Reduced	Signal to receiver that sender has slowed down (ECN)
ECE	ECN-Echo	Signal to sender to slow down when receiver gets congestion indication
URG	Urgent	Urgent pointer is valid (seldom used)
ACK	Acknowledgement	Acknowledgement number is valid (nearly all packets)

PSH	Push	Deliver data to application immediately, don't buffer
RST	Reset	Abruptly reset or reject a confused connection
SYN	Synchronize	Establish connection. SYN=1,ACK=0 = request; SYN=1,ACK=1 = reply
FIN	Finish	Release connection (sender has no more data)

Window Size and Flow Control

TCP uses a **variable-sized sliding window** for flow control. The Window size field tells how many bytes may be sent starting at the acknowledged byte.

Window Size = 0

A window size of 0 is **legal**. It says: bytes up to (Ack number - 1) have been received, but the receiver is not ready for more data yet. The receiver can later grant permission by sending a segment with the same ACK number and a nonzero window size.

Key difference from Chap 3 protocols: Acknowledgements and permission to send are **completely decoupled** in TCP, giving additional flexibility.

Checksum

Mandatory in TCP (unlike UDP where it's optional). Computed the same way as UDP except: protocol number = **6** for TCP.

TCP Options

Table 6.12: Common TCP Options

Option	Purpose	Key Detail
MSS	Maximum Segment Size	Default 536-byte payload; hosts must accept 556 bytes
Window Scale	Extend window beyond 64 KB	Shift window up to 14 bits left, allowing up to 2^{30} bytes
Timestamp	RTT estimation + PAWS	Echoed by receiver; prevents sequence number wraparound

SACK Selective Acknowledgement Reports received ranges beyond cumulative ack (RFC 2108)

Numerical Tip

On an **OC-12 line (600 Mbps)**, a full 64-KB window takes < **1 msec** to output. With a 50 msec RTT, the sender is idle > **98%** of the time without window scaling. Window scale enables up to **2³⁰ bytes (~1 GB)** windows.

Chapter 7: DNS, SMTP & The World Wide Web

7.1 The Domain Name System (DNS)

Motivation for DNS

The Problem

IP addresses are **hard for people to remember**. If a company moves its server to a different machine with a different IP, everyone needs to be told the new address.

The Solution

High-level, readable names **decouple machine names from machine addresses**. Example: `www.cs.washington.edu` works regardless of its current IP address. DNS provides the mechanism to convert names to network addresses.

Historical Evolution

ARPANET era: A single file `hosts.txt` listed all computer names and IP addresses. Every night, all hosts fetched it.

Table 7.1: Why `hosts.txt` Failed

Problem	Explanation
File size	Would be too large for millions of PCs
Name conflicts	Would occur constantly without central management
Central management	Infeasible for a huge international network

DNS was invented in 1983 and has been a key part of the Internet ever since.

Essence of DNS

1. A **hierarchical, domain-based naming scheme**

2. A **distributed database system** implementing this naming scheme

Primary use: Mapping host names to IP addresses. Defined in **RFCs 1034, 1035, 2181**.

The Resolver Mechanism

Table 7.2: DNS Resolution Steps

Step	Action
1	Application calls the resolver library, passing the name
2	Resolver sends a query to a local DNS server
3	Local DNS server looks up the name and returns the IP address
4	Resolver returns the IP address to the caller

Transport: **UDP packets**. With the IP address, the program establishes a TCP connection or sends UDP packets.

7.1.1 The DNS Name Space

DNS uses a **hierarchical naming scheme** similar to the postal system (country, state, city, street, name).

ICANN (Internet Corporation for Assigned Names and Numbers) manages the top of the hierarchy, created in 1998.

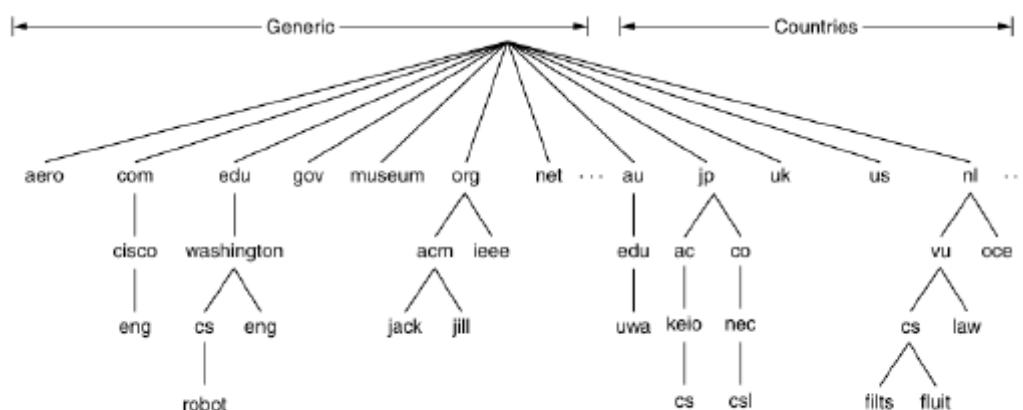


Figure 7-1. A portion of the Internet domain name space.

Figure 7.1: A portion of the Internet domain name space. The tree has generic top-level domains (com, edu, gov, org, net, etc.) and country domains (au, jp, uk, us, nl, etc.). Leaves represent domains with hosts.

Top-Level Domain Types

Table 7.3: TLD Categories

Type	Description
Generic domains	Original domains from 1980s + domains via ICANN applications
Country domains	One entry per country (ISO 3166). Internationalized domains (Arabic, Cyrillic, Chinese) introduced in 2010.

Domain	Intended use	Start date	Restricted?
com	Commercial	1985	No
edu	Educational institutions	1985	Yes
gov	Government	1985	Yes
int	International organizations	1988	Yes
mil	Military	1985	Yes
net	Network providers	1985	No
org	Non-profit organizations	1985	No
aero	Air transport	2001	Yes
biz	Businesses	2001	No
coop	Cooperatives	2001	Yes
info	Informational	2002	No
museum	Museums	2002	Yes
name	People	2002	No
pro	Professionals	2002	Yes
cat	Catalan	2005	Yes
jobs	Employment	2005	Yes
mobli	Mobile devices	2005	Yes
tel	Contact details	2005	Yes
travel	Travel industry	2005	Yes
xxx	Sex industry	2010	No

Figure 7-2. Generic top-level domains.

Figure 7.2: *Generic top-level domains. Some are restricted (edu, gov, mil, aero, etc.) and some are open (com, net, org, biz, info, etc.).*

Domain Name Syntax

- Each domain is named by the **path upward** from it to the (unnamed) root
- Components separated by **periods** ("dot")
- Example: `eng.cisco.com` (vs. UNIX-style `/com/cisco/eng`)
- **Case-insensitive:** edu, Edu, EDU are the same
- Component names: up to **63 characters**
- Full path names: must not exceed **255 characters**

Table 7.4: Absolute vs Relative Names

Type	Syntax	Characteristic
------	--------	----------------

Relative	No trailing period (eng.cisco.com)	Must be interpreted in context
-----------------	------------------------------------	--------------------------------

Domain Placement

US organizations typically under **generic domains**; organizations outside the US typically under **their country domain**. Large companies often register under multiple TLDs (e.g., sony.com, sony.net, sony.nl).

Exam Tip

Naming follows **organizational boundaries**, not physical networks. Two departments on the same LAN can have different domains. One department split across buildings normally shares the same domain.

7.1.2 Domain Resource Records

Fundamental Concept

Every domain has a **set of resource records** associated with it. These records **are** the DNS database. The primary function of DNS is to map domain names onto resource records.

Resource Record Format

A resource record is a **five-tuple**:

Domain_name	Time_to_live	Class	Type	Value
-------------	--------------	-------	------	-------

Table 7.5: RR Field Descriptions

Field	Description
Domain name	The domain this record applies to (primary search key)
Time to live (TTL)	How stable the record is. Large value (e.g., 86400=1 day) = stable; small value (e.g., 60=1 min) = volatile
Class	For Internet: always IN
Type	What kind of record (SOA, A, AAAA, MX, NS, CNAME, etc.)

Value Number, domain name, or ASCII string (semantics depend on type)

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

Figure 7-3. The principal DNS resource record types.

Figure 7.3: The principal DNS resource record types. SOA (zone authority), A (IPv4 address), AAAA (IPv6 address), MX (mail exchange), NS (name server), CNAME (canonical name/alias), PTR (pointer/reverse lookup), SPF (sender policy), SRV (service), TXT (text).

Record Type Descriptions

Table 7.6: DNS Resource Record Types in Detail

Type	Meaning	Value
SOA	Start of Authority	Parameters for this zone (admin email, serial number, flags, timeouts)
A	IPv4 address of a host	32-bit integer. Every Internet host must have at least one.
AAAA	IPv6 address of a host	128-bit integer ("Quad A" = 4 A's)
MX	Mail exchange	Priority + domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Alias for a domain name
PTR	Pointer	Alias for IP address (reverse lookup)
SPF	Sender Policy Framework	Text encoding of mail sending policy
SRV	Service	Host that provides a given service
TXT	Text	Descriptive ASCII text (often encodes SPF info)

CNAME vs PTR

CNAME creates an alias for a **domain name** (e.g., `www.cs.vu.nl` is an alias for `star.cs.vu.nl`). **PTR** associates a name with an **IP address** for reverse lookups. Both point to another name, but their interpretations differ.

```
; Authoritative data for cs.vu.nl
cs.vu.nl.      86400 IN  SOA  star.boss (9527,7200,7200,241920,86400)
cs.vu.nl.      86400 IN  MX   1 zephyr
cs.vu.nl.      86400 IN  MX   2 top
cs.vu.nl.      86400 IN  NS   star

star          86400 IN  A    130.37.56.205
zephyr        86400 IN  A    130.37.20.10
top           86400 IN  A    130.37.20.11
www           86400 IN  CNAME star.cs.vu.nl
ftp           86400 IN  CNAME zephyr.cs.vu.nl

flits         86400 IN  A    130.37.16.112
flits         86400 IN  A    192.31.231.165
flits         86400 IN  MX   1 flits
flits         86400 IN  MX   2 zephyr
flits         86400 IN  MX   3 top

rowboat       IN  A    130.37.56.201
              IN  MX   1 rowboat
              IN  MX   2 zephyr

little-sister IN  A    130.37.62.23
laserjet      IN  A    192.31.231.216
```

Figure 7-4. A portion of a possible DNS database for `cs.vu.nl`.

Figure 7.4: A portion of a possible DNS database for `cs.vu.nl`. Shows SOA, MX, NS, A, and CNAME records for various hosts including aliases for `www` and `ftp`.

7.1.3 Name Servers

Why Not a Single Name Server?

Table 7.7: Problems with Single Name Server

Problem	Consequence
Overloading	Server would be so overloaded as to be useless
Single point of failure	If it ever went down, the entire Internet would be crippled

Zones

The DNS name space is divided into **nonoverlapping zones**. Each zone contains some part of the tree. Zone boundaries are decided by each zone's administrator.

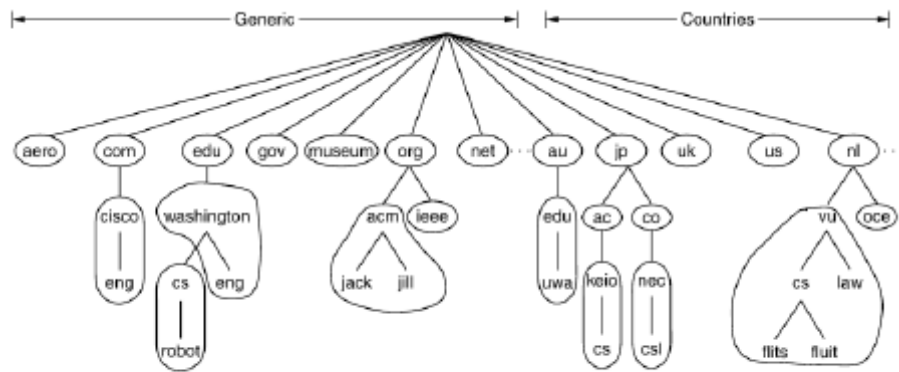


Figure 7-5. Part of the DNS name space divided into zones (which are circled).

Figure 7.5: Part of the DNS name space divided into zones (circled). *washington.edu* handles *eng.washington.edu* but not *cs.washington.edu* -- CS has its own zone with its own name servers.

Name Server Types

Table 7.8: Name Server Types

Type	Description
Primary	Gets information from a file on its disk
Secondary	Gets information from the primary name server

Name Resolution Process

1. Resolver passes query to a **local name server**
2. If the domain falls under the server's jurisdiction: returns **authoritative resource records**
3. If the domain is **remote**: begins a **remote query**

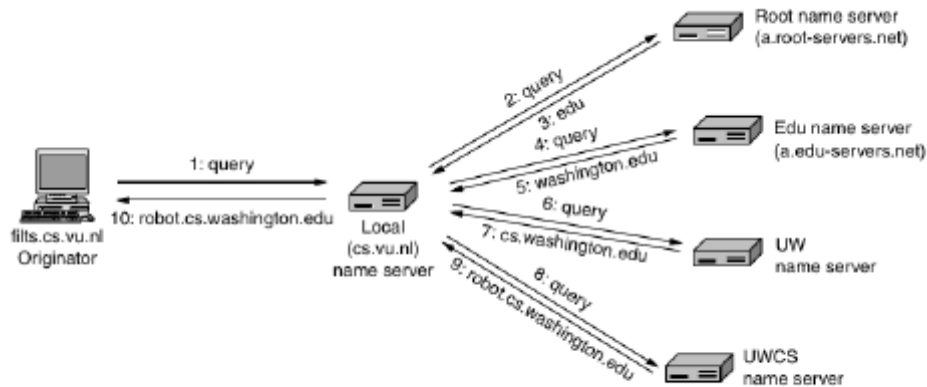


Figure 7-6. Example of a resolver looking up a remote name in 10 steps.

Figure 7.6: Example of a resolver looking up a remote name in 10 steps. *flits.cs.vu.nl* queries its local name server, which iteratively queries root, .edu, washington.edu, and finally cs.washington.edu name servers.

10-Step Remote Query Example

Table 7.9: DNS Resolution Walkthrough (flits.cs.vu.nl -> robot.cs.washington.edu)

Step	Action	Detail
1	Originator -> Local NS	Query contains domain name, type=A, class=IN
2	Local NS -> Root NS	Start at top of hierarchy
3	Root -> Local NS	Returns .edu name server IP
4	Local NS -> .edu NS	Sends full query to .edu server
5	.edu -> Local NS	Returns washington.edu name server
6	Local NS -> UW NS	Sends query to UW name server
7	UW -> Local NS	Returns UW CS name server (cs.washington.edu)
8	Local NS -> UWCS NS	Queries CS name server
9	UWCS -> Local NS	Authoritative answer: robot.cs.washington.edu
10	Local NS -> Originator	Forwards final answer

Two Query Mechanisms

Table 7.10: Recursive vs Iterative Queries

Mechanism	Description	Used By
Recursive	Name server handles full resolution; returns only final answer	Local name server (on behalf of hosts)
Iterative	Returns partial answer (referral); requester continues	Root, TLD, and remote name servers

Caching

All answers, including partial answers, are cached. This greatly reduces query steps and improves performance. The 10-step example is actually the **worst case** (no cached info).

Cache Validity

Cached answers are **not authoritative**. Changes at the source won't propagate automatically. The **TTL field** in each RR tells remote servers how long to cache: stable info = days; volatile info = seconds/minutes.

DNS Transport

Queries and responses use **UDP**. Reliability: if no response arrives, DNS repeats the query, then tries another server. A 16-bit identifier matches answers to queries.

7.2.4 Message Transfer: SMTP

SMTP Overview

SMTP (Simple Mail Transfer Protocol) is used by **Message Transfer Agents (MTAs)** to relay messages from originator to recipient. Two uses: **mail submission** (UA -> MTA) and **message transfer** (MTA -> MTA).

Basic SMTP Operation

- Transport: **TCP connection to port 25**
- SMTP is an **ASCII protocol** (easy to develop, test, debug)
- Server listens on port 25, accepts connections, accepts messages, returns error reports for undeliverable mail

SMTP Message Transfer Walkthrough

1. Client establishes TCP connection to port 25; **waits for server to talk first**
2. Server sends greeting with its identity and readiness
3. If unwilling: client disconnects and retries later
4. If willing: mail exchange begins

```

S: 220 ee.uwa.edu.au SMTP service ready
C: HELO abc123.com
S: 250 os.washington.edu says hello to ee.uwa.edu.au
C: MAIL FROM: <alice@cs.washington.edu>
S: 250 sender ok
C: RCPT TO: <bob@ee.uwa.edu.au>
S: 250 recipient ok
C: DATA
S: 354 Send mail; end with "." on a line by itself
C: From: alice@cs.washington.edu
C: To: bob@ee.uwa.edu.au
C: MIME-Version: 1.0
C: Message-Id: <004700B41.AA00747@ee.uwa.edu.au>
C: Content-Type: multipart/alternative; boundary=qwertypasdfghjklzxcvbnm
C: Subject: Earth orbits sun integral number of times
C:
C: This is the preamble. The user agent ignores it. Have a nice day.
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: text/html
C:
C: <p>Happy birthday to you
C: Happy birthday to you
C: Happy birthday dear <bob> Bob <bob>
C: Happy birthday to you
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: message/rfc822; body:
C:   content-type="text/html";
C:   title="kaggle.cs.washington.edu/";
C:   directory="pub/";
C:   name="birthday.and"
C:
C: content-type: audio/basic
C: content-transfer-encoding: base64
C: --qwertyuiopasdfghjklzxcvbnm
C:
S: 250 message accepted
C: QUIT
S: 221 ee.uwa.edu.au closing connection

```

Figure 7.15: Sending a message from `alice@cs.washington.edu` to `bob@ee.uwa.edu.au`.

Figure 7.15: Sample SMTP dialog. *C* = client commands, *S* = server replies with numeric codes (220=ready, 250=ok, 354=send data, 221=closing).

Command	Purpose
HELO / EHLO	Identify sending host; EHLO initiates ESMTP negotiation
MAIL FROM:	Specify sender's email address
RCPT TO:	Specify recipient (can repeat for multiple recipients)
DATA	Request permission to send message body
. (dot alone)	Signal end of message body
QUIT	Terminate the SMTP session

Table 7.12: SMTP Reply Codes

Code	Meaning	Examples
220	Service ready	220 ee.uwa.edu.au SMTP service ready
221	Service closing	221 ee.uwa.edu.au closing connection
250	Request completed OK	250 sender ok; 250 message accepted
354	Start mail input	354 Send mail; end with "." on a line by itself

Limitations of Basic SMTP

Table 7.13: SMTP Limitations

Limitation	Description	Consequence
No authentication	FROM command can give any sender	Useful for spam
ASCII-only	Transfers ASCII, not binary data	Requires base64 MIME encoding
No encryption	Sends messages in the clear	No privacy

ESMTP (Extended SMTP)

SMTP was revised with an **extension mechanism** (RFC 5321). Clients send **EHLO** instead of **HELO**; if accepted, server replies with supported extensions.

Keyword	Description
AUTH	Client authentication
BINARYMIME	Server accepts binary messages
CHUNKING	Server accepts large messages in chunks
SIZE	Check message size before trying to send
STARTTLS	Switch to secure transport (TLS; see Chap. 8)
UTF8SMTP	Internationalized addresses

Figure 7-16. Some SMTP extensions.

Figure 7.16: Some SMTP extensions. *AUTH* (client authentication), *BINARYMIME* (binary messages), *CHUNKING* (large messages in chunks), *SIZE* (check size before sending), *STARTTLS* (secure transport), *UTF8SMTP* (internationalized addresses).

Table 7.14: ESMTP Extensions

Keyword	Description
AUTH	Client authentication
BINARYMIME	Server accepts binary messages
CHUNKING	Server accepts large messages in chunks
SIZE	Check message size before trying to send
STARTTLS	Switch to secure transport (TLS)
UTF8SMTP	Internationalized addresses

Mail Submission vs Message Transfer

Table 7.15: Mail Submission vs Message Transfer

Aspect	Mail Submission	Message Transfer
Purpose	User agent -> MTA	MTA -> MTA
Port	587 (preferred)	25
Authentication	Required (AUTH)	Not always required
Message checking	Server checks/corrects format	Less checking
Standard	RFC 4409	RFC 5321

Multipart MIME Messages

```
From: alice@cs.washington.edu
To: bob@ee.uwa.edu.au
MIME-Version: 1.0
Message-Id: <0704760941_AA00747@cs.washington.edu>
Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
Subject: Earth orbits sun integral number of times

This is the preamble. The user agent ignores it. Have a nice day.

--qwertyuiopasdfghjklzxcvbnm
Content-Type: text/html

<p>Happy birthday to you<br>
Happy birthday to you<br>
Happy birthday dear <b>Bob </b><br>
Happy birthday to you</p>

--qwertyuiopasdfghjklzxcvbnm
Content-Type: message/external-body;
  access-type="anon-http";
  site="bicycle.cs.washington.edu";
  directory="pub";
  name="birthday.snd"

content-type: audio/basic
content-transfer-encoding: base64
--qwertyuiopasdfghjklzxcvbnm--
```

Figure 7-14. A multipart message containing HTML and audio alternatives.

Figure 7.14: A multipart message containing HTML and audio alternatives. Uses boundary string to separate parts. Content-Type: multipart/alternative lets the recipient choose the best format.

7.3 The World Wide Web

Overview and History

Definition

The Web is an **architectural framework for accessing linked content** spread over millions of machines. Invented at **CERN in 1989** by **Tim Berners-Lee**.

Table 7.16: Web Timeline

Year	Milestone
1989	Proposal made at CERN
~1990	First text-based prototype operational
1991	Public demonstration at Hypertext '91
1993	Mosaic -- first graphical browser (Marc Andreessen)
1994	Netscape formed; W3C established
1994-1997	Browser War: Netscape Navigator vs Internet Explorer

7.3.1 Architectural Overview

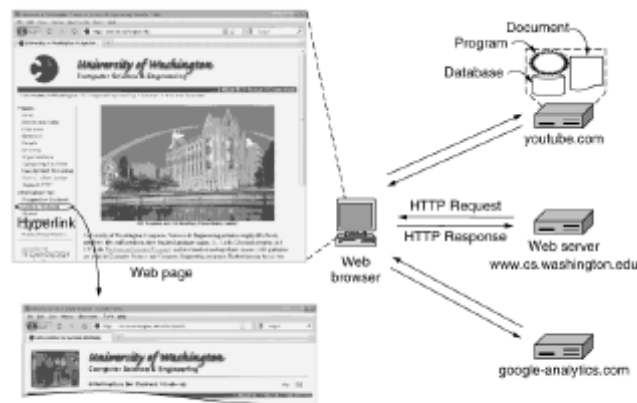


Figure 7-18. Architecture of the Web.

Figure 7.18: *Architecture of the Web.* The browser contacts multiple servers (main page server, YouTube for video, Google Analytics for tracking) and integrates content for display. Users cannot tell where content originates.

Steps in Fetching a Web Page

Table 7.17: Page Fetch Steps

Step	Action
1	Browser determines the URL from the clicked hyperlink
2	Browser asks DNS for IP address of <code>www.cs.washington.edu</code>
3	DNS replies with <code>128.208.3.88</code>
4	Browser makes TCP connection to <code>128.208.3.88</code> on port 80
5	Sends HTTP request asking for <code>/index.html</code>
6	Server sends page as HTTP response
7	Browser fetches other embedded URLs
8	Browser displays the complete page
9	TCP connections released if idle

URLs (Uniform Resource Locators)

Table 7.18: URL Components

Part	Description	Example
Protocol (scheme)	Protocol to use	http, https, ftp
DNS name of host	Machine hosting the page	www.cs.washington.edu
Path	Specific page identifier	index.html

Name	Used for	Example
http	Hypertext (HTML)	http://www.ee.uwa.edu/~rob/
https	Hypertext with security	https://www.bank.com/accounts/
ftp	FTP	ftp://ftp.cs.vu.nl/pub/minix/README
file	Local file	file:///usr/suzanne/prog.c
mailto	Sending email	mailto:JohnUser@acm.org
rtsp	Streaming media	rtsp://youtube.com/montypython.mpg
sip	Multimedia calls	sip:eve@adversary.com
about	Browser information	about:plugins

Figure 7-19. Some common URL schemes.

Figure 7.19: Some common URL schemes. *http* (hypertext), *https* (secure), *ftp* (file transfer), *file* (local file), *mailto* (email), *rtsp* (streaming), *sip* (multimedia calls), *about* (browser info).

Static vs Dynamic Pages

Table 7.19: Page Types

Type	Definition	Example
Static	Same every time displayed	Simple HTML file
Dynamic	Generated on demand by a program	Personalized store front page

URI vs URL vs URN

Table 7.20: URI/URL/URN Distinction

Term	Full Name	Function
URL	Uniform Resource Locator	Tells how to locate a resource (protocol + host + path)

URN	Uniform Resource Name	Tells the name only -- no location info
URI	Uniform Resource Identifier	General term encompassing both URLs and URNs

7.3.4 HTTP -- The HyperText Transfer Protocol

HTTP Overview

HTTP (RFC 2616) is a **simple request-response protocol** that runs over **TCP**. Request/response headers in **ASCII**; contents in **MIME-like format**.

Connections: HTTP 1.0 vs HTTP 1.1

Table 7.21: HTTP Version Comparison

Aspect	HTTP 1.0	HTTP 1.1
Connections	One request per connection, then close	Persistent connections (connection reuse)
Pipelining	No	Yes
Performance	Poor for pages with many objects	Better -- amortizes TCP overhead
Congestion control	Multiple connections compete	Single connection avoids competition

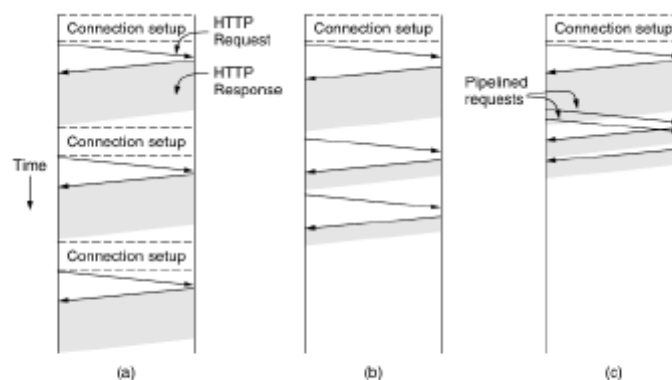


Figure 7-36. HTTP with (a) multiple connections and sequential requests. (b) A persistent connection and sequential requests. (c) A persistent connection and pipelined requests.

Figure 7.36: HTTP connection strategies. (a) Multiple connections, sequential requests -- slowest. (b) Persistent connection, sequential requests -- faster. (c) Persistent connection with pipelined requests -- fastest.

Why Persistent Connections Are Faster

1. **No connection setup overhead:** Each TCP connection needs at least one RTT to establish
2. **Better congestion control:** Multiple short TCP connections each go through slow-start; one longer connection learns the network path once

HTTP Methods

Method	Description
GET	Read a Web page
HEAD	Read a Web page's header
POST	Append to a Web page
PUT	Store a Web page
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Connect through a proxy
OPTIONS	Query options for a page

Figure 7-37. The built-in HTTP request methods.

Figure 7.37: The built-in HTTP request methods. *GET* (read), *HEAD* (read header), *POST* (append/submit), *PUT* (store), *DELETE* (remove), *TRACE* (echo/debug), *CONNECT* (proxy), *OPTIONS* (query capabilities).

Table 7.22: HTTP Methods Detail

Method	Description	Safe?	Idempotent?
GET	Read a Web page -- vast majority of requests	Yes	Yes
HEAD	Read a page's header only (for indexing/testing)	Yes	Yes
POST	Append/submit data (forms, SOAP)	No	No
PUT	Store a page on remote server	No	Yes
DELETE	Remove a page (needs auth)	No	Yes
TRACE	Echo the incoming request (debugging)	Yes	Yes
CONNECT	Connect through a proxy	No	No
OPTIONS	Query options for a page	Yes	Yes

Safe vs Idempotent

Safe methods do not modify server state (GET, HEAD, TRACE, OPTIONS). **Idempotent methods** produce the same result if executed multiple times (GET, HEAD, PUT, DELETE, TRACE, OPTIONS). POST is neither safe nor idempotent.

HTTP Status Codes

Code	Meaning	Examples
1xx	Information	100 = server agrees to handle client's request
2xx	Success	200 = request succeeded; 204 = no content present
3xx	Redirection	301 = page moved; 304 = cached page still valid
4xx	Client error	403 = forbidden page; 404 = page not found
5xx	Server error	500 = internal server error; 503 = try again later

Figure 7-38. The status code response groups.

Figure 7.38: The status code response groups. 1xx=Information, 2xx=Success, 3xx=Redirection, 4xx=Client error, 5xx=Server error.

Table 7.23: HTTP Status Code Groups

Code	Meaning	Examples
1xx	Information	100 = server agrees to handle request
2xx	Success	200 = request succeeded; 204 = no content
3xx	Redirection	301 = page moved; 304 = cached page still valid
4xx	Client error	403 = forbidden; 404 = page not found
5xx	Server error	500 = internal server error; 503 = try again later

Quick Reference Summary Tables

Transport vs Network Layer

Summary 1: Transport Layer vs Network Layer

Aspect	Transport Layer	Network Layer
Primary goal	Efficient, reliable service to application processes	Packet delivery across networks
Entity location	User machines (end systems)	Routers (carrier-operated)
User control	High	Low / None
Service types	Connection-oriented, Connectionless	Connection-oriented, Connectionless
Key improvement	Can be more reliable than underlying network	Best-effort delivery
Boundary role	Provider (layers 1-4) vs User (layers 5+)	Part of service provider

UDP vs TCP

Summary 2: UDP vs TCP Comparison

Feature	UDP	TCP
Type	Connectionless	Connection-oriented
Header size	8 bytes	20 bytes (fixed) + up to 40 bytes options
Flow control	No	Yes (sliding window)
Congestion control	No	Yes
Reliability	No retransmission	Retransmission with timers
Checksum	Optional	Mandatory
Sequencing	No	32-bit per-byte sequence numbers

Demultiplexing	Source + Destination ports	Source + Destination ports
Typical use	DNS, streaming, games	HTTP, FTP, email

DNS Record Types Quick Reference

Summary 3: DNS Resource Records Summary

Type	Purpose	Value
SOA	Zone authority info	Zone parameters
A	IPv4 address	32-bit integer
AAAA	IPv6 address	128-bit integer
MX	Mail server	Priority + domain
NS	Name server	Server name
CNAME	Alias	Canonical domain name
PTR	Reverse lookup	Alias for IP address
SRV	Service location	Host name
TXT	Text/SPF	ASCII string

HTTP Connection Strategies

Summary 4: HTTP Connection Strategies Compared

Strategy	Connections	Requests	Pros	Cons
Multiple sequential	Multiple (one per object)	Sequential	Simple	High overhead; slow-start penalty per connection
Persistent sequential	One	Sequential	Reduced overhead	Server idle between requests
Persistent pipelined	One	Pipelined	Fastest; minimal idle time	More complex

Parallel	Multiple simultaneous	One per connection	Hides some latency	Connections compete; discouraged
----------	--------------------------	-----------------------	-----------------------	-------------------------------------

Exam Practice: Questions & Answers

How to Use This Section

These questions cover both **theoretical concepts** and **numerical problems** commonly asked in exams. Try answering each question on your own before checking the answer. Numerical problems include step-by-step solutions.

Section A: Theoretical Questions

1 What is the primary objective of the transport layer?

Answer: The transport layer aims to provide efficient, reliable, and cost-effective data transmission service to application-layer processes. It achieves this by using the services of the network layer while isolating upper layers from network imperfections.

2 Why do we need a separate transport layer if the network layer already provides similar services?

Answer: (1) The transport layer runs on users' machines where users have full control; the network layer runs on carrier-operated routers where users have no control. (2) The transport layer can compensate for poor network service through retransmissions and connection recovery. (3) It isolates applications from network technology differences -- applications work across Ethernet, WiMAX, etc. without modification.

3 What is the transport entity, and where can it be located?

Answer: The transport entity is the software/hardware that performs data transmission in the transport layer. Locations: (1) OS kernel -- most common, (2) Library package bound into applications -- most common, (3) Separate user process, (4) Network interface card -- less common.

4 Differentiate between connection-oriented and connectionless transport services.

Answer: Connection-oriented has three phases: establishment, data transfer, and release. It uses addressing and flow control similar to connection-oriented network service. Connectionless is very similar to connectionless network service -- each packet is independent. It is difficult to provide connectionless transport over connection-oriented networks because setting up a connection for a single packet is inefficient.

5 Why was DNS invented? What were the problems with hosts.txt?

Answer: DNS was invented in 1983 because hosts.txt failed to scale: (1) File size became too large for millions of hosts, (2) Name conflicts occurred without central management, (3) Central management was infeasible for a global network due to load and latency. DNS uses a hierarchical, distributed approach that scales to the entire Internet.

6 Explain the DNS resolver mechanism with steps.

Answer: (1) Application calls the resolver library procedure, passing the domain name. (2) Resolver sends a UDP query containing the name to a local DNS server. (3) Local DNS server looks up the name and returns the IP address. (4) Resolver returns the IP address to the application. The application can then establish a TCP connection or send UDP packets.

7 What is the difference between recursive and iterative DNS queries?

Answer: In a **recursive query**, the name server handles the full resolution on behalf of the client and returns only the final answer. Used by local name servers. In an **iterative query**, the name server returns a partial answer (referral to another server), and the requester must continue the resolution. Used by root and TLD servers to avoid overloading them.

8 Explain why caching is important in DNS and how TTL controls it.

Answer: Caching greatly reduces query steps and improves performance. All answers, including partial referrals, are cached. However, cached answers may become out of date. The **TTL (Time To Live)** field in each resource record tells remote servers how long to cache it: stable information (e.g., a machine's IP that hasn't changed in years) can have large TTL (e.g., 86400 = 1 day); volatile information should have small TTL (seconds or minutes).

9 What is the purpose of MX records? Why are they needed?

Answer: MX (Mail Exchange) records specify which host is prepared to accept email for a domain. They are needed because not every machine in a domain is configured to receive email. MX records include a priority value so multiple mail servers can be listed in order of preference.

10 Differentiate between CNAME and PTR records.

Answer: **CNAME (Canonical Name)** creates an alias for a domain name (e.g., www.cs.vu.nl -> star.cs.vu.nl). It is primarily used for convenience when services move. **PTR (Pointer)** is used for reverse lookups -- mapping an IP address back to a domain name. Both point to another name but serve different purposes.

11 Describe the SMTP mail transfer process step by step.

Answer: (1) Sending machine establishes TCP connection to port 25 of receiving machine. (2) Server sends greeting first (220 code). (3) Client sends HELO/EHLO to identify itself. (4) Client sends MAIL FROM: with sender address. (5) Client sends RCPT TO: with recipient address. (6) Client sends DATA command. (7) Server responds with 354 (ready for data). (8) Client sends message body, ending with a dot on its own line. (9) Server acknowledges with 250. (10) Client sends QUIT; server closes connection with 221.

12 What are the limitations of basic SMTP, and how does ESMTP address them?

Answer: Basic SMTP limitations: (1) No authentication -- FROM can be faked (spam), (2) ASCII-only -- requires base64 for binary, (3) No encryption -- no privacy. ESMTP addresses these through extensions: AUTH (authentication), BINARYMIME (binary data), STARTTLS (encryption), SIZE (message size checking), CHUNKING (large messages), UTF8SMTP (internationalized addresses).

13 Compare mail submission and message transfer in terms of port, authentication, and purpose.

Answer: **Mail submission** (UA to MTA): uses port 587, requires AUTH authentication, server checks/corrects message format, defined in RFC 4409. **Message transfer** (MTA to MTA): uses port 25, authentication not always required, less format checking, defined in RFC 5321.

14 Explain the difference between static and dynamic web pages.

Answer: A **static page** is the same every time it is displayed (e.g., a simple HTML file). A **dynamic page** is generated on demand by a program running on the server, and its content may vary per user or per request (e.g., a personalized storefront showing different products to different customers).

15 Explain URL, URI, and URN with differences.

Answer: **URL (Uniform Resource Locator)** specifies both the name and location of a resource (protocol + host + path). **URN (Uniform Resource Name)** specifies only the name without location information, allowing content to be found from any available copy. **URI (Uniform Resource Identifier)** is the general term encompassing both URLs and URNs.

16 What is HTTP pipelining, and why is it faster?

Answer: HTTP pipelining allows a client to send multiple HTTP requests on a persistent connection without waiting for each response. It is faster because: (1) No connection setup overhead per object, (2) TCP congestion control (slow-start) only happens once on a single connection rather than for multiple short connections, (3) Server idle time is minimized.

17 What are safe and idempotent HTTP methods? Give examples.

Answer: **Safe methods** do not modify server state: GET, HEAD, TRACE, OPTIONS. **Idempotent methods** produce the same result when executed multiple times: GET, HEAD, PUT, DELETE, TRACE, OPTIONS. POST is neither safe nor idempotent. PUT and DELETE are idempotent but not safe.

18 Explain the 5-tuple that uniquely identifies a TCP connection.

Answer: The 5-tuple consists of: (1) Protocol (TCP), (2) Source IP address, (3) Source port number, (4) Destination IP address, (5) Destination port number. Together these five values uniquely identify a single TCP connection between two hosts.

19 What is the three-way handshake in TCP, and which flags are used?

Answer: TCP connection establishment uses a three-way handshake: (1) Client sends SYN=1, ACK=0 (connection request), (2) Server replies with SYN=1, ACK=1 (connection accepted + acknowledgement), (3) Client sends ACK=1 to acknowledge the server's SYN. The SYN flag means "synchronize sequence numbers," and ACK means "acknowledgement number is valid."

20 What is the purpose of the TCP Window Scale option? When is it needed?

Answer: The Window Scale option allows TCP to overcome the 64 KB limit of the 16-bit Window Size field. It lets both sides negotiate a scale factor (up to 14 bits left shift), enabling window sizes up to 2^{30} bytes (~1 GB). It is needed on high-bandwidth, high-delay networks (long fat pipes) where the bandwidth-delay product exceeds 64 KB.

Section B: Numerical Problems

N1 A TCP connection uses the Window Scale option with a scale factor of 5. If the Window Size field in the TCP header shows 4000, what is the actual advertised window size in bytes?

Answer: Actual window = Window Size field $\times 2^{\text{scale_factor}}$ = $4000 \times 2^5 = 4000 \times 32 = 128,000$ bytes.

N2 A host wants to send data at 1 Gbps over a connection with 100 ms RTT. What minimum window size is needed to keep the pipe full? Does the basic TCP window (16-bit, max 64 KB) suffice? If not, what scale factor is needed?

Answer: Bandwidth-delay product = $1 \text{ Gbps} \times 100 \text{ ms} = 10^9 \times 0.1 = 12.5 \text{ MB}$.

Max basic window = $2^{16} - 1 = 65,535$ bytes = ~64 KB. This is insufficient.

Scale factor needed: $12.5 \text{ MB} / 64 \text{ KB} = \sim 200$, so we need $2^{\text{scale}} \geq 200$. Since $2^7 = 128$ and $2^8 = 256$, we need a **scale factor of at least 8**.

N3 A UDP datagram has 2000 bytes of data. What is the total size of the UDP segment? What is the value in the UDP Length field?

Answer: UDP header = 8 bytes. Data = 2000 bytes. Total segment size = 2008 bytes. The UDP Length field contains **2008** (header + data).

N4 What is the maximum amount of data that can be carried in a single TCP segment? Show the calculation.

Answer: Maximum IP packet = 65,535 bytes. IP header (minimum) = 20 bytes. TCP header (minimum) = 20 bytes. Max TCP data = $65,535 - 20 - 20 = 65,495$ bytes.

N5 A DNS A record has a TTL of 86400 seconds. How long will a caching name server keep this record? Convert this to hours and explain what it indicates about the stability of the record.

Answer: $\text{TTL} = 86400 \text{ seconds} = 86400 / 3600 = 24 \text{ hours (1 day)}$. This large TTL indicates the information is **highly stable** -- the IP address is unlikely to change, so it is safe to cache for a long duration.

N6 What is the maximum length of a domain name component? What is the maximum length of a full domain path name?

Answer: Each component (label) can be up to **63 characters**. The full domain name path must not exceed **255 characters** (including dots).

N7 A TCP sender has sent bytes 0 through 9999. It receives an acknowledgement with Ack number = 8000 and Window size = 5000. How many more bytes can it send? Explain.

Answer: Ack number 8000 means bytes 0-7999 are acknowledged. The sender can send from byte 8000 onward. Window size 5000 means up to 5000 bytes starting at 8000. So the sender can send bytes **8000 through 12999 = 5000 bytes** total.

N8 An HTTP 1.0 client needs to fetch a page with the main HTML file and 5 embedded images, all from the same server. How many TCP connections are needed? If each connection setup takes 1 RTT, what is the total connection setup time in RTTs?

Answer: HTTP 1.0 uses one connection per object. Total objects = 1 HTML + 5 images = 6 objects. So **6 TCP connections** are needed. Total connection setup time = $6 \times 1 \text{ RTT} = 6 \text{ RTTs}$. (In HTTP 1.1 with persistent connections, this would be just 1 RTT.)

N9 A UDP segment has a checksum field value of 0xFFFF (all 1s). What does this indicate? What if the value were 0x0000?

Answer: 0xFFFF (all 1s) means the checksum **was computed** and the result happened to be 0 (in 1's complement, 0 is stored as all 1s). 0x0000 means the checksum **was not computed** -- the sender skipped checksum calculation. This is unambiguous due to the properties of 1's complement arithmetic.

N10 At 56 kbps (original ARPANET speed), how long does it take to cycle through all 32-bit TCP sequence numbers? At 1 Gbps, how long does it take?

Answer: Total sequence numbers = $2^{32} = 4,294,967,296$ bytes.

At 56 kbps = 7,000 bytes/sec. Time = $4,294,967,296 / 7,000 = 613,567$ seconds = **~7.1 days (~1 week)**.

At 1 Gbps = 125,000,000 bytes/sec. Time = $4,294,967,296 / 125,000,000 = \text{~34.4 seconds}$.

This shows why sequence number wraparound (PAWS, timestamps) became critical at modern speeds.

N11 A DNS name server receives a query for www.example.com. It has cached the .com name server address but nothing else. How many query steps are needed to resolve the name? List them.

Answer: Since the .com server is cached, we skip root. Steps: (1) Local NS queries .com NS, (2) .com NS returns example.com NS, (3) Local NS queries example.com NS, (4) example.com NS returns the A record for www.example.com, (5) Local NS returns answer to client. Total = **5 steps** (vs. 10 steps with no cached info).

N12 If a TCP connection uses the MSS option with value 1460, and the IP header is 20 bytes, what is the total size of each segment? How many segments are needed to send 100 KB of data?

Answer: MSS = 1460 bytes (payload). TCP header = 20 bytes. IP header = 20 bytes. Total per segment = $1460 + 20 + 20 = \text{1500 bytes}$.

Data to send = 100 KB = 102,400 bytes. Number of segments = $\text{ceil}(102,400 / 1460) = \text{ceil}(70.14) = \text{71 segments}$. The last segment carries $102,400 - (70 \times 1460) = 102,400 - 102,200 = 200$ bytes.

Q13 Calculate the UDP checksum for a packet where the 16-bit words (including header, zero-padded data, and pseudoheader) sum to 0x12345 in 1's complement arithmetic. What value goes in the checksum field?

Answer: In 1's complement, a 5-digit sum 0x12345 wraps around: add the carry back: $0x2345 + 0x1 = 0x2346$. Checksum = 1's complement of 0x2346 = **0xDCB9**. The receiver adds all 16-bit words including checksum; the result should be 0xFFFF (which represents 0 in 1's complement).

Q14 An SMTP server responds with code "354". What does this mean? A client then sends a dot on a line by itself. What response code should it expect?

Answer: Code **354** means "Start mail input; end with CRLF.CRLF" -- the server is ready to receive the message body. After the client sends the terminating dot on its own line, the server should respond with **250** (message accepted/requested action completed OK).

Q15 A web page has 1 HTML file, 8 images, 2 CSS files, and 1 JavaScript file -- all from the same server. Compare the total connection setups needed for HTTP 1.0 vs HTTP 1.1 (persistent).

Answer: Total objects = $1 + 8 + 2 + 1 = 12$ objects.

HTTP 1.0: 12 TCP connections (one per object) = 12 connection setups.

HTTP 1.1: 1 TCP connection (persistent, reused) = 1 connection setup.

Savings = 11 connection setups. Each setup requires at least 1 RTT, so HTTP 1.1 saves at least **11 RTTs** of setup time.

Final Revision Checklist

- Transport entity locations and the provider-user distinction (layers 1-4 vs 5+)
- UDP header format (8 bytes: 4 fields of 16 bits each) and checksum calculation
- IPv4 pseudoheader composition (src IP, dst IP, pad, protocol, UDP length)
- TCP header all fields, especially the 8 flags and their meanings
- 5-tuple connection identification
- TCP options: MSS, Window Scale, Timestamp, SACK
- DNS hierarchical name space, zone delegation, ICANN role
- DNS resolution: recursive vs iterative, 10-step example
- All RR types: SOA, A, AAAA, MX, NS, CNAME, PTR, SRV, TXT
- SMTP command sequence with numeric codes (220, 250, 354, 221)
- ESMTP extensions and their purposes
- Mail submission (port 587, AUTH) vs message transfer (port 25)
- URL components, URI/URN distinction
- HTTP methods (safe, idempotent), status code groups (1xx-5xx)
- HTTP connection strategies: multiple, persistent, pipelined, parallel